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APPENDICES

Appendix A Yokogawa Process Control Tables
Appendix B MOSCAD RTU Process Control Tables

ACRONYMS AND DEFINITIONS

Acronym	Description
CRO	Control room operator
CV	Control variable
DCS	Distributed control system
FCS	Field control station
HP	High pressure
HMI	Human machine interface
IBC	Intermediate bulk container
LCP	Local control panel
LCS	Local control station
~~Owner~~	Owner Resources Limited
OCS	Operator control station
OIS	Operator interface system
~~Owner~~	~~Owner Resources Limited~~
P&ID	Piping and instrumentation diagram
PCS	Plant control system; Yokogawa DCS
PID	Proportional, integral and differential
PLC	Programmable logic controller
PV	Process variable
SCADA	Supervisory control and data acquisition; General Electric iFIX
SG	Specific gravity
SP	Setpoint variable
TBC	To be confirmed
TSF	Tailings storage facility
VSD	Variable speed drive
WSCS	Water services control system; Motorola PLC hardware and MOSCAD software

~~HOLDS~~

~~Number~~	~~Description~~
~~002~~	~~Require vendor data for the Seal Water Self Cleaning Filter~~

1 INTRODUCTION

This document provides a description of the Owner FIM TSF Project process control philosophy, which is intended to provide a basis for the Process Control System (PCS) programming. The control of the new tailings infrastructure will be integrated with the existing plant control system and follow current site practises and standards.

This document may use slightly different nomenclature to that currently in use at Owner Given this project is to be integrated into the existing plant and control system, the nomenclature currently in use will take precedence.

The tailings disposal system requires upgrading as part of the wider Owner Growth Project. The tailings disposal system will be upgraded to include:

- A new final tails tank and associated agitator (by EPC
- Four new primary final tails pumps (by EPC
- Provision for flushing the new primary final tails pumps with high pressure process water to the pump discharge and low pressure saline water to the suction side (by EPC
- Further upgrades to the FIM II Extension TSF, including:
  - A new valve station (valve station No.1) downstream of the FIM II booster station (by Owner Operations).
  - A new valve station (valve station No.2) downstream of valve station No.1 (by Owner Operations).
  - A new valve station (valve station No.3) downstream of valve station No.2 (by Owner Operations).
  - A new cell (cell G) including a central decant turret and decant pump.
  - FIM II Extension cell G underdrainage system and underdrainage pump.
  - FIM II Extension cell G toe drain system and two toe drain pumps.
- A new TSF, FIM III consisting of:
  - FIM III valve station downstream of the FIM III booster station.
  - Provision to flush the FIM III valve station with decant water from Decant Dam 4.
  - Hydraulic powerpack at the FIM III valve station.
  - Three separate cells, each fitted with a central decant turret and decant pump.
  - Underdrainage system for cell 1 including underdrainage pit and underdrainage pump.
  - Underdrainage systems for cells 2 and 3 with a common underdrainage pit and underdrainage pump.
  - Four FIM III seepage bores and pumps.
  - Toe drain system and toe drain pump.
  - Decant water collection pond, Decant Dam 4, and three decant return pumps in a duty/duty/standby configuration.
- A fourth tailings line (tailings Line D) which will deposit tailings to FIM I (via existing piping) and FIM III.
- A new booster station at FIM III consisting of:
  - Two booster pumps.
  - Provision to flush the booster pumps with decant water from Decant Dam 4.
  - Seal water tank, supplied from the Eastern Borefields Saline Dam via a self-cleaning filter.
  - Duty/standby seal water pumps for each booster pump.
  - A booster station washdown pump.
  - A hydraulic powerpack.
  - A self-contained safety shower including tank and chiller.
- Extension of tailings line B to deposit tailings to FIM III.

- Decant return water systems for both existing and new decant water systems with cyanide destruction by hydrogen peroxide (described elsewhere) into new inline mixers at the southern part of the FIM plant. The decant streams include:
  - Kaltails decant.
  - A combined stream of water from offsite roaster Saline Dam and FIM I decant.
  - A combined stream of water from Decant Dam 3 and Decant Dam 4.

2 REFERENCE DOCUMENTS

This document must be read in conjunction with Appendix A - Control & Instrumentation Tables (Yokogawa DCS) and Appendix B - Control & Instrumentation Tables (MOSCAD RTU), which provide details of individual instruments, control loops, calculations etcetera, and the latest revisions of the drawings and documents listed in Section 2.1. A table of any vendor documents will be included when the documentation is available.

2.1 Client/Project Documentation

- 2000-IC001-M07-20001_4 DCS Template Specification
- 2000-IC001-J02-20001_0 DCS Templates User Manual
- Owner FIM TSF Project Tailings Sequence Flowcharts 2000-F00-LST-PR-40001.
- Owner FIM TSF Project FIM III Valve Station Sequence Flowcharts 2000-F00-LST-PR-40002.
- Kaltails Tailings Booster Pump Station Control Philosophy.
- FIM 1 Tailings Line & Booster Pump Upgrade Operation and Control Philosophy CTRL-OPS-2021-C10799_R0.
- FIM 2 Tailings Booster Pump Station Control Philosophy.
- Owner Growth Project Piping and Instrumentation Diagrams (P&IDs) listed in Table 2.1

Table 2.1 Piping & Instrumentation Diagrams

Number	Name
2000-F23-PID-PR-10001	Final Tails Tank Piping & Instrumentation Diagram
2000-F23-PID-PR-10002	Final Tails Pumps Piping & Instrumentation Diagram
2000-F23-PID-PR-40002	FIM III Booster Station - Sheet 1 of 2 Piping & Instrumentation Diagram
2000-F23-PID-PR-40003	FIM III Booster Station - Sheet 2 of 2 Piping & Instrumentation Diagram
2000-F23-PID-PR-40005	FIM III Valve Station Piping & Instrumentation Diagram
2000-F23-PID-PR-40006	FIM III Deposition Cell 1 Piping & Instrumentation Diagram
2000-F23-PID-PR-40007	FIM III Deposition Cell 2 Piping & Instrumentation Diagram
2000-F23-PID-PR-40008	FIM III Deposition Cell 3 Piping & Instrumentation Diagram
2000-F30-PID-PR-40003	FIM II Extension Deposition Cell G Deposition Piping & Instrumentation Diagram
2000-F75-PID-PR-25007	Tailings and Water Services – Sheet 1 of 3 Piping & Instrumentation Diagram
2000-F75-PID-PR-25008	Tailings and Water Services – Sheet 2 of 3 Piping & Instrumentation Diagram
2000-F75-PID-PR-25010	Tailings and Water Services – Sheet 3 of 3 Piping & Instrumentation Diagram
2000-F75-PID-PR-40001	FIM II Cell G Decant Piping & Instrumentation Diagram
2000-F75-PID-PR-40002	FIM III Decant - Sheet 1 of 2 Piping & Instrumentation Diagram
2000-F75-PID-PR-40003	FIM III Decant - Sheet 2 of 2 Piping & Instrumentation Diagram
2000-F75-PID-PR-40004	Decant Dam 4 Piping & Instrumentation Diagram
2000-F75-PID-PR-40006	FIM II Cell G Seepage Water Piping & Instrumentation Diagram
2000-F75-PID-PR-40007	FIM II Cell G Underdrainage and Toe Drain Pumps Piping & Instrumentation Diagram
2000-F75-PID-PR-40008	FIM III Typical Seepage Pump Arrangement Piping & Instrumentation Diagram
2000-F75-PID-PR-40009	FIM III Seepage Water Piping & Instrumentation Diagram
2000-F75-PID-PR-40011	FIM III Toe Drain Pump Piping & Instrumentation Diagram
2000-F75-PID-PR-40012	FIM III Underdrainage Pumps Piping & Instrumentation Diagram

Number	Name
2000-F75-PID-PR-40014	FIM III Booster Station Seal Water Piping & Instrumentation Diagram
2000-F75-PID-PR-40016	FIM II Valve Station No. 1 Piping & Instrumentation Diagram
2000-F75-PID-PR-40020	FIM I Booster Pump Piping & Instrumentation Diagram
2000-F75-PID-PR-40021	Kearns Corner Piping & Instrumentation Diagram
2000-F75-PID-PR-40022	FIM II Booster Station Upgrade Piping & Instrumentation Diagram
2000-F75-PID-PR-40023	Kaltails Booster Station Upgrade Piping & Instrumentation Diagram
2000-F75-PID-PR-40024	FIM II Booster Station – Hub Piping & Instrumentation Diagram
2000-F75-PID-PR-40025	Valve Station No. 2 Piping & Instrumentation Diagram
2000-F75-PID-PR-40026	Valve Station No. 3 Piping & Instrumentation Diagram
2000-F75-PID-PR-40029	Seepage Headers Piping & Instrumentation Diagram
2000-F75-PID-PR-40031	Decant Dam 3 Pump Upgrade Sheet 1 of 2 Piping & Instrumentation Diagram
2000-F75-PID-PR-40032	Decant Dam 3 Pump Upgrade Sheet 2 of 2 Piping & Instrumentation Diagram
2000-F175-PID-PR-40002	Eastern Borefields Saline Dam (SEBDT01HK) – Transfer Pumps Piping & Instrumentation Diagram

3 TAILINGS DISPOSAL

Plant tailings will be pumped from the new final tails tank by the new primary final tails pumps to either of the following TSFs:

- Kaltails (East or West).
- FIM I.
- FIM II cells A/B, C and D and FIM II Extension cells E, F and G.
- FIM III cells 1, 2 and 3.

At the full Growth Project throughput rate, three tailings lines will be required at any given time. However, in normal operating conditions all four tailings lines will be used, thereby configured as n-1. In the event that a primary final tails pump unexpectedly shuts down, the other three pumps will ramp up in speed to maintain the level in the final tails tank. The tailings lines will be configured as follows:

- Line A: Kaltails, FIM II, FIM II Extension.
- Line B: Kaltails, FIM III.
- Line C: Kaltails, FIM II, FIM II Extension.
- Line D: FIM I, FIM III.

3.1 Possible Destinations and Design Limitations

A summary of the four tails lines and the limitations of each line are summarised in Table 3.1

Table 3.1 Possible Tails Line Destinations & Flowrates (m^3/h)

Line	FIM I	FIM II			FIM II Ext			Kaltails		FIM III		
		A/B	C	D	E	F	G	West	East	Cell 1	Cell 2	Cell 3
MINIMUM FLOWRATES												
A	-	750	-	-	750			750		-		
B	-	-	-	-	-			750		1,000		
C	-	750	-	-	750			750		-		
D	500	-	-	-	-			-		1,000		
MAXIMUM FLOWRATES												
A	-	1,000[1]	-	-	1,300[1,2]	1,300[1,2]	1,300[1]	1,000[3]		-		
B	-	-	-	-	-			1,000[3]		1,500[4]		
C	-	1,000[1]	-	-	1,300[1,2]	1,300[1,2]	1,300[1]	1,000[3]		-		
D	800	-			-			-		1,500[4]		

3.2 Control and Operating Requirements

The selection of which TSFs are used will be as per the current operation, with personnel from the site services team required to open and close manual valves at various locations to achieve the desired flow path. The duty for each of the final tails pumps will be allocated through the PCS which will set the minimum and maximum flowrate and maximum allowable discharge pressure for each of the pumps according to Table 3.1 above.

Limitations may be required when operating above nominal plant throughput rates in order to not overload the tailings area switchroom transformer. Preliminary calculations have determined that the

[1] Both pump tip speeds will operate above 32 m/s.
[2] Due to pressure limitations, existing ~~pipe~~ piping on top of the TSF will need to be upgraded within 3 years to maintain the maximum 1,300 m^3/h flowrate.
[3] Tip speed of booster pump will be above 32 m/s.
[4] Tip speed will be above 32 m/s at 1,500 m^3/h for line B primary pump and Line D pending delivery location.

maximum load case will be 3 final tailings pumps operating at approximately 4,300 m^3/h with a load of approximately 2.1 MW. It is recommended that the output of LIC-23401B is capped at 4,300 m^3/h (TBC during commissioning).

~~Only once this has been completed will the primary final tails pumps and booster pumps be started from the PCS.~~

The control and operating requirements of the new tailings infrastructure are as follows:

- Modification of the existing FIM tailings disposal pump control logic and new final tails tank (located at the ~~west~~ east end of CIL~~4~~3) (by EPC
- Control and sequence logic for the new final tails pumps (~~also referred to as the primary final tailings pumps which will be~~ located at the east end of CIL4).
- Control and sequence logic for the new FIM III booster pumps and associated seal water pumps.
- Control and sequence logic for the new FIM III valve station (located downstream of the FIM III booster station).
- Modification of existing control ~~and sequence~~ logic and interlocks for the existing Kaltails booster pumps and associated seal water pumps.
- Modification of existing control ~~and sequence~~ logic and interlocks for the existing FIM I booster pump and associated seal water pump.
- Modification of existing control and sequence logic and interlocks for the existing FIM II booster pumps and associated seal water pumps, with~~.~~ ~~Bypass~~ bypass capabilities for lines A and C to FIM II Extension.

Flushing of the tailings lines can be completed at the following locations, by the following methods:

- Addition of saline water to the suction line of the final tails pumps (automated sequence initiated by the CRO from the PCS).
- Addition of high pressure process water to the discharge line of final tails pumps (actuated valves opened/closed by the CRO from the PCS).
- At Kearn's corner using decant water (manual valves, opened/closed locally by site services team).
- At the Kaltails booster station using decant water from Kaltails Decant Dam (manual valves, opened/closed by the site services team).
- At the Kaltails valve station using decant water from Kaltails Decant Dam (manual valves, opened/closed by the site services team).
- At the FIM II booster station using decant water from ~~d~~Decant ~~d~~Dam 3 (manual valves, opened/closed by the site services team).
- At the FIM II valve station No.1 using decant water from ~~d~~Decant ~~d~~Dam 3 (manual valves, opened/closed by the site services team).
- At the FIM II valve station No.2 using decant water from ~~d~~Decant ~~d~~Dam 3 (manual valves, opened/closed by the site services team).
- At the FIM II valve station No.3 using decant water from ~~d~~Decant ~~d~~Dam 3 (manual valves, opened/closed by the site services team).
- At the FIM III booster station (upstream of the booster pumps in the suction lines) using decant water from ~~d~~Decant ~~d~~Dam 4 (manual and hydraulically actuated valves, opened/closed locally by the site services team or from the PCS).
- At the FIM III valve station using decant water from ~~d~~Decant ~~d~~Dam 4 (manual and hydraulically actuated valves, opened/closed locally by the site services team via a vendor PLC ~~or from the PCS~~).

3.3 Control System

The control system hardware for the existing Kaltails, FIM I and FIM II booster stations will remain unchanged.

The existing Yokogawa DCS hardware and software (PCS) at the FIM plant that is being used to facilitate the integration of the FIM III booster station is being upgraded (by EPC A new node at the FIM III booster station will be installed. A fibre optic cable (via a buried fibre) will be run to the new switchroom installed at the FIM III booster station to facilitate this. As a result, the new FIM III booster station pumping system will be configured as a remote ~~I/O node~~field control station (FCS) for the Yokogawa plant DCS, unlike the existing booster station pumping systems.

The existing locally installed Yokogawa PLCs will be retained:

- Kaltails booster station PLC.
- FIM I booster station PLC.
- FIM II booster station PLC.

These PLCs are controlled by telemetry signals from the PCS.

The following is a high level summary of the new tailings infrastructure controlled by the PCS:

- Final ~~t~~Tails ~~t~~Tank ~~a~~Agitator.
- Final ~~t~~Tails ~~p~~Pumps 1 – 4 (including ancillary drives).
- FIM III ~~b~~Booster ~~p~~Pumps 1 & 2 (including ancillary drives).
- FIM III ~~b~~Booster ~~p~~Pump 1 ~~s~~Seal ~~w~~Water ~~p~~Pumps 1 & 2.
- FIM III ~~b~~Booster ~~p~~Pump 2 ~~s~~Seal ~~w~~Water ~~p~~Pumps 1 & 2.
- FIM III ~~b~~Booster ~~p~~Pumps ~~s~~Seal ~~w~~Water ~~s~~Self-~~c~~Cleaning ~~f~~Filter.
- FIM III ~~b~~Booster ~~s~~Station ~~w~~Washdown ~~p~~Pump.
- FIM III ~~b~~Booster ~~s~~Station ~~s~~Safety ~~s~~Shower.

3.4 Final Tailings Pumps Process Control (2000-F23-PID-PR-10001 / 10002)

The final tails tank level element will provide input to a level controller that will provide input into two independent level control loops. Level control loop A will be used to control water addition to the final tails tank, while level control loop B will be used to control the operating final tails pumps' speed. The output of the level control loop B will cascade to a calculation block which will split the output from the level controller to each of the operating final tails pumps' flow controllers. The split will not necessarily be equal and will depend on which TSFs are selected and how many tailings lines are in operation. The flow controllers will in turn adjust the speed of each of the final tails pumps to maintain the required tails flowrate. If one pump were to trip, the calculation block would adjust the split of the level controller output, and the other three pumps would increase in speed (limited to the maximum flowrate based on the pumping destination) to meet the flowrate requirements. If a booster pump were to trip, the associated final tails pump will also trip.

In order for this arrangement to work correctly, the level setpoint for control loop A (saline water addition) must be lower than the level setpoint for control loop B (final tails pump speed).

The control loops will operate as follows to maintain the final tails tank level:

On low slurry flow to the final tails tank, the operating final tails pumps' speed will reduce to maintain tank level, until the flow reaches the minimum allowable flow. Should the incoming flow to the tank be below the minimum allowable discharge flow, the level in the tank will fall below the setpoint for level controller B.

As the final tails tank level reduces further, it will fall below the setpoint for level controller A, meaning the saline water addition control loop will open the water addition valve in order to maintain the tank

level at the (lower) setpoint for control loop A. The operating final tails pumps speed will not alter as the tank level will still be below the setpoint for level control loop B.

Once slurry flow to the final tails tanks is increased, the tank level will rise above that for control loop A, causing the saline water addition rate to decrease and the tank level will again be controlled by level control loop B adjusting the speed of the operating final tails pumps. This arrangement will provide seamless control of the final tails tank level.

The final tails tank level / final tails flow can be controlled as follows:

- Cascade mode: The level element on the final tails tank will provide input to a level controller (level control loop B), with the level setpoint entered by the operator. The output of the level controller will cascade to a flow controller (via a ratio calculation block), which in turn will adjust the speed of the relevant final tails pump to maintain the required tails flowrate.
- Auto: The final tails pump flow controller will maintain an operator input flowrate by adjusting the speed of the relevant final tails pump. The flowrate will be measured by a flowmeter on the tails line.
- Manual mode: The operator will enter a speed percentage for the relevant final tails pump.

Level control loop A is outside the scope of this document. Due to legacy issues with existing tailings piping, very slow ramp-up times (minutes) for the final tails pumps will be required.

3.5 Tailings Booster Pump Process Control

3.5.1 Kaltails Tailings Booster Pumps (2000-F75-PID-PR-40021 / 40023)

Two booster pumps are currently installed at the Kaltails TSF, Kaltails ~~b~~Booster ~~p~~Pumps 1 and 2 (F30-PP-78A and F30-PP-78B). Kaltails ~~b~~Booster ~~p~~Pump 1 can be configured to operate with tails lines A and B, while Kaltails ~~b~~Booster ~~p~~Pump 2 can only be used with tails line C.

Control for both Kaltails booster pumps will be modified to follow the logic used to control the existing FIM I ~~b~~Booster ~~p~~Pump. Each booster pump will be controlled as follows:

Cascade mode: The reference speed of the final tails pump selected to pump to the Kaltails TSF via the relevant Kaltails booster pump will be used as the speed setpoint for the Kaltails booster pump with an offset (negative, nominally -10%, TBC during commissioning). The booster pump speed controller will adjust the speed of the Kaltails booster pump to maintain the required speed offset.

Auto mode: The pressure controller will adjust the speed of the Kaltails booster pump to maintain an operator input suction pressure setpoint, with the pressure measured by a pressure transmitter on the booster pump suction line.

Manual mode: The operator enters a speed percentage for Kaltails booster pump.

The existing booster pump control strategy will be retained in the DCS and will be used to control the respective booster pump only if conditions dictate.

Any drive interlocking for the Kaltails booster pumps associated with the existing tailings disposal pumps F30-PP-62A, F30-PP-62B, F30-PP-62C and F30-PP-62D will ~~require modification to no longer consider these pumps and~~be modified to ~~instead~~ reference ~~to~~ the new final tails pumps F23-PP-467, F23-PP-468 (F30-PP-78A) and F23-PP-469 (F30-PP-78B), respectively. All other aspects of control for the Kaltails booster pumps will remain unchanged.

3.5.2 FIM I Tailings Booster Pump (2000-F75-PID-PR-40020)

A single booster pump is currently installed at the FIM I TSF, FIM I ~~b~~Booster ~~p~~Pump (F30-PP-79). The FIM I ~~b~~Booster ~~p~~Pump will be reconfigured so that it can only be used with tails line D. Any drive interlocking for the FIM I ~~b~~Booster ~~p~~Pump associated with the existing tailings disposal pumps F30-PP-

62B and F30-PP-62D will ~~require modification to no longer consider these pumps and instead~~be modified to reference ~~to~~ the new final tails pump F23-PP-466. All other aspects of control for the FIM I ~~b~~Booster ~~p~~Pump will remain unchanged.

3.5.3 FIM II Tailings Booster Pumps (2000-F75-PID-PR-40021 / 40022)

Two booster pumps are currently installed at the FIM II TSF, FIM II ~~b~~Booster ~~p~~Pumps 1 and 2 (F30-PP-77A and F30-PP-77B). The configuration of the pumps will remain unchanged, with FIM II ~~b~~Booster ~~p~~Pump 1 dedicated to tails line C and FIM II ~~b~~Booster ~~p~~Pump 2 dedicated to tails line A.

Control for both FIM II booster pumps will be modified to follow the logic used to control the existing FIM I ~~b~~Booster ~~p~~Pump. Each booster pump will be controlled as follows:

Cascade mode: The reference speed of the final tails pump selected to pump to the FIM II TSF via the relevant FIM II booster pump will be used as the speed setpoint for the FIM II booster pump with an offset (negative, nominally -10%, TBC during commissioning). The booster pump speed controller will adjust the speed of the FIM II booster pump to maintain the required speed offset.

Auto mode: The pressure controller will adjust the speed of the FIM II booster pump to maintain an operator input suction pressure setpoint, with the pressure measured by a pressure transmitter on the booster pump suction line.

Manual mode: The operator enters a speed percentage for FIM II booster pump.

The existing control strategy will be retained in the DCS and will be used to control the respective booster pump only if conditions dictate.

Any drive interlocking for the FIM II booster pumps associated with the existing tailings disposal pumps F30-PP-62A, F30-PP-62B, F30-PP-62C and F30-PP-62D will ~~require modification to no longer consider these pumps, and instead~~be modified to reference ~~to~~ the new final tails pumps, F23-PP-469 (F30-PP-77A) and F23-PP-468 (F30-PP-77B), respectively. All other aspects of control for the FIM II booster pumps will remain unchanged.

~~Any~~ All existing sequences for the FIM II booster pumps will ~~need to~~ be revised to include a pause in the start sequence to confirm that ~~new~~ manual valves installed at the new FIM II valve station No.1 are in the correct position prior to the pumps starting.

3.5.4 FIM III Tailings Booster Pumps (2000-F23-PID-PR-40002 / 40003)

Two new booster pumps will be installed at the FIM III TSF, FIM III ~~b~~Booster ~~p~~Pumps 1 and 2 (F23-PP-651 and F23-PP-652). The booster pumps will be installed approximately 5.6 km from the plant and will operate in a duty/duty configuration. FIM III ~~b~~Booster ~~p~~Pump 1 will be dedicated to tails line D and FIM III ~~b~~Booster ~~p~~Pump 2 dedicated to tails line B. The FIM III booster pump~~s~~' start will be permitted once the pressure in the respective booster pump~~s~~' suction line is above the low alarm setpoint.

In normal operation if two tails lines are required to discharge tails to FIM III, lines B and D will operate as follows:

- Line B operating using ~~f~~Final ~~t~~Tails ~~p~~Pump 2 and FIM III ~~b~~Booster ~~p~~Pump 2.
- Line D operating using ~~f~~Final ~~t~~Tails ~~p~~Pump 1 and FIM III ~~b~~Booster ~~p~~Pump 1.

The valves at the FIM III booster station will be hydraulically actuated.

Control for both FIM III booster pumps will follow the logic used to control the existing FIM I ~~b~~Booster ~~p~~Pump. A flowmeter installed on the discharge line of each booster pump may be used to control the respective booster pump if desired. Each booster pump will be controlled as follows:

- Cascade mode: The reference speed of the final tails pump selected to pump to the FIM III TSF via the relevant FIM III booster pump will be used as the speed setpoint for the FIM III booster pump with an offset (negative, nominally -10%, TBC during commissioning). The booster pump speed controller will adjust the speed of the FIM III booster pump to maintain the required speed offset.
- Auto mode: The flow controller will adjust the speed of the FIM III booster pump to maintain an operator input flow setpoint, with the flowrate measured by a flowmeter on the booster pump discharge line.
- Manual mode: The operator enters a speed percentage for FIM III booster pump.

3.5.5 Sequence Logic

The following sequences are detailed in the Owner FIM TSF Project Tailings Sequence Flowcharts (2000-F00-LST-PR-40001):

- Final Tails Pump 1 (F23-PP-466) Start.
- Final Tails Pump 1 (F23-PP-466) Stop.
- Final Tails Pump 1 (F23-PP-466) Flush.
- Final Tails Pump 2 (F23-PP-467) Start.
- Final Tails Pump 2 (F23-PP-467) Stop.
- Final Tails Pump 2 (F23-PP-467) Flush.
- Final Tails Pump 3 (F23-PP-468) Start.
- Final Tails Pump 3 (F23-PP-468) Stop.
- Final Tails Pump 3 (F23-PP-468) Flush.
- Final Tails Pump 4 (F23-PP-469) Start.
- Final Tails Pump 4 (F23-PP-469) Stop.
- Final Tails Pump 4 (F23-PP-469) Flush.
- FIM III Booster Pump 1 (F23-PP-651) Start.
- FIM III Booster Pump 1 (F23-PP-651) Stop.
- FIM III Booster Pump 1 (F23-PP-651) Flush.
- FIM III Booster Pump 2 (F23-PP-652) Start.
- FIM III Booster Pump 2 (F23-PP-652) Stop.
- FIM III Booster Pump 2 (F23-PP-652) Flush.

3.5.6 Tailings Slurry Differential Flow

Flowmeters at either end of each of the four tailings lines will monitor the slurry flowrate down each line from the final tails tank to the discharge of the relevant booster pump. Any significant differential between the two measured flow values will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the relevant final tails pump. The following differential flow calculations in the PCS will raise an alarm on the PCS:

- Line A to Kaltails using F23-PP-468 and F30-PP-78A.
- Line A to FIM II using F23-PP-468 and F30-PP-77B.
- Line A to FIM II Extension using F23-PP-468 and F30-PP-77B.
- Line B to Kaltails using F23-PP-467 and F30-PP-78A.
- Line B to FIM III using F23-PP-467 and F23-PP-652.
- Line C to Kaltails using F23-PP-469 and F30-PP-78B.
- Line C to FIM II using F23-PP-469 and F30-PP-77A.
- Line C to FIM II Extension using F23-PP-469 and F30-PP-77A.
- Line D to FIM I using F23-PP-466 and F30-PP-79.
- Line D to FIM III using F23-PP-466 and F23-PP-651.

3.5.7 TSF Selection

Selection of which TSF cell (or paddock) is used for tailings deposition will be by Owner Operations which will be informed by the Deposition Plan developed by WSP. There are existing manual valves at the following locations that will require opening or closing by the site services team as required to send tailings to the selected TSF:

- ~~Tails Station 6.~~
- Kearn's Corner.
- FIM II Bypass Valve Station.
- FIM II Valve Station Hub.
- FIM II Extension Valve Station Hub.

These valves have no position feedback to the PCS.

3.5.8 FIM III Cell Selection (2000-F23-PID-PR-40005 to 40008)

FIM III will consist of three separate cells. Selection of the active cell will be made by opening or closing hydraulically actuated valves located at the FIM III valve station~~, either locally from a local control panel or~~ via a sequence manually initiated from a local vendor control panel~~the PCS~~. There are also various manual valves, without position feedback, on top of the TSF that will need to be opened or closed, by the field operator as required based on the cell selection.

~~Spigotting~~ Spigot management into each tailings cell will be fully manual, with the location and number of operating spigots selected and monitored by the field operator.

For details of the ~~flushing~~ deposition sequences refer to the Owner FIM TSF Project FIM III Valve Station Tailings Sequence Flowcharts (2000-F00-LST-PR-40002).

3.6 Final Tails Pump Flushing (2000-F23-PR-PID-10002 / 40002 / 40003)

If a tailings line is to be taken out of service for an extended period, it should be flushed to enable the line to be brought back online without issue. Flushing will be completed by adding saline water to the suction line of each ~~primary~~ final tails pump (if the respective pump is available to run) or using high pressure (HP) process water to the discharge line of each ~~primary~~ final tails pump if the respective pump is inoperable. Flushing mode~~s~~ will be ~~able to be~~ selected from the DCS for each of the final tails pumps. In this mode the interlocks associated with the downstream booster pumps will be bypassed, so that the final tails pump can be used to flush the section of tailings line to the booster station.

3.6.1 Flushing with Saline Water

When flushing a tailings line with saline water, the respective ~~primary~~ final tails pump and booster pump should first be shut down using the associated shutdown sequences. The ~~primary~~ final tails pump will then be restarted using a dedicated flushing sequence and if available the relevant booster pump will be restarted using the startup sequence. The tailings line will be flushed from the final tails tank to the relevant TSF with the flushing timer set to 90 minutes (TBC during commissioning). ~~until the site services team are satisfied that the line is sufficiently flushed. This will require v~~ Visual confirmation at the TSF by the site services team~~.~~ will also be required to ensure the line is sufficiently flushed.

If the relevant booster pump is not available, the tailings line ~~will~~ can be flushed from the final tails tank to the scour pit at the relevant booster station. ~~The tailings line will be flushed until the site services team are satisfied that the line is sufficiently flushed. This will require~~The flushing timer will be set to 90 minutes (TBC during commissioning) and visual confirmation at the relevant booster station scour pit ~~by the site services team.~~ by will also be required to ensure the line is sufficiently flushed.

Up to two tailings lines can be flushed at any given time based on the capacity of the saline water pumps.

Saline water addition to each of the final tails pump suction lines can be controlled as follows:

- Auto mode: The pressure controller will adjust the open percentage of the pressure control valve to maintain an operator input pressure setpoint, with the pressure measured by a pressure transmitter on the saline water addition line.
- Manual mode: The operator enters an open percentage for the pressure control valve.

For details of the sequences refer to the Owner FIM TSF Project Tailings Sequence Flowcharts (2000-F00-LST-PR-40001).

3.6.2 Flushing with High Pressure Process Water

If a ~~primary~~ final tails pump becomes inoperable and a tailings line is shut down unexpectedly, the line may require ~~to be~~ flush<u>inged</u>~~ed~~ using HP process water. The inoperable pump will be isolated, with the pump suction and discharge valves closed, prior to the HP process water addition valve being opened from the PCS by the CRO. If available, the relevant booster pump will be restarted using the startup sequence, and the tailings line will be flushed from the discharge of the final tails pump to the relevant TSF until the site services team are satisfied that the line is sufficiently flushed. This will require visual confirmation at the TSF by the site services team.

If the relevant booster pump is not available, the tailings line will be flushed from the discharge of the final tails pump to the scour pit at the relevant booster station. The tailings line will be flushed until the site services team are satisfied that the line is sufficiently flushed. This will require visual confirmation at the relevant booster station scour pit by the site services team.

<u>Flushing with HP process water is started manually and the duration of flushing will be determined by visual confirmation at either the relevant booster station scour pit or TSF destination by the site services team, who will determine when the line is sufficiently flushed to stop the manual flushing process.</u>

3.7 Booster Station Pump Flushing

3.7.1 Kaltails

Flushing of the tailings lines from the Kaltails booster station to the Kaltails TSF will be unchanged from the existing operation.

3.7.2 FIM II

Flushing of the tailings lines from the FIM II booster station to the FIM II TSF will be unchanged from the existing operation.

3.7.3 FIM III <u>(2000-F23-PID-PR-40002 / 40003)</u>

Flushing of the tailings lines from the FIM III booster station to the FIM III TSF will be by a sequence manually initiated from the PCS if the relevant FIM III booster pump is available. Flushing mode will be able to be selected from the DCS for each of the FIM III booster pumps. In this mode the interlocks associated with the upstream final tails pumps will be bypassed, so that the FIM III booster pump can be used to flush the section of tailings line from the booster station to the TSF. This will likely only be undertaken if the final tails pumps are inoperable and both saline water and high pressure process water are unavailable. The suction line of the FIM III booster pump<u>, as well as the decant return line,</u> will be isolated and decant water from Decant Dam 4 will be added using two Decant Dam 4 pumps. The section of tailings line from the booster station to the TSF will be flushed using the FIM III booster pump<u>, with the flushing timer set to 90 minutes (TBC during commissioning).</u> ~~until the site services team are satisfied that the line is sufficiently flushed. This will require v~~<u>V</u>isual confirmation at the TSF by the site services team<u>. will also be required to ensure the line is sufficiently flushed.</u>

Flushing of the tailings lines from the FIM III booster station to the FIM III TSF ~~will~~ can also be manually initiated from the PCS ~~and will only be required if~~when a FIM III booster pump is inoperable. In this case, ~~F~~flushing of the tailings lines from the FIM III booster station to the FIM III TSF will be completed once a tailings line has been flushed from the discharge of the relevant final tails pump to the scour pit at the FIM III booster station. Once this has been completed, the section of tailings line from the booster station to the TSF will be flushed through the shut-down booster pump to the relevant TSF.

Flushing water for the FIM III booster station will be taken from Decant Dam 4. All valves associated with the flushing water arrangement can be opened and closed either locally from a local control panel or from the PCS and will have position feedback.

For details of the flushing sequences refer to the Owner FIM TSF Project Tailings Sequence Flowcharts (2000-F00-LST-PR-40001).

3.7.4 Tailings Cell Pipeline Flushing (2000-F23-PID-PR-40005 to 40008)

Flushing of the tailings cell pipelines from the FIM III valve station to the FIM III TSF will be required at the end of a valve station transfer sequence and will be performed by an independent sequence manually initiated from the Water Services Control System (WSCS) ~~PCS~~. The decant return line to the plant will be isolated and decant water from Decant Dam 4 will be added to the required tailings line using two Decant Dam 4 pumps. The decant water will be added after the H section at the FIM III valve station through hydraulically actuated valves that will have position feedback. The section of tailings line from the valve station to the TSF will be flushed using the Decant Dam 4 pumps, with the flushing timer set to 60 minutes (TBC during commissioning). On completion of the flushing timer, the operator will be required to confirm the flushing is completed at the local vendor control panel before the sequence will proceed to the final step. ~~until the site services team are satisfied that the line is sufficiently flushed. This will require v~~Visual confirmation at the TSF by the site services team~~.~~ will also be required to ensure the line is sufficiently flushed.

Flushing water for the FIM III valve station will be taken from Decant Dam 4. All valves associated with the flushing water arrangement at Decant Dam 4 can be opened and closed either locally from a local control panel or from the PCS and will have position feedback.

For details of the flushing sequences refer to the Owner FIM TSF Project FIM III Valve Station Tailings Sequence Flowcharts (2000-F00-LST-PR-40002).

3.8 Seal Water (2000-F75-PID-PR-40014 / 2000-F175-PID-PR-40002)

A dedicated seal water tank will be installed to supply seal water to the FIM III booster pumps. The tank will be fed with saline water from the Eastern Borefields Saline Dam that will be filtered before it enters the seal water tank.

3.8.1 FIM III Booser Pumps Seal Water Filter ~~(HOLD 002)~~

The filter will be self-cleaning and will be controlled by a local control panel. The filter will periodically discharge to grade during back-flushing. The filter may be bypassed when offline to ensure continuous supply of water to the seal water tank. When back-flushing is required, the pressure in the supply line from the Eastern Borefields Saline Dam pumps will be increased to 500 kPa. A new pressure control loop on the discharge line of the Eastern Borefields Saline Dam pumps will be ~~required~~ installed to facilitate this. The pressure control loop can be controlled in the following modes:

- Auto mode: The pressure controller will adjust the open percentage of the pressure control valve to maintain an operator input pressure setpoint, with the pressure measured by a pressure transmitter on the Eastern Borefields Saline Dam pumps discharge line.
- Manual mode: The operator enters an open percentage for the pressure control valve.

The filter PLC will have the following outputs to the PCS:

- Fault.
- Flush cycle on.

The flush cycle on output will be used to increase the pressure setpoint for the control loop to the back-flush pressure.

3.8.2 Seal Water Tank

The seal water tank will be fitted with a level transmitter to provide monitoring and input to a level controller. Flow of filtered saline water into the seal water tank can be controlled in the following modes:

- Auto mode: The seal water tank level controller will maintain an operator input setpoint by adjusting the position of the level control valve.
- Manual mode: The operator enters an open percentage for the seal water tank level control valve.

The level in the FIM III Seal Water Tank is controlled with the use of a level control valve to ensure stable operation of the FIM III booster pumps. To maintain the tank level, operation of the Eastern Borefields Saline Dam pumps will be automatically controlled via the WSCS, with pump start and stop requests based on a combination of:

- FIM III Seal Water Tank level deviation from setpoint (SP)
- Level control valve (LCV-75300) position (% open)

This combined control strategy ensures that pumps are only operated when sufficient valve opening exists to allow flow, preventing conditions such as pump deadheading and improving overall system stability. To further protect the pumps from deadheading, the Eastern Borefields Saline Dam pumps discharge pressure controller (PIC-17504) will raise an alarm on high high pressure and on activation of this alarm the Eastern Borefields pumps are interlocked to trip.

If the level in the seal water tank falls below the setpoint **and** the level control valve is more than 10% open, a start signal will be sent to the WSCS to start one of the Eastern Borefields Saline Dam pumps. If the level in the tank continues to fall **and** the level control valve is more than 30% open, a request will be sent to start a second Eastern Borefields pump.

A stop signal will be sent to the WSCS to stop one of the Eastern Borefields pumps if the tank level rises above the setpoint **or** the level control valve closes to less than 10% open. A second stop signal will be sent if the tank level continues to rise **or** the level control valve closes to below 5% open.

The start and stop request parameters for the Eastern Borefields Saline Dam pumps are summarised in Table 3.8. These setpoints are to be confirmed during commissioning and will be able to be modified from SCADA with the appropriate access level.

Table 3.8.1 Eastern Borefields Saline Dam Pump Setpoints

	FIM III Seal Water Tank Level Start	LCV-75300 Valve Position Start	FIM III Seal Water Tank Level Stop	LCV-75300 Valve Position Stop
Duty Pump	< SP - 5%	> 10% Open	> SP + 5%	< 5% Open
Standby Pump	< SP - 10%	> 30% Open	> SP	< 10% Open

~~If the seal water level control valve opens to 100% and the level in the tank does not increase, a start signal will be sent to the WSCS to start one of the Eastern Borefields Saline Dam pumps. If on starting the pump the level still does not increase, a second pump will be started.~~

3.8.3 Seal Water Pumps

Seal water will be supplied to ~~the~~ each FIM III booster pump~~s~~ by ~~two sets of~~dedicated duty/standby seal water pumps. The new seal water pumps will ~~normally~~ operate as duty/standby and will include actuated suction isolation valves. Given seal water is critical to the operation of the FIM III booster pumps, the standby seal water pump will start automatically on a trip of the duty pump. Whether a booster pump is running or not, if there is slurry flow through the booster pump the duty seal water pump must be in operation.

The seal water pumps can be controlled in the following modes:

- Cascade mode: The measured discharge pressure of relevant FIM III booster pump will be input to a calculation block to calculate the required seal water pressure. The output of the calculation will be passed to the seal water pressure controller which will adjust the speed of duty seal water pump to maintain the calculated setpoint.
- Auto mode: The seal water pressure controller will maintain an operator input setpoint by adjusting the speed of the duty seal water pump.
- Manual mode: The operator enters a speed percentage for the duty seal water pump.

Flow transmitters will be included on the seal water feed to each of the FIM III booster pumps. The flow transmitters will have low low flow alarms that will be activated just above the minimum safe seal water flow for the pump. A low seal water flow will raise an alarm on the PCS and prevent the relevant booster pump from starting. A low low flow condition will raise an alarm on the PCS, and if not cleared within a given time, the standby seal water pump will start. If, after an additional time period the low low flow condition has still not cleared, the relevant FIM III booster pump will trip. A field operator should inspect the seal water assembly in the event of low seal water flow alarm.

A high seal water flow will raise an alarm on the PCS indicating that the respective FIM III booster pumps mechanical seal is worn and requires inspection.

3.9 Washdown Water

Washdown water will be provided at the FIM III booster station by a duty only washdown water pump, fed from the seal water tank. The fixed speed washdown water pump will draw water from the seal water tank, at a height above the seal water pump suction so as to preserve water in the tank for the seal water pumps. The pump will be started and stopped by the site services team as required. To prevent the pump from potentially running unattended and consuming water unnecessarily, a timer will stop the pump after it has been running for 10 minutes. The pump will trip if the level in the tank falls below 50%.

3.10 Potable Water

Potable water will be provided ~~at the FIM III booster station by~~ to a standalone safety shower at the FIM III booster station which includes:

- A storage tank of nominally 1,500 L capacity and low level alarm.
- A chiller designed to maintain the storage tank water at 22 °C.

The level in the storage tank will be maintained by the existing potable water distribution system with a manual float valve in the storage tank. ~~Periodic manual dosing of the tank is required to maintain potable water quality.~~

3.11 FIM III Booster Station Actuated Valves

Actuated valves at the new FIM III booster station will be hydraulically actuated and powered by a dedicated hydraulic power pack (F23-HY-420). The hydraulic power pack ~~will~~can be started and stopped from the local vendor control panel or from the PCS and will provide run and fault status to the PCS. All hydraulically actuated valves at the FIM III booster station will be able to be opened and closed either

locally from athe local vendor control panel or from the PCS. All hydraulically actuated valves at the FIM III booster station will have open and closed proximity switches that provide feedback to the PCS.

3.12 FIM III Valve Station Actuated Valves

Actuated valves at the new FIM III valve station will be hydraulically actuated and powered by a dedicated hydraulic power pack (F23-HY-421). The hydraulic power pack will be started and stopped from the vendor control panel and will provide run and fault status to the PCS. All hydraulically actuated valves at the FIM III valve station will be able to be opened and closed locally from athe local vendor control panel ~~or from the PCS~~. All hydraulically actuated valves at the FIM III valve station will have open and closed proximity switches that provide feedback to the PCS.

4 SALINE WATER RETURN

4.1 Control and Operating Requirements

The control and operating requirements of the existing saline water infrastructure will remain unchanged and operate as per the current operation.

The control and operating requirements of the new saline water infrastructure are as follows:

- Control logic for the FIM II Extension cell G decant pump, underdrainage pump and toe drain pumps.
- Control logic for the FIM III decant pumps, underdrainage pumps, seepage bore pumps and toe drain pump.
- Modification of existing control logic for the MOSCAD water PLC.
- Inclusion of the new Decant Dam 4 pumps in the MOSCAD water PLC to allow coordination with the Decant Dam 3 pumps.

4.2 Control System

The existing Wwater Sservices ~~c~~Control Ssystem (WSCS) hardware at FIM will be unchanged. ~~A~~ ~~w~~Water services control stations are ~~is~~ located in:

- The FIM control room.
- The FIM site services office.
- Black street.

A new Motorola PLC will be required at FIM III to control the new saline water infrastructure such that it can be integrated with the existing WSCS. Infrastructure that will be controlled by the new PLC includes:

- Decant Dam 4 ~~d~~Decant ~~p~~Pumps 1, 2 & 3.
- Decant Dam 4 ~~v~~Vacuum ~~p~~Pump.
- FIM III ~~c~~Cells 1, 2 & 3 ~~d~~Decant ~~p~~Pumps.
- FIM III ~~s~~Seepage ~~p~~Pumps 1, 2, 3 & 4.
- FIM III ~~t~~Toe ~~d~~Drain ~~p~~Pump.
- FIM III ~~u~~Underdrainage ~~p~~Pumps 1 & 2.
- FIM III ~~s~~Seal ~~w~~Water ~~t~~Tank level using water from the Eastern Borefields Saline Dam.

~~A new Motorola PLC will be required at saline dam #3 to control the new saline water infrastructure such that it can be integrated with the existing WSCS. Infrastructure that will be controlled by the new PLC includes:~~

- ~~Saline dam #3 water pumps 1, 2 & 3.~~

The existing Motorola PLC at FIM II will require modification to accommodate the new saline water infrastructure. Infrastructure that will be controlled by this PLC includes:

- FIM II ~~c~~Cell G ~~d~~Decant ~~p~~Pump.
- FIM II ~~c~~Cell G ~~u~~Underdrainage ~~p~~Pump.
- FIM II ~~c~~Cell G ~~t~~Toe ~~d~~Drain ~~p~~Pumps 1 & 2.

4.3 FIM II Extension Cell G

4.3.1 Decant Pump (2000-F75-PID-PR-40001 / 40031)

FIM II Extension cell G will have a floating turret that will ensure that cell G's decant pump will draw clean water from the supernatant water in the TSF. The decant pump will have an integral vacuum priming pump and will be self-priming with no external control required. The decant pump will have a local control panel for operator control of the self-priming decant pump.~~the start of the vacuum pump initiated by a start request for the decant pump. The vacuum pump will operate until the suction line for the decant pump has been primed, which will be detected by a level switch in the decant pump suction line which will stop the vacuum pump. The decant pump in cell G will be variable speed and can be controlled in the following modes:~~

~~Auto mode: A level controller will adjust the speed of the FIM II cell G decant pump to maintain an operator input setpoint.~~

~~Manual mode: The operator enters a speed percentage for FIM II cell G decant pump.~~

The turret will be fitted with a bed-level transmitter for monitoring of the slurry level within the TSF supernatant pond. This will facilitate control of the decant pump to minimise pumping of solids.

A flowmeter will be fitted to the decant return water line from the decant pump for monitoring of flow.

The FIM II cell G underdrainage line ~~decant return water line~~ will be combined with the FIM II cell G ~~underdrainage~~ decant return water line for transfer to Decant Dam 3. A flowmeter at the end of the combined return pipeline near Decant Dam 3 will monitor the return water flowrate. Any significant differential between the measured flow value at Decant Dam 3 and the sum of the two flowmeters at the FIM II cell G decant and underdrainage pumps will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the FIM II cell G decant and underdrainage pumps.

4.3.2 Underdrainage Pump (2000-F75-PID-PR-40007)

FIM II Extension cell G will have an underdrainage collection system and underdrainage pump. The underdrainage pump will be ~~locally~~ started and stopped automatically by the level in the underdrainage pit, or via a local start/stop station as required.

The underdrainage return line will be combined with the FIM II cell G decant return line for transfer to Decant Dam 3. A flowmeter at the end of the combined return pipeline near Decant Dam 3 will monitor the return water flowrate. Any significant differential between the measured flow value at Decant Dam 3 and the sum of the two flowmeters at the FIM II cell G decant and underdrainage pumps will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the FIM II cell G decant and underdrainage pumps.

4.3.3 Toe Drain Pumps (2000-F75-PID-PR-40006 / 40007)

FIM II Extension cell G will have a toe drain system and two toe drain pumps. The toe drain pumps will be started and stopped automatically by the level in the toe drain pit, or from the local start/stop stations as required. A flowmeter will be fitted to each of the toe drain return water lines for monitoring of flow.

The two toe drain pump discharge lines will be combined into a ~~single~~ common~~line~~ header for transfer to the Eastern Borefields Saline Dam via the existing FIM II seepage header.

4.4 FIM III

4.4.1 Decant Pumps (2000-F75-PID-PR-40002 to 40004)

Each cell in FIM III will have a floating turret that will ensure that the respective cell's decant pump will draw clean water from the supernatant water in the TSF. Each cell's decant pump will have an integral vacuum priming pump. The decant pumps will be self-priming with no external control required. Each decant pump will have a local control panel for operator control of the self-priming decant pump.

The turrets will be fitted with a bed-level transmitter for monitoring of the slurry level within the TSF supernatant pond. This will facilitate control of the decant pump to minimise pumping of solids.

~~Each cell will be fitted with a level transmitter for monitoring and provide input to the respective decant pump.~~ A flowmeter will be fitted to the decant return water line from each decant pump for monitoring of flow.

The three individual decant return lines will be combined into a ~~single line~~common header for transfer to Decant Dam 4. A flowmeter at the end of the decant return pipeline near Decant Dam 4 will monitor the return water flowrate. Any significant differential between the measured flow value at Decant Dam 4 and the sum of the three flowmeters at the decant cells will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the FIM III decant pumps.

4.4.2 Underdrainage Pumps (2000-F75-PID-PR-40004 / 40012)

Each cell in FIM III will have an underdrainage system; cell 1 will have a dedicated collection manifold and underdrainage pump whereas cells 2 and 3 will have a common collection manifold and underdrainage pump.

Each underdrainage pump will be ~~locally~~ started and stopped automatically by the level in the underdrainage pit, or via a local start/stop station as required. Each underdrainage pump discharge will have a flow transmitter for monitoring purposes. The two underdrainage pump discharge lines will be combined into a ~~single line~~ common header for transfer to Decant Dam 4. Any significant differential between the measured flow value at Decant Dam 4 and the sum of the flowmeters at the underdrainage pumps will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the FIM III underdrainage pumps.~~The differential flow alarms will be masked for a period after both start and stop of the underdrainage pumps.~~

4.4.3 Seepage Pumps (2000-F75-PID-PR-40008 / 40009 / 2000-F175-PR-PID-40002)

The FIM III TSF will have four seepage bores and associated seepage pumps. Each seepage pump will be started and stopped either from the PCS or via the local start/stop station. Each seepage pump discharge will be fitted with a flow transmitter and pressure transmitter for control and monitoring purposes. On detection of low flow, the seepage pump will stop, and a timer started which will delay the re-start of the pump. ~~If o~~On completion of the timer the pump will re-start and if the low flow alarm is once again activated, the pump will stop and the timer will be reset and will recommence.

The discharge of the four seepage bore pumps will be combined into a single line for transfer to the ~~e~~Eastern ~~b~~Borefields ~~s~~Saline ~~d~~Dam. The discharge of the FIM III ~~t~~Toe ~~d~~Drain ~~p~~Pump will be combined with the discharge of the seepage bore pumps. Any significant differential between the measured flow value at the ~~e~~Eastern ~~b~~Borefields ~~s~~Saline ~~d~~Dam and the sum of the ~~four~~ flowmeters at the four seepage bore pumps and the toe drain pump will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the FIM III toe drain and seepage pumps.~~The differential flow alarms will be masked for a period after both start and stop of the pumps.~~

4.4.4 Toe Drain Pump (2000-F75-PID-PR-40009 / 40011 / 2000-F175-PID-PR-40002)

FIM III will have a toe drain system and a toe drain pump. The toe drain pump will be ~~locally~~ started and stopped automatically by the level in the toe drain pit, or from the local start/stop station as required. The toe drain pump discharge will have a flow transmitter for monitoring purposes.

The discharge of the toe drain pump will be combined with the discharge of the FIM III seepage bore pumps. Any significant differential between the measured flow value at the Eastern Borefields Saline Dam and the sum of the flowmeters at the four seepage bore pumps and the toe drain pump will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the FIM III toe drain and seepage pumps.

4.4.5 Decant Return Water (2000-F75-PID-PR-40004 / 40032)

Decant water from all three FIM III cells, as well as underdrainage water will be collected in a common pond, Decant Dam 4. The dam will be configured with duty/standby settling ponds and an emergency overflow pond in addition to the main dam. Water from the settling ponds will overflow to the main dam by gravity, which will additionally overflow into the overflow pond by gravity.

Three Nnew ~~duty/duty/standby~~ centrifugal pumps will transfer the water to FIM in a new pipeline that will also be used to transfer water from the existing ~~d~~Decant ~~d~~Dam 3. The number of duty pumps required will be dependent on flow demand.

The decant pumps will share a common suction header that is fed by three shared floating suction turrets in Decant Dam 4 to ensure the pumps draw clean water. All three pumps will be able to draw water from the main decant pond as well as the overflow pond. If pumping from the overflow pond only one pump will be allowed to operate due to flow limitations of the floating suction turrets.

The pumps will share a common vacuum priming pump, through 3 shared turrets (2 in the main pond and 1 in the overflow pond). The ~~pumps will be self-priming with the~~ start of the vacuum pump will be initiated by a start request for any of the decant pumps. The vacuum pump will operate until the suction line for the decant pumps has been primed, which will be detected by a level switch in the decant pumps suction line which will stop the vacuum pump. Typically, two pumps will only be required when flushing pipelines and~~,~~ in normal operation only one pump will operate. When flushing, the valve in the return line to the FIM North Dam (XV-75340) will be closed ~~in order to~~to ensure sufficient flowrate and pressure is available for flushing.

Two operating philosophies will be required; decant return and flushing.

When returning decant water to FIM at flowrates below 1,100 m^3/h only one pump should operate. At flowrates above 1,100 m^3/h two pumps should be used.

The pumps will be variable speed and can be controlled in the following modes for decant return:

- Auto mode: A level controller will adjust the speed of the duty Decant Dam 4 pump(s) to maintain an operator input setpoint.
- Manual mode: The operator enters a speed percentage for the duty Decant Dam 4 pump(s).

When flushing the pumps can be controlled in the following modes:

- Auto mode: A flow controller will adjust the speed of the duty Decant Dam 4 pumps to maintain an operator input setpoint.
- Manual mode: The operator enters a speed percentage for the duty Decant Dam 4 pumps.

When Decant Dam 4 water is required for flushing duties, the speed reference selector (HS-75333) will be switched to flow control and the flowrate setpoint for the flow controller will be set to 1,100 m^3/h. This

will automatically trigger the start of a second Decant Dam 4 pump to achieve the required flowrate and pressure for flushing to the TSF.

Once flushing is complete, HS-75333 will be switched back to level control, and the operator will enter the required setpoint into the level control loop. This will cause the flowrate to ramp down below 1,100 m^3/h which will automatically trigger one of the Decant Dam 4 pumps to stop. The WSCS will select which pumps to start and stop based on run hours.

The main pond and overflow pond at Decant Dam 4 will be fitted with a level transmitter for monitoring and control.

The Decant Dam 4 decant return line will be combined with the Decant Dam 3 decant return line for transfer to the plant. A flowmeter just prior to ~~h~~Hydrogen ~~p~~Peroxide ~~i~~Inline ~~m~~Mixer 3 will monitor the return water flowrate. Any significant differential between the measured flow value at the inline mixer and the sum of the two flowmeters at Decant Dam 3 and Decant Dam 4 will raise an alarm on the PCS. On start-up there will be a 10 minute (TBC during commissioning) delay timer before leak detection becomes active. Activation of the leak detection alarm will interlock the Decant Dam 3 and Decant Dam 4 pumps.

The decant return lines are also used for various flushing duties. During periods where flushing water is being used the differential flow alarms will be masked in order to prevent invalid alarms.

Both Decant Dam 3 and Decant Dam 4 have manual valves installed downstream of the pump discharge, which introduces a potential risk of maloperation leading to overpressure in the decant return line. To mitigate this risk, a pressure relief valve has been installed on each return water line to divert water back to Decant Dam 3 or 4 in the event of an overpressure event. ~~When either of these valves is open and returning water to a decant dam, the differential flow alarm on the return water line will be disabled.~~

In the event of a differential flow alarm, confirmation with the site services team is required to determine whether manual flushing duties are currently in operation or if either pressure relief valves are open. In this case the differential flow alarm on the return water line will be disabled. If all manual flushing and pressure relief valves are closed, further investigation into the cause of the differential alarm is required.

4.5 Antiscalant Dosing

Antiscalant will be dosed to the respective turrets for each of the following decant pumps and decant return pumps:

- FIM II Cell G Decant Pump.
- FIM III Cell 1 Decant Pump.
- FIM III Cell 2 Decant Pump.
- FIM III Cell 3 Decant Pump.
- Decant Dam 4 Pumps 1, 2 & 3.

The antiscalant dosing system will comprise of an intermediate bulk container (IBC) of antiscalant and a dosing pump at each location. Level elements in the IBC's will provide feedback to the PCS.

The antiscalant dosing pumps will be solar powered, 24V units and will have no communication to the PCS.

Appendix A Yokogawa Process Control Tables

Owner **FIM TSF PROJECT**

PROCESS CONTROL TABLES - YOKOGAWA DCS

2000-F00-PHL-PR-40001 - APPENDIX A

Table of Contents		Total	On Hold	Complete (%)	FAT Complete (%)	SAT Complete (%)
1	**Analogue**	85	0	100.0	30.6	0.0
2	**Digital**	75	2	97.3	37.3	0.0
3	**Totalisers**	5	0	100.0	80.0	0.0
4	**Open/Close Valves**	18	0	100.0	44.4	0.0
5	**Control Valves**	1	0	100.0	0.0	0.0
6	**PID Controllers**	9	0	100.0	44.4	0.0
7	**Miscellaneous Controllers**	19	0	100.0	21.1	0.0
8	**Interlocks**	378	1	99.7	43.7	0.0
9	**Calculations**	25	0	100.0	52.0	0.0
	Total	615	3	99.5	41.0	0.0

Alarms	Priority	Count	%
1	Urgent (Critical Failure)	10	5.0
2	High (Fault/Error)	67	33.5
3	Low (Warning)	64	32.0
4	Logging	59	29.5
	Total	200	100

Abbreviations	
FC	Fail closed
FO	Fail open
FL	Fail last
LCP	Local control panel
LCS	Local control station
A_L	Low alarm
A_H	High alarm
A_LL	Low-Low alarm
A_HH	High-High alarm
P_L	Low process switchpoint
P_H	High process switchpoint
P_LL	Low-Low process switchpoint
P_HH	High-High process switchpoint
OIT	Operator interface terminal
Range	Instrument range
SEQ	Sequence
TBA	To be advised
v	Vendor controlled
VTA	Vendor to advise
x	PCS controlled
*	Denotes a sequence stop in place of an interlock

ANALOGUE INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	Trend (Sec)	Range			Process Switchpoints				Alarms								Alarm Masking	Control System	Comments	Rev
							Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)				
UT23575A	UI23575A	2000-F23-TER-IC-40008	FIM III BOOSTER STATION 11/0.433KV TRANSFORMER ROOM TEMPERATURE TRANSMITTER	F23SR601 TRF ROOM TEMP		1	0	50	℃	-	-	-	-	-	-	-	-	35	5	38	5	-	Yokogawa DCS	Display and alarm	4
UT23576A	UI23576A	2000-F23-TER-IC-40008	FIM III BOOSTER STATION HV SWITCHROOM TEMPERATURE TRANSMITTER	F23SR601 HV ROOM TEMP		1	0	50	℃	-	-	-	-	-	-	-	-	35	5	38	5	-	Yokogawa DCS	Display and alarm	4
UT23577A	UI23577A	2000-F23-TER-IC-40008	FIM III BOOSTER STATION SWITCHROOM TEMPERATURE TRANSMITTER	F23SR601 LV ROOM TEMP		1	0	50	℃	-	-	-	-	-	-	-	-	35	5	38	5	-	Yokogawa DCS	Display and alarm	4
F23CALC40003	WI23417	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE DRY TONNES	F23PP466 DISCH TONNES		1	0	1,500	t/h	-	-	-	-	-	-	-	-	-	-	-	-	F23-PP-466 NOT RUN	Yokogawa DCS		2
DIT23418	DI23418	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE DENSITY TRANSMITTER	F23PP466 DISCH DENSITY		1	1.00	2.00	t/m³	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS		1
F23CALC40004	DNI23418	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE DENSITY	F23PP466 DISCH DENSITY		1	0	100	%	-	-	-	-	-	-	50	5	60	5	-	-	F23-PP-466 NOT RUN	Yokogawa DCS	A_H = Warning A_L = Warning	2
PIT23419	PI23419A	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE PRESSURE TRANSMITTER [FIM I]	F23PP466 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	850	5	900	5	F23-PP-466 NOT RUN OR XV75609 CLOSED	Yokogawa DCS	Alarm setpoints for FIM I pumping duty A_HH = Trip F23-PP-466 A_H = Warning A_L = Warning	1
PIT23419	PI23419B	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE PRESSURE TRANSMITTER [FIM III]	F23PP466 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	950	5	1,000	5	F23-PP-466 NOT RUN OR XV75608 CLOSED	Yokogawa DCS	Alarm setpoints for FIM III pumping duty A_HH = Trip F23-PP-466 A_H = Warning A_L = Warning	1
F23CALC40006	WI23430	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE DRY TONNES	F23PP467 DISCH TONNES		1	0	1,500	t/h	-	-	-	-	-	-	-	-	-	-	-	-	F23-PP-467 NOT RUN	Yokogawa DCS		2
DIT23431	DI23431	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE DENSITY TRANSMITTER	F23PP467 DISCH DENSITY		1	1.00	2.00	t/m³	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS		1
F23CALC40007	DNI23431	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE DENSITY	F23PP467 DISCH DENSITY		1	0	100	%	-	-	-	-	-	-	50	5	60	5	-	-	F23-PP-467 NOT RUN	Yokogawa DCS	A_H = Warning A_L = Warning	2
PIT23432	PI23432A	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE PRESSURE TRANSMITTER [KALTAILS]	F23PP467 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	650	5	690	5	F23-PP-467 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Alarm setpoints for Kaltails pumping duty A_HH = Trip F23-PP-467 A_H = Warning A_L = Warning	1
PIT23432	PI23432B	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE PRESSURE TRANSMITTER [FIM III]	F23PP467 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	950	5	1,000	5	F23-PP-467 NOT RUN OR XV23532 CLOSED	Yokogawa DCS	Alarm setpoints for FIM III pumping duty A_HH = Trip F23-PP-467 A_H = Warning A_L = Warning	1
F23CALC40009	WI23441	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE DRY TONNES	F23PP468 DISCH TONNES		1	0	1,500	t/h	-	-	-	-	-	-	-	-	-	-	-	-	F23-PP-468 NOT RUN	Yokogawa DCS		2
DIT23442	DI23442	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE DENSITY TRANSMITTER	F23PP468 DISCH DENSITY		1	1.00	2.00	t/m³	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS		1
F23CALC40010	DNI23442	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE DENSITY	F23PP468 DISCH DENSITY		1	0	100	%	-	-	-	-	-	-	50	5	60	5	-	-	F23-PP-468 NOT RUN	Yokogawa DCS	A_H = Warning A_L = Warning	2
PIT23443	PI23443A	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE PRESSURE TRANSMITTER [KALTAILS]	F23PP468 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	650	5	690	5	F23-PP-468 NOT RUN OR KV30933 CLOSED	Yokogawa DCS	Alarm setpoints for Kaltails pumping duty A_HH = Trip F23-PP-468 A_H = Warning A_L = Warning	1
PIT23443	PI23443B	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE PRESSURE TRANSMITTER [FIM II / FIM II EXT]	F23PP468 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	1,150	5	1,200	5	F23-PP-468 NOT RUN OR KV30901 CLOSED	Yokogawa DCS	Alarm setpoints for FIM II / FIM II EXT pumping duty A_HH = Trip F23-PP-468 A_H = Warning A_L = Warning	2
F23CALC40012	WI23452	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE DRY TONNES	F23PP469 DISCH TONNES		1	0	1,500	t/h	-	-	-	-	-	-	-	-	-	-	-	-	F23-PP-469 NOT RUN	Yokogawa DCS		2
DIT23453	DI23453	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE DENSITY TRANSMITTER	F23PP469 DISCH DENSITY		1	1.00	2.00	t/m³	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS		1
F23CALC40013	DNI23453	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE DENSITY	F23PP469 DISCH DENSITY		1	0	100	%	-	-	-	-	-	-	50	5	60	5	-	-	F23-PP-469 NOT RUN	Yokogawa DCS	A_H = Warning A_L = Warning	2
PIT23454	PI23454A	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE PRESSURE TRANSMITTER [KALTAILS]	F23PP469 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	650	5	690	5	F23-PP-469 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Alarm setpoints for Kaltails pumping duty A_HH = Trip F23-PP-469 A_H = Warning A_L = Warning	1
PIT23454	PI23454B	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE PRESSURE TRANSMITTER [FIM II / FIM II EXT]	F23PP469 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	1,150	5	1,200	5	F23-PP-469 NOT RUN OR KV30900 CLOSED	Yokogawa DCS	Alarm setpoints for FIM II / FIM II EXT pumping duty A_HH = Trip F23-PP-469 A_H = Warning A_L = Warning	2
TT23470	**TI23470**	**2000-F23-PID-PR-10002**	**F23-PP-466 FINAL TAILS PUMP 1 GEARBOX TEMPERATURE TRANSMITTER**	**F23PP466 GEARBOX TEMP**		**1**	**0**	**100**	**℃**	-	-	-	-	-	-	-	-	**85**	**5**	**90**	**5**	-	**Yokogawa DCS**	**A_HH = Trip F23-PP-466 A_H = Warning**	**5**
TT23472	**TI23472**	**2000-F23-PID-PR-10002**	**F23-PP-467 FINAL TAILS PUMP 2 GEARBOX TEMPERATURE TRANSMITTER**	**F23PP467 GEARBOX TEMP**		**1**	**0**	**100**	**℃**	-	-	-	-	-	-	-	-	**85**	**5**	**90**	**5**	-	**Yokogawa DCS**	**A_HH = Trip F23-PP-467 A_H = Warning**	**5**
TT23474	**TI23474**	**2000-F23-PID-PR-10002**	**F23-PP-468 FINAL TAILS PUMP 3 GEARBOX TEMPERATURE TRANSMITTER**	**F23PP468 GEARBOX TEMP**		**1**	**0**	**100**	**℃**	-	-	-	-	-	-	-	-	**85**	**5**	**90**	**5**	-	**Yokogawa DCS**	**A_HH = Trip F23-PP-468 A_H = Warning**	**5**
TT23476	**TI23476**	**2000-F23-PID-PR-10002**	**F23-PP-469 FINAL TAILS PUMP 4 GEARBOX TEMPERATURE TRANSMITTER**	**F23PP469 GEARBOX TEMP**		**1**	**0**	**100**	**℃**	-	-	-	-	-	-	-	-	**85**	**5**	**90**	**5**	-	**Yokogawa DCS**	**A_HH = Trip F23-PP-469 A_H = Warning**	**5**
PIT23511	PI23511	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SUCTION PRESSURE TRANSMITTER	F23PP651 SUCT PRESSURE		1	0	1,000	kPa	-	-	-	-	-	-	20	5	800	5	-	-	-	Yokogawa DCS	A_H = Warning A_L = Warning	1
PIT23520	PI23520	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE PRESSURE TRANSMITTER	F23PP651 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	1,000	5	1,100	5	F23-PP-651 NOT RUN	Yokogawa DCS	A_HH = Trip F23-PP-651 A_H = Warning A_L = Warning	2
F23CALC40014	FDI23521	2000-F23-PID-PR-40002	LINE D TO FIM III TAILINGS DIFFERENTIAL FLOW CALCULATION	LINE D FIM III DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	**600**	50	**600**	-	-	**F23-PP-466 NOT RUN**	Yokogawa DCS	**Alarm masked for 600 seconds on startup of F23-PP-466** A_H = Warning A_L = Warning	**5**
FIT23523	FI23523	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SEAL WATER FLOW TRANSMITTER	F23PP651 SEAL WATER FLOW		1	0	50	L/min	-	-	-	-	6.5	25	13	5	20	5	-	-	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	A_H = Warning A_L = Warning A_LL = Trip F23-PP-651	1
TE23651A	TI23651A	2000-F23-PID-PR-40002	F23-PP-651 FIM III BOOSTER PUMP 1 WINDING TEMPERATURE 1	F23PP651 WINDING 1 RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	120	5	130	5	-	Yokogawa DCS	~~**Vendor Information to be confirmed**~~	**5**
TE23651B	TI23651B	2000-F23-PID-PR-40002	F23-PP-651 FIM III BOOSTER PUMP 1 WINDING TEMPERATURE 2	F23PP651 WINDING 2 RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	120	5	130	5	-	Yokogawa DCS	~~**Vendor Information to be confirmed**~~	**5**
TE23651C	TI23651C	2000-F23-PID-PR-40002	F23-PP-651 FIM III BOOSTER PUMP 1 WINDING TEMPERATURE 3	F23PP651 WINDING 3 RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	120	5	130	5	-	Yokogawa DCS	~~**Vendor Information to be confirmed**~~	**5**
TE23651D	TI23651D	2000-F23-PID-PR-40002	F23-PP-651 FIM III BOOSTER PUMP 1 DRIVE END BEARING TEMPERATURE	F23PP651 DE BEARING RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	90	5	95	5	-	Yokogawa DCS	~~**Vendor Information to be confirmed**~~	**5**
TE23651E	TI23651E	2000-F23-PID-PR-40002	F23-PP-651 FIM III BOOSTER PUMP 1 NON-DRIVE END BEARING TEMPERATURE	F23PP651 NDE BEARING RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	90	5	95	5	-	Yokogawa DCS	~~**Vendor Information to be confirmed**~~	**5**
TE23651F	TI23651F	2000-F23-PID-PR-40002	F23-PP-651 FIM III BOOSTER PUMP 1 INTERNAL AMBIENT TEMPERATURE	F23PP651 AMBIENT RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	45	5	50	5	-	Yokogawa DCS	~~**Vendor Information to be confirmed**~~	**5**

ANALOGUE INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	Trend (Sec)	Range			Process Switchpoints				Alarms									Alarm Masking	Control System	Comments	Rev
							Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)				
IT23579	II23579	2000-F23-PID-PR-40002	FIM III HYDRAULIC POWER PACK CURRENT STATUS SIGNAL	F23HY420 CURRENT		1	0	13.4	A	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS	Display only	4
PIT23531	PI23531	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SUCTION PRESSURE TRANSMITTER	F23PP652 SUCT PRESSURE		1	0	1,000	kPa	-	-	-	-	-	-	20	5	800	5	-	-	-	Yokogawa DCS	A_H = Warning A_L = Warning	1
PIT23540	PI23540	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE PRESSURE TRANSMITTER	F23PP652 DISCH PRESSURE		1	0	2,000	kPa	-	-	-	-	-	-	100	5	1,000	5	1,100	5	F23-PP-652 NOT RUN	Yokogawa DCS	A_HH = Trip F23-PP-652 A_H = Warning A_L = Warning	2
F23CALC40016	FDI23541	2000-F23-PID-PR-40003	LINE B TO FIM III TAILINGS DIFFERENTIAL FLOW CALCULATION	LINE B FIM III DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	**F23-PP-467 NOT RUN**	Yokogawa DCS	**Alarm masked for 600 seconds on startup of F23-PP-467** A_H = Warning A_L = Warning	5
FIT23543	FI23543	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SEAL WATER FLOW TRANSMITTER	F23PP652 SEAL WATER FLOW		1	0	50	L/min	-	-	-	-	6.5	25	13	5	20	5	-	-	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	A_H = Warning A_L = Warning A_LL = Trip F23-PP-652	1
TE23652A	TI23652A	2000-F23-PID-PR-40003	F23-PP-652 FIM III BOOSTER PUMP 2 WINDING TEMPERATURE 1	F23PP652 WINDING 1 RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	120	5	130	5	-	Yokogawa DCS	~~Vendor Information to be confirmed~~	5
TE23652B	TI23652B	2000-F23-PID-PR-40003	F23-PP-652 FIM III BOOSTER PUMP 2 WINDING TEMPERATURE 2	F23PP652 WINDING 2 RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	120	5	130	5	-	Yokogawa DCS	~~Vendor Information to be confirmed~~	5
TE23652C	TI23652C	2000-F23-PID-PR-40003	F23-PP-652 FIM III BOOSTER PUMP 2 WINDING TEMPERATURE 3	F23PP652 WINDING 3 RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	120	5	130	5	-	Yokogawa DCS	~~Vendor Information to be confirmed~~	5
TE23652D	TI23652D	2000-F23-PID-PR-40003	F23-PP-652 FIM III BOOSTER PUMP 2 DRIVE END BEARING TEMPERATURE	F23PP652 DE BEARING RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	90	5	95	5	-	Yokogawa DCS	~~Vendor Information to be confirmed~~	5
TE23652E	TI23652E	2000-F23-PID-PR-40003	F23-PP-652 FIM III BOOSTER PUMP 2 NON-DRIVE END BEARING TEMPERATURE	F23PP652 NDE BEARING RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	90	5	95	5	-	Yokogawa DCS	~~Vendor Information to be confirmed~~	5
TE23652F	TI23652F	2000-F23-PID-PR-40003	F23-PP-652 FIM III BOOSTER PUMP 2 INTERNAL AMBIENT TEMPERATURE	F23PP652 AMBIENT RTD		1	-20	200	℃	-	-	-	-	-	-	-	-	45	5	50	5	-	Yokogawa DCS	~~Vendor Information to be confirmed~~	5
FIT23592	**FI23592**	**2000-F23-PID-PR-40005**	**FIM III LINE 1 SOUTH FLOW TRANSMITTER**	**FIM III LINE 1 SOUTH FLOW**		**1**	**0**	**2,000**	**m³/h**	-	-	-	-	-	-	**100**	**5**	-	-	-	-	-	**Yokogawa DCS**	**NOTE: Duplication of tag from MOSCAD RTU. Low level alarm to be made available via RTU comms and configured in DCS.** **A_L = Warning**	**5**
F23CALC40018	**FDI23592A**	**2000-F23-PID-PR-40005**	**LINE D TO FIM III LINE 1 SOUTH DIFFERENTIAL FLOW CALCULATION**	**LINE D TO FIM III 1S DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-651** **A_H = Warning** **A_L = Warning**	**5**
F23CALC40019	**FDI23592B**	**2000-F23-PID-PR-40005**	**LINE B TO FIM III LINE 1 SOUTH DIFFERENTIAL FLOW CALCULATION**	**LINE B TO FIM III 1S DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-652** **A_H = Warning** **A_L = Warning**	**5**
FIT23593	**FI23593**	**2000-F23-PID-PR-40005**	**FIM III LINE 1 NORTH FLOW TRANSMITTER**	**FIM III LINE 1 NORTH FLOW**		**1**	**0**	**2,000**	**m³/h**	-	-	-	-	-	-	**100**	**5**	-	-	-	-	-	**Yokogawa DCS**	**NOTE: Duplication of tag from MOSCAD RTU. Low level alarm to be made available via RTU comms and configured in DCS.** **A_L = Warning**	**5**
F23CALC40020	**FDI23593A**	**2000-F23-PID-PR-40005**	**LINE B TO FIM III LINE 1 NORTH DIFFERENTIAL FLOW CALCULATION**	**LINE B TO FIM III 1N DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-652** **A_H = Warning** **A_L = Warning**	**5**
F23CALC40021	**FDI23593B**	**2000-F23-PID-PR-40005**	**LINE D TO FIM III LINE 1 NORTH DIFFERENTIAL FLOW CALCULATION**	**LINE D TO FIM III 1N DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-651** **A_H = Warning** **A_L = Warning**	**5**
FIT23594	**FI23594**	**2000-F23-PID-PR-40005**	**FIM III LINE 3 WEST FLOW TRANSMITTER**	**FIM III LINE 3 WEST FLOW**		**1**	**0**	**2,000**	**m³/h**	-	-	-	-	-	-	**100**	**5**	-	-	-	-	-	**Yokogawa DCS**	**NOTE: Duplication of tag from MOSCAD RTU. Low level alarm to be made available via RTU comms and configured in DCS.** **A_L = Warning**	**5**
F23CALC40022	**FDI23594A**	**2000-F23-PID-PR-40005**	**LINE B TO FIM III LINE 3 WEST DIFFERENTIAL FLOW CALCULATION**	**LINE B TO FIM III 3W DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-652** **A_H = Warning** **A_L = Warning**	**5**
F23CALC40023	**FDI23594B**	**2000-F23-PID-PR-40005**	**LINE D TO FIM III LINE 3 WEST DIFFERENTIAL FLOW CALCULATION**	**LINE D TO FIM III 3W DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-651** **A_H = Warning** **A_L = Warning**	**5**
FIT23595	**FI23595**	**2000-F23-PID-PR-40005**	**FIM III LINE 3 EAST FLOW TRANSMITTER**	**FIM III LINE 3 EAST FLOW**		**1**	**0**	**2,000**	**m³/h**	-	-	-	-	-	-	**100**	**5**	-	-	-	-	-	**Yokogawa DCS**	**NOTE: Duplication of tag from MOSCAD RTU. Low level alarm to be made available via RTU comms and configured in DCS.** **A_L = Warning**	**5**
F23CALC40024	**FDI23595A**	**2000-F23-PID-PR-40005**	**LINE D TO FIM III LINE 3 EAST DIFFERENTIAL FLOW CALCULATION**	**LINE D TO FIM III 3E DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-651** **A_H = Warning** **A_L = Warning**	**5**
F23CALC40025	**FDI23595B**	**2000-F23-PID-PR-40005**	**LINE B TO FIM III LINE 3 EAST DIFFERENTIAL FLOW CALCULATION**	**LINE B TO FIM III 3E DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-652** **A_H = Warning** **A_L = Warning**	**5**
FIT23596	**FI23596**	**2000-F23-PID-PR-40005**	**FIM III LINE 2 WEST FLOW TRANSMITTER**	**FIM III LINE 2 WEST FLOW**		**1**	**0**	**2,000**	**m³/h**	-	-	-	-	-	-	**100**	**5**	-	-	-	-	-	**Yokogawa DCS**	**NOTE: Duplication of tag from MOSCAD RTU. Low level alarm to be made available via RTU comms and configured in DCS.** **A_L = Warning**	**5**
F23CALC40026	**FDI23596A**	**2000-F23-PID-PR-40005**	**LINE B TO FIM III LINE 2 WEST DIFFERENTIAL FLOW CALCULATION**	**LINE B TO FIM III 2W DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-652** **A_H = Warning** **A_L = Warning**	**5**
F23CALC40027	**FDI23596B**	**2000-F23-PID-PR-40005**	**LINE D TO FIM III LINE 2 WEST DIFFERENTIAL FLOW CALCULATION**	**LINE D TO FIM III 2W DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-651** **A_H = Warning** **A_L = Warning**	**5**
FIT23597	**FI23597**	**2000-F23-PID-PR-40005**	**FIM III LINE 2 EAST FLOW TRANSMITTER**	**FIM III LINE 2 EAST FLOW**		**1**	**0**	**2,000**	**m³/h**	-	-	-	-	-	-	**100**	**5**	-	-	-	-	-	**Yokogawa DCS**	**NOTE: Duplication of tag from MOSCAD RTU. Low level alarm to be made available via RTU comms and configured in DCS.** **A_L = Warning**	**5**
F23CALC40028	**FDI23597A**	**2000-F23-PID-PR-40005**	**LINE D TO FIM III LINE 2 EAST DIFFERENTIAL FLOW CALCULATION**	**LINE D TO FIM III 2E DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-651** **A_H = Warning** **A_L = Warning**	**5**
F23CALC40029	**FDI23597B**	**2000-F23-PID-PR-40005**	**LINE B TO FIM III LINE 2 EAST DIFFERENTIAL FLOW CALCULATION**	**LINE B TO FIM III 2E DIFF FLOW**		**1**	**-100**	**100**	**m³/h**	-	-	-	-	-	-	**-50**	**600**	**50**	**600**	-	-	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Alarm masked for 600 seconds on startup of F23-PP-652** **A_H = Warning** **A_L = Warning**	**5**
F75CALC40006	FDI75902	2000-F75-PID-PR-40020	LINE D TO FIM I DIFFERENTIAL FLOW CALCULATION	D TO FIM1 DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	**600**	50	**600**	-	-	**F23-PP-466 NOT RUN**	Yokogawa PLC	**Alarm masked for 600 seconds on startup of F23-PP-466** A_H = Warning A_L = Warning	**5**
F75CALC40007	FDI30909	2000-F75-PID-PR-40022	LINE A TO FIM II DIFFERENTIAL FLOW CALCULATION	A TO FIM2 DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	**600**	50	**600**	-	-	**F23-PP-468 NOT RUN**	Yokogawa PLC	**Alarm masked for 600 seconds on startup of F23-PP-468** A_H = Warning A_L = Warning	**5**
F75CALC40008	FDI30908	2000-F75-PID-PR-40022	LINE C TO FIM II DIFFERENTIAL FLOW CALCULATION	C TO FIM2 DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	**600**	50	**600**	-	-	**F23-PP-469 NOT RUN**	Yokogawa PLC	**Alarm masked for 600 seconds on startup of F23-PP-469** A_H = Warning A_L = Warning	**5**

2000-F00-PHL-PR-40001 - APPENDIX A

ANALOGUE INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	Trend (Sec)	Range			Process Switchpoints				Alarms								Alarm Masking	Control System	Comments	Rev
							Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)				
F75CALC40009	FDI30940A	2000-F75-PID-PR-40023	LINE A TO KALTAILS DIFFERENTIAL FLOW CALCULATION	A TO KALTAILS DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F23-PP-468 NOT RUN	Yokogawa PLC	Alarm masked for 600 seconds on startup of F23-PP-468 A_H = Warning A_L = Warning	5
F75CALC40010	FDI30940B	2000-F75-PID-PR-40023	LINE B TO KALTAILS DIFFERENTIAL FLOW CALCULATION	B TO KALTAILS DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F23-PP-467 NOT RUN	Yokogawa PLC	Alarm masked for 600 seconds on startup of F23-PP-467 A_H = Warning A_L = Warning	5
F75CALC40011	FDI30941	2000-F75-PID-PR-40023	LINE C TO KALTAILS DIFFERENTIAL FLOW CALCULATION	C TO KALTAILS DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F23-PP-469 NOT RUN	Yokogawa PLC	Alarm masked for 600 seconds on startup of F23-PP-469 A_H = Warning A_L = Warning	5
FIT23481	FI23481	2000-F24-PID-PR-10004	F23-PP-466 FINAL TAILS PUMP 1 GLAND WATER FLOW INDICATING TRANSMITTER	F23PP466 GW FLOW		1	0	9	m³/h	-	-	-	-	1.6	25	1.9	5	-	-	-	-	F23-PP-466 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	A_L = Warning A_LL = Trip F23-PP-466	3
FIT23485	FI23485	2000-F24-PID-PR-10004	F23-PP-467 FINAL TAILS PUMP 2 GLAND WATER FLOW INDICATING TRANSMITTER	F23PP467 GW FLOW		1	0	9	m³/h	-	-	-	-	1.6	25	1.9	5	-	-	-	-	F23-PP-467 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	A_L = Warning A_LL = Trip F23-PP-467	3
FIT23489	FI23489	2000-F24-PID-PR-10004	F23-PP-468 FINAL TAILS PUMP 3 GLAND WATER FLOW INDICATING TRANSMITTER	F23PP468 GW FLOW		1	0	9	m³/h	-	-	-	-	1.6	25	1.9	5	-	-	-	-	F23-PP-468 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	A_L = Warning A_LL = Trip F23-PP-468	3
FIT23493	FI23493	2000-F24-PID-PR-10004	F23-PP-469 FINAL TAILS PUMP 4 GLAND WATER FLOW INDICATING TRANSMITTER	F23PP469 GW FLOW		1	0	9	m³/h	-	-	-	-	1.6	25	1.9	5	-	-	-	-	F23-PP-469 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	A_L = Warning A_LL = Trip F23-PP-469	3
FIT75610	FI75610	2000-F75-PID-PR-25008	F75-PW-82A/B DECANT WATER DISCHARGE TO F75-TK-010 CYANIDE WATER TANK	KALTAILS DECANT FLOW		1	0	1,000	m³/h	-	-	-	-	-	-	50	5	-	-	-	-	-	Yokogawa DCS	A_L = Warning	1
FIT75611	FI75611	2000-F75-PID-PR-25008	DECANT DAM 3 AND 4 DECANT WATER DISCHARGE TO BORE WATER DAM #1	DD3&4 RETURN FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	50	5	-	-	-	-	-	Yokogawa DCS	A_L = Warning	2
~~F75CALC40001~~	~~FDI75611~~	~~2000-F75-PID-PR-25008~~	~~DECANT DAM 3 AND 4 RETURN DIFFERENTIAL FLOW CALCULATION~~	~~DD3&4 RETURN DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~600~~	~~50~~	~~600~~	-	-	~~F75-PW-74 NOT RUN AND F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN~~	~~Yokogawa DCS~~	~~Alarm masked for 600 seconds on startup of decant pumps A_H = Warning A_L = Warning~~	5
FIT75612	FI75612	2000-F75-PID-PR-25008	FIM I DECANT RETURN & OFFSITE SALINE WATER FLOW TRANSMITTER	FIM I DECANT & OFFSITE SALT FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	50	5	-	-	-	-	-	Yokogawa DCS	A_L = Warning	1
LIT75620	LI75620	2000-F75-PID-PR-25010	EVENT POND SILT TRAP LEVEL TRANSMITTER	F00PS465 LEVEL		1	0	100	%	-	10	90	-	-	-	-	-	95	5	-	-	-	Yokogawa DCS	A_H = Warning	1
LIT75621	LI75621	2000-F75-PID-PR-25010	EVENT POND LEVEL TRANSMITTER	F00PN401 LEVEL		1	0	100	%	-	10	90	-	-	-	-	-	95	5	-	-	-	Yokogawa DCS	A_H = Warning	1
FIT75321	FI75321	2000-F75-PID-PR-40014	BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER 2	F75TK651 INFLOW 2		1	0	25	m³/h	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS		1
~~F75CALC40005~~	~~FDI75321~~	~~2000-F75-PID-PR-40014~~	~~BORE WATER TO FIM III SEAL WATER TANK DIFFERENTIAL FLOW~~	~~F75TK651 INFLOW DIFF~~		~~1~~	~~-25~~	~~25~~	~~m³/h~~	-	-	-	-	-	-	~~-10~~	~~600~~	~~10~~	~~600~~	-	-	~~SECDT01PP1 NOT RUN AND SECDT01PP2 NOT RUN AND SECDT01PP3 NOT RUN~~	~~Yokogawa DCS~~	~~Alarm masked for 600 seconds on startup of pumps A_H = Warning A_L = Warning~~	5
ZT75300	ZI75300	2000-F75-PID-PR-40014	BORE WATER TO FIM III SEAL WATER TANK LEVEL CONTROL VALVE FEEDBACK	LCV75300 POSITION		1	0	100	%	-	-	-	-	-	-	-	-	-	-	-	-	-	Yokogawa DCS		4
F75CALC40012	FDI30115	2000-F75-PR-PID-40025	LINE C TO FIM II CELL F DIFFERENTIAL FLOW CALCULATION	LINE C TO FIM II F DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F30-PP-77A NOT RUN	Yokogawa DCS	Alarm masked for 600 seconds on startup of F30-PP-77A A_H = Warning	5
F75CALC40013	FDI30116	2000-F75-PR-PID-40025	LINE A TO FIM II CELL F DIFFERENTIAL FLOW CALCULATION	LINE A TO FIM II F DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F30-PP-77B NOT RUN	Yokogawa DCS	Alarm masked for 600 seconds on startup of F30-PP-77B A_H = Warning	5
F75CALC40014	FDI30117	2000-F75-PR-PID-40026	LINE C TO FIM II CELL E DIFFERENTIAL FLOW CALCULATION	LINE C TO FIM II E DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F30-PP-77A NOT RUN	Yokogawa DCS	Alarm masked for 600 seconds on startup of F30-PP-77A A_H = Warning	5
F75CALC40015	FDI30118	2000-F75-PR-PID-40026	LINE C TO FIM II CELL G DIFFERENTIAL FLOW CALCULATION	LINE C TO FIM II G DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F30-PP-77A NOT RUN	Yokogawa DCS	Alarm masked for 600 seconds on startup of F30-PP-77A A_H = Warning	5
F75CALC40016	FDI30119	2000-F75-PR-PID-40026	LINE A TO FIM II CELL E DIFFERENTIAL FLOW CALCULATION	LINE A TO FIM II E DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F30-PP-77B NOT RUN	Yokogawa DCS	Alarm masked for 600 seconds on startup of F30-PP-77B A_H = Warning	5
F75CALC40017	FDI30120	2000-F75-PR-PID-40026	LINE A TO FIM II CELL G DIFFERENTIAL FLOW CALCULATION	LINE A TO FIM II G DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	F30-PP-77B NOT RUN	Yokogawa DCS	Alarm masked for 600 seconds on startup of F30-PP-77B A_H = Warning	5

DIGITAL INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	State 0	State 1	Alarm State	Source	Alarm Delay	Alarm Priority	Alarm Masking	Control System	Comments	Rev
F23SW602HTY	F23SW602_HTY	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC INCOMER ACB SHUNT TRIP 24VDC OK	F23SW602 ACB SHUNT OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23SW602AF	F23SW602_AF	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC INCOMER ARC FLASH DETECTION SYSTEM ALARM	F23SW602 ARC FLASH ALM		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23SW602OP	F23SW602_OP	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC INCOMER ACB OPENED	F23SW602 ACB OPEN		Not Activated	Activated	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	4
F23SW602CL	F23SW602_CL	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC INCOMER ACB CLOSED	F23SW602 ACB CLOSE		Not Activated	Activated	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	4
F23SW602TRP	F23SW602_TRP	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC INCOMER ACB TRIPPED	F23SW602 ACB TRIP		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23SW602SD	F23SW602_SD	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC INCOMER SURGE DIVERTER OK	F23SW602 INC SRG OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23SR601ESTP	F23SR601_ESTP	2000-F23-PID-PR-10002	F23-SR-601 HV/LV SWITCHROOM EMERGENCY SHUTDOWN 1	F23SW601 EMERG SD1		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23COM601PS1	F23COM601_PS1	2000-F23-PID-PR-10002	F23-COM-601 LV SWITCHROOM COMMUNICATIONS RACK UPS PS1 HEALTHY	F23COM601 UPS PS1 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23COM601PS2	F23COM601_PS2	2000-F23-PID-PR-10002	F23-COM-601 LV SWITCHROOM COMMUNICATIONS RACK UPS PS2 HEALTHY	F23COM601 UPS PS2 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23COM601PS3	F23COM601_PS3	2000-F23-PID-PR-10002	F23-COM-601 LV SWITCHROOM COMMUNICATIONS RACK 48VDC PS3 HEALTHY	F23COM601 48V PS3 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23BC601FL1	F23BC601_FL1	2000-F23-PID-PR-10002	F23-BC-601 48VDC BATTERY CHARGER MCR RELAY CARD MAINS FAIL	F23BC601 MCR BATT CHR FL		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23BC601FL2	F23BC601_FL2	2000-F23-PID-PR-10002	F23-BC-601 48VDC BATTERY CHARGER COMMON ALARM	F23BC601 BATT CHR ALM		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23BC601FL3	F23BC601_FL3	2000-F23-PID-PR-10002	F23-BC-601 48VDC BATTERY CHARGER MCR RELAY EARTH FAULT	F23BC601 MCR BATT CHR EF		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23SW602PS1	F23SW602_PS1	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC 24V DISTRIBUTION UPS PS1 HEALTHY	F23SW602 24V UPS PS1 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23SW602PS2	F23SW602_PS2	2000-F23-PID-PR-10002	F23-SW-602 415VAC MCC 24V DISTRIBUTION UPS PS2 HEALTHY	F23SW602 24V UPS PS2 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23DCS601PS1	F23DCS601_PS1	2000-F23-PID-PR-10002	F23-DCS-601 PS1 24VDC HEALTHY	F23DCS601 24V PS1 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23DCS601PS2	F23DCS601_PS2	2000-F23-PID-PR-10002	F23-DCS-601 PS2 24VDC HEALTHY	F23DCS601 24V PS2 OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23TR602ALM	F23TR602_ALM	2000-F23-PID-PR-10002	F23-TR-602 11/0.433KV TRANSFORMER TEMPERATURE MONITOR ALARM HEALTHY	F23TR602 TRF TEMP OK		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23TR602TRP	F23TR602_TRP	2000-F23-PID-PR-10002	F23-TR-602 11/0.433KV TRANSFORMER TEMPERATURE MONITOR TRIP HEALTHY	F23TR602 TRF TRIP		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
F23TR602FLT	F23TR602_FLT	2000-F23-PID-PR-10002	F23-TR-602 11/0.433KV TRANSFORMER TEMPERATURE MONITOR FAULT HEALTHY	F23TR602 TRF FAULT		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	2
ZSC23420	ZIC23420	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE HAND VALVE POSITION SWITCH CLOSED	F23PP466 DIS HND VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23420	ZIO23420	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE HAND VALVE POSITION SWITCH OPEN	F23PP466 DIS HND VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23433	ZIC23433	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE HAND VALVE POSITION SWITCH CLOSED	F23PP467 DIS HND VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23433	ZIO23433	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE HAND VALVE POSITION SWITCH OPEN	F23PP467 DIS HND VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23444	ZIC23444	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE HAND VALVE POSITION SWITCH CLOSED	F23PP468 DIS HND VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23444	ZIO23444	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE HAND VALVE POSITION SWITCH OPEN	F23PP468 DIS HND VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23455	ZIC23455	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE HAND VALVE POSITION SWITCH CLOSED	F23PP469 DIS HND VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23455	ZIO23455	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE HAND VALVE POSITION SWITCH OPEN	F23PP469 DIS HND VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
PSL23471	**PAL23471**	**2000-F23-PID-PR-10002**	**F23-PP-466 FINAL TAILS PUMP 1 GEARBOX PRESSURE SWITCH LOW**	**F23PP466 GEARBOX PRES LO**		**Alarm**	**Normal**	**Alarm**	**Hardwired to PCS**	**30**	**-**	**F23-PP-466 NOT RUN**	**Yokogawa DCS**	**Trips F23-PP-466**	**5**
PSL23473	**PAL23473**	**2000-F23-PID-PR-10002**	**F23-PP-467 FINAL TAILS PUMP 2 GEARBOX PRESSURE SWITCH LOW**	**F23PP467 GEARBOX PRES LO**		**Alarm**	**Normal**	**Alarm**	**Hardwired to PCS**	**30**	**-**	**F23-PP-467 NOT RUN**	**Yokogawa DCS**	**Trips F23-PP-467**	**5**
PSL23475	**PAL23475**	**2000-F23-PID-PR-10002**	**F23-PP-468 FINAL TAILS PUMP 3 GEARBOX PRESSURE SWITCH LOW**	**F23PP468 GEARBOX PRES LO**		**Alarm**	**Normal**	**Alarm**	**Hardwired to PCS**	**30**	**-**	**F23-PP-468 NOT RUN**	**Yokogawa DCS**	**Trips F23-PP-468**	**5**
PSL23477	**PAL23477**	**2000-F23-PID-PR-10002**	**F23-PP-469 FINAL TAILS PUMP 4 GEARBOX PRESSURE SWITCH LOW**	**F23PP469 GEARBOX PRES LO**		**Alarm**	**Normal**	**Alarm**	**Hardwired to PCS**	**30**	**-**	**F23-PP-469 NOT RUN**	**Yokogawa DCS**	**Trips F23-PP-469**	**5**
ZSC23512	ZIC23512	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SUCTION VALVE CLOSED	F23PP651 SUC VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23512	ZIO23512	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SUCTION VALVE OPEN	F23PP651 SUC VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23514	ZIC23514	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 FLUSHING VALVE CLOSED	F23PP651 FLSH VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23514	ZIO23514	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 FLUSHING VALVE OPEN	F23PP651 FLSH VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC	ZIC	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE VALVE CLOSED	F23PP651 DIS VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO	ZIO	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE VALVE OPEN	F23PP651 DIS VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23518	ZIC23518	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE DRAIN VALVE CLOSED	F23PP651 DIS DRN VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23518	ZIO23518	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE DRAIN VALVE OPEN	F23PP651 DIS DRN VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23527	ZIC23527	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SUCTION DRAIN VALVE CLOSED	F23PP651 SUC DRN VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23527	ZIO23527	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SUCTION DRAIN VALVE OPEN	F23PP651 SUC DRN VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
US23519	UA23519	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 BURSTING DISC RELIEF INDICATOR	F23PP651 BRST DISC		Alarm	Normal	Alarm	Hardwired to PCS	3	High	-	Yokogawa DCS		1
XS23570	XI23570	2000-F23-PID-PR-40002	FIM III BOOSTER STATION HYDRAULIC POWER PACK RUN	F23HY420 RUNNING		Inactive (N_Running)	Active (Running)	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	5
XS23571	XI23571	2000-F23-PID-PR-40002	FIM III BOOSTER STATION HYDRAULIC POWER PACK FAULT	F23HY420 FAULT		Fault	Normal	Fault	Hardwired to PCS	3	Med	-	Yokogawa DCS		4
XS23572	XY23572	2000-F23-PID-PR-40002	FIM III HYDRAULIC POWER PACK START SIGNAL	F23HY420 START CMD		Off	On	-	Hardwired to PCS	0	-	-	Yokogawa DCS	No alarm. Indication only	5
XS23573	XI23573	2000-F23-PID-PR-40002	FIM III BOOSTER STATION HYDRAULIC POWER PACK READY	F23HY420 READY		Inactive (N_Ready)	Active (Ready)	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	5
XS23574	XI23574	2000-F23-PID-PR-40002	FIM III HYDRAULIC POWER PACK FIELD E-STOP SIGNAL	F23HY420 PB STOP		Pressed	Not Pressed	-	Hardwired to PCS	0	-	-	Yokogawa DCS	No alarm. Indication only	5
XS23578	XI23578	2000-F23-PID-PR-40002	FIM III HYDRAULIC POWER PACK FIELD START SIGNAL	F23HY420 PB START		Not Pressed	Pressed	-	Hardwired to PCS	0	-	-	Yokogawa DCS	No alarm. Indication only	5
ZSC23532	ZIC23532	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SUCTION VALVE CLOSED	F23PP652 SUC VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23532	ZIO23532	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SUCTION VALVE OPEN	F23PP652 SUC VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1

DIGITAL INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	State 0	State 1	Alarm State	Source	Alarm Delay	Alarm Priority	Alarm Masking	Control System	Comments	Rev
ZSC23534	ZIC23534	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 FLUSHING VALVE CLOSED	F23PP652 FLSH VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23534	ZIO23534	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 FLUSHING VALVE OPEN	F23PP652 FLSH VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23537	ZIC23537	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE VALVE CLOSED	F23PP652 DIS VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23537	ZIO23537	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE VALVE OPEN	F23PP652 DIS VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23538	ZIC23538	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE DRAIN VALVE CLOSED	F23PP652 DIS DRN VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23538	ZIO23538	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE DRAIN VALVE OPEN	F23PP652 DIS DRN VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSC23545	ZIC23545	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SUCTION DRAIN VALVE CLOSED	F23PP652 SUC DRN VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
ZSO23545	ZIO23545	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SUCTION DRAIN VALVE OPEN	F23PP652 SUC DRN VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	1
US23539	UA23539	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 BURSTING DISC RELIEF INDICATOR	F23PP652 BRST DISC		Alarm	Normal	Alarm	Hardwired to PCS	3	High	-	Yokogawa DCS		1
F23FIP601ALM	F23FIP601_ALM	2000-F23-TER-IC-40009	F23-FIP-601 LV SWITCHROOM FIRE INDICATOR PANEL ALARM	F23FIP601 FIRE ALARM		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	4
F23FIP601FLT	F23FIP601_FLT	2000-F23-TER-IC-40009	F23-FIP-601 LV SWITCHROOM FIRE INDICATOR PANEL FAULT	F23FIP601 FIRE FAULT		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	4
F23FIP601ISL	F23FIP601I_SL	2000-F23-TER-IC-40009	F23-FIP-601 LV SWITCHROOM FIRE INDICATOR PANEL DISABLED	F23FIP601 FIRE ISO		Alarm	Normal	Alarm	Hardwired to PCS	-	-	-	Yokogawa DCS	Alarm and indication	4
ZSC23550	ZIC23550	2000-F23-TER-IC-40009	F23-SR-601 TRANSFORMER ROOM INTERNAL DOOR POSITION SWITCH - DOOR CLOSED	F23TR602 INT DOOR CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	4
ZSC23551	ZIC23551	2000-F23-TER-IC-40009	F23-SR-601 TRANSFORMER ROOM EXTERNAL DOOR POSITION SWITCH - DOOR CLOSED	F23TR602 EXT DOOR CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	Yokogawa DCS	No alarm. Indication only	4
XS75366	XI75366	2000-F75-PID-PR-40014	FIM III BOOSTER PUMPS SEAL WATER SELF CLEANING FILTER FAULT	F75FI600 FAULT	~~002~~	Fault	Normal	Fault	Hardwired to PCS	3	Med	-	Yokogawa DCS		5
XS75367	XI75367	2000-F75-PID-PR-40014	FIM III BOOSTER PUMPS SEAL WATER SELF CLEANING FILTER FLUSH CYCLE ON	F75FI600 FLUSH	~~002~~	Inactive (N_Flushing)	Active (Flushing)	-	Hardwired to PCS	-	-	-	Yokogawa DCS		5
ZSC75301	ZIC75301	2000-F75-PID-PR-40014	FIM III BOOSTER PUMP 1 SEAL WATER PUMPS 1 & 2 SUPPLY VALVE CLOSED	F23PW651/652 SUC VLV CLS		N_Closed	Closed	-	Hardwired to PCS	0	-	-	Yokogawa DCS	Digital Ouput	4
ZSO75301	ZIO75301	2000-F75-PID-PR-40014	FIM III BOOSTER PUMP 1 SEAL WATER PUMPS 1 & 2 SUPPLY VALVE OPEN	F23PW651/652 SUC VLV OP		N_Open	Open	-	Hardwired to PCS	0	-	-	Yokogawa DCS	Digital Ouput	4
ZSC75302	ZIC75302	2000-F75-PID-PR-40014	FIM III BOOSTER PUMP 2 SEAL WATER PUMPS 1 & 2 SUPPLY VALVE CLOSED	F23PW653/654 SUC VLV CLS		N_Closed	Closed	-	Hardwired to PCS	0	-	-	Yokogawa DCS	Digital Ouput	4
ZSO75302	ZIO75302	2000-F75-PID-PR-40014	FIM III BOOSTER PUMP 2 SEAL WATER PUMPS 1 & 2 SUPPLY VALVE OPEN	F23PW653/654 SUC VLV OP		N_Open	Open	-	Hardwired to PCS	0	-	-	Yokogawa DCS	Digital Ouput	4
HS75316SRT	HS75316_SRT	2000-F75-PID-PR-40014	FIM III BOOSTER STATION WASHDOWN PUMP ADDITIONAL START BUTTON 1	F75PW665 PB1 START		Not Pressed	Pressed	-	Hardwired to PCS	0	-	-	Yokogawa DCS		2
HS75316STP	HS75316_STP	2000-F75-PID-PR-40014	FIM III BOOSTER STATION WASHDOWN PUMP ADDITIONAL STOP BUTTON 1	F75PW665 PB1 STOP		Pressed	Not Pressed	-	Hardwired to PCS	0	-	-	Yokogawa DCS		2
HS75317SRT	HS75317_SRT	2000-F75-PID-PR-40014	FIM III BOOSTER STATION WASHDOWN PUMP ADDITIONAL START BUTTON 2	F75PW665 PB2 START		Not Pressed	Pressed	-	Hardwired to PCS	0	-	-	Yokogawa DCS		2
HS75317STP	HS75317_STP	2000-F75-PID-PR-40014	FIM III BOOSTER STATION WASHDOWN PUMP ADDITIONAL STOP BUTTON 2	F75PW665 PB2 STOP		Pressed	Not Pressed	-	Hardwired to PCS	0	-	-	Yokogawa DCS		2

TOTALISERS

Tag Source	Tag Name	Type	P&ID	Source Description	Abbreviated Description (Faceplate)	Hold	Units	Reset Period			Control System	Comments	Rev
								Shift	Day	Cont			
WI23417	WQI23417	WQI	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE DRY TONNES	F23PP466 TOTALISER DRY		t/h	x	x	x	Yokogawa DCS		2
WI23430	WQI23430	WQI	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE DRY TONNES	F23PP467 TOTALISER DRY		t/h	x	x	x	Yokogawa DCS		2
WI23441	WQI23441	WQI	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE DRY TONNES	F23PP468 TOTALISER DRY		t/h	x	x	x	Yokogawa DCS		2
WI23452	WQI23452	WQI	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE DRY TONNES	F23PP469 TOTALISER DRY		t/h	x	x	x	Yokogawa DCS		2
FIT75321	FQI75321	FQI	2000-F75-PID-PR-40014	BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER 2	F75TK651 TOTAL INFLOW 2		m^3	x	x	x	Yokogawa DCS		1

OPEN-CLOSE VALVES

Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Valve Detail	Fail Position	Manual	Auto	Hold	Open Condition (Auto Mode)	Close Condition (Auto Mode)	Condition	Interlocks								Control System	Comments	Rev
												Description	Delay On	Delay Off	Open Permissive	Close Permissive	Open Trip	Close Trip	Bypass	Reset Required			
XV23414	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE VALVE	F23PP466 DISCHARGE	Detail 3	FL	x	x		As per F23-SEQ-1001 As per F23-SEQ-1003	As per F23-SEQ-1002	F23-PP-466 NOT RUN	F23-PP-466 NOT RUN	5			x			x		Yokogawa DCS		3
XV23415	2000-F23-PID-PR-10002	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE FLUSHING WATER VALVE	F23PP466 DISCHARGE FLUSH	Detail 4	FC	x	x			As per F23-SEQ-1001	SUM(ZIC23428, ZIC23439, ZIC23450) = 1				x						Yokogawa DCS	Up to two tailings lines can be flushed at the same time	4
XV23427	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE VALVE	F23PP467 DISCHARGE	Detail 3	FL	x	x		As per F23-SEQ-1004 As per F23-SEQ-1006	As per F23-SEQ-1005	F23-PP-467 NOT RUN	F23-PP-467 NOT RUN	5			x			x		Yokogawa DCS		3
XV23428	2000-F23-PID-PR-10002	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE FLUSHING WATER VALVE	F23PP467 DISCHARGE FLUSH	Detail 4	FC	x	x			As per F23-SEQ-1004	SUM(ZIC23415, ZIC23439, ZIC23450) = 1				x						Yokogawa DCS	Up to two tailings lines can be flushed at the same time	4
XV23438	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE VALVE	F23PP468 DISCHARGE	Detail 3	FL	x	x		As per F23-SEQ-1007 As per F23-SEQ-1009	As per F23-SEQ-1008	F23-PP-468 NOT RUN	F23-PP-468 NOT RUN	5			x			x		Yokogawa DCS		3
XV23439	2000-F23-PID-PR-10002	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE FLUSHING WATER VALVE	F23PP468 DISCHARGE FLUSH	Detail 4	FC	x	x			As per F23-SEQ-1007	SUM(ZIC23415, ZIC23428, ZIC23450) = 1				x						Yokogawa DCS	Up to two tailings lines can be flushed at the same time	4
XV23449	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE VALVE	F23PP469 DISCHARGE	Detail 3	FL	x	x		As per F23-SEQ-1010 As per F23-SEQ-1012	As per F23-SEQ-1011	F23-PP-469 NOT RUN	F23-PP-469 NOT RUN	5			x			x		Yokogawa DCS		3
XV23450	2000-F23-PID-PR-10002	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE FLUSHING WATER VALVE	F23PP469 DISCHARGE FLUSH	Detail 4	FC	x	x			As per F23-SEQ-1010	SUM(ZIC23415, ZIC23428, ZIC23450) = 1				x						Yokogawa DCS	Up to two tailings lines can be flushed at the same time	4
XV23512	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 SUCTION VALVE	F23PP651 SUCTION	Detail 5	FL	x	x		As per F23-SEQ-2001	As per F23-SEQ-2002 As per F23-SEQ-2003	F23-PP-651 NOT RUN	F23-PP-651 NOT RUN	5			x			x		Yokogawa DCS		3
XV23514	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 FLUSHING VALVE	F23PP651 FLUSH	Detail 5	FL	x	x		As per F23-SEQ-2003												Yokogawa DCS		3
XV	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE VALVE	F23PP651 DISCHARGE	Detail 5	FL	x	x		As per F23-SEQ-2001 As per F23-SEQ-2003	As per F23-SEQ-2002	F23-PP-651 NOT RUN	F23-PP-651 NOT RUN	5			x			x		Yokogawa DCS		3
XV23518	2000-F23-PID-PR-40002	FIM III BOOSTER PUMP 1 DISCHARGE DRAIN VALVE	F23PP651 DISCHARGE DRAIN	Detail 5	FL	x	x			As per F23-SEQ-2001 As per F23-SEQ-2003											Yokogawa DCS		3
XV23532	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 SUCTION VALVE	F23PP652 SUCTION	Detail 5	FL	x	x		As per F23-SEQ-2004	As per F23-SEQ-2005 As per F23-SEQ-2006	F23-PP-652 NOT RUN	F23-PP-652 NOT RUN	5			x			x		Yokogawa DCS		3
XV23534	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 FLUSHING VALVE	F23PP652 FLUSH	Detail 5	FL	x	x		As per F23-SEQ-2006												Yokogawa DCS		3
XV23537	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE VALVE	F23PP652 DISCHARGE	Detail 5	FL	x	x		As per F23-SEQ-2004 As per F23-SEQ-2006	As per F23-SEQ-2005	F23-PP-652 NOT RUN	F23-PP-652 NOT RUN	5			x			x		Yokogawa DCS		3
XV23538	2000-F23-PID-PR-40003	FIM III BOOSTER PUMP 2 DISCHARGE DRAIN VALVE	F23PP652 DISCHARGE DRAIN	Detail 5	FL	x	x			As per F23-SEQ-2004 As per F23-SEQ-2006											Yokogawa DCS		3
XV75301	2000-F75-PID-PR-40014	FIM III BOOSTER PUMP 1 SEAL WATER PUMPS 1 & 2 ISOLATION VALVE	F23PW651/652 SUCTION	Detail 14	FL	x	x		As per F23-SEQ-2001 As per F23-SEQ-2003	As per F23-SEQ-2002	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	5			x			x		Yokogawa DCS		4
XV75302	2000-F75-PID-PR-40014	FIM III BOOSTER PUMP 2 SEAL WATER PUMPS 1 & 2 ISOLATION VALVE	F23PW653/654 SUCTION	Detail 14	FL	x	x		As per F23-SEQ-2004 As per F23-SEQ-2006	As per F23-SEQ-2005	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	5			x			x		Yokogawa DCS		4

2000-F00-PHL-PR-40001 - APPENDIX A

CONTROL VALVES

Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Valve Type	Fail Position	Manual	Auto	Hold	Open Condition (Auto Mode)	Close Condition (Auto Mode)	Condition	Interlocks									Control System	Comments	Rev
												Description	Delay On	Delay Off	Open Permissiv	Close Permissiv e	Open Trip	Close Trip	Bypass	Reset Required			
LCV75300	2000-F75-PID-PR-40014	FIM III BOOSTER STATION SEAL WATER TANK LEVEL CONTROL VALVE	F75TK651 LEVEL CONTROL	Detail 13	N/A	x	x				LIT75300 > A_HH		5					x	x		Yokogawa DCS		1

PID CONTROLLERS

Tag Source	Tag Name	P&ID	Abbreviated Description (Faceplate)	Hold	Normal Operating Mode	Modes			Process Value Tag	Setpoint Variable (SP)				Range			Process Switchpoints				Alarms								Alarm Masking	Manipulated Variable (MV)							Control System	Comment	Rev
						Manual	Auto	Cascade		Min	Max	Nom / Peak	Unit	Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)		Range Min	Range Max	Lower Limit	Upper Limit	Unit	Destination	Rev Actino			
FIT23417	FIC23417	2000-F23-PID-PR-10002	F23PP466 FLOWRATE		Cascade	X	X	X	FY23417	500	1,500	974	m³/h	0	2,000	m³/h	-	-	-	-	50	5	100	5	-	-	-	-	F23-PP-466 NOT RUN	0	100	30	80	% spd	F23-PP-466	Yes	Yokogawa DCS	Controller output to be limited to not exceed PIT-23419 A_H A_L = Warning A_LL = Trip F23-PP-466	4
FIT23430	FIC23430	2000-F23-PID-PR-10002	F23PP467 FLOWRATE		Cascade	X	X	X	FY23430	750	1,500	974	m³/h	0	2,000	m³/h	-	-	-	-	50	5	100	5	-	-	-	-	F23-PP-467 NOT RUN	0	100	30	80	% spd	F23-PP-467	Yes	Yokogawa DCS	Controller output to be limited to not exceed PIT-23432 A_H A_L = Warning A_LL = Trip F23-PP-467	4
FIT23441	FIC23441	2000-F23-PID-PR-10002	F23PP468 FLOWRATE		Cascade	X	X	X	FY23441	750	1,500	974	m³/h	0	2,000	m³/h	-	-	-	-	50	5	100	5	-	-	-	-	F23-PP-468 NOT RUN	0	100	30	80	% spd	F23-PP-468	Yes	Yokogawa DCS	Controller output to be limited to not exceed PIT-23443 A_H A_L = Warning A_LL = Trip F23-PP-468	4
FIT23452	FIC23452	2000-F23-PID-PR-10002	F23PP469 FLOWRATE		Cascade	X	X	X	FY23452	750	1,500	974	m³/h	0	2,000	m³/h	-	-	-	-	50	5	100	5	-	-	-	-	F23-PP-469 NOT RUN	0	100	30	80	% spd	F23-PP-469	Yes	Yokogawa DCS	Controller output to be limited to not exceed PIT-23454 A_H A_L = Warning A_LL = Trip F23-PP-469	4
FIT23521	FIC23521	2000-F23-PID-PR-40002	F23PP651 FLOWRATE		Automatic	X	X	~~X~~	~~FY23521~~	750	1,500	974	m³/h	0	2,000	m³/h	-	-	-	-	50	5	100	5	-	-	-	-	F23-PP-651 NOT RUN	0	100	30	80	% spd	F23-PP-651	No	Yokogawa DCS	Controller output to be limited to not exceed PIT-23520 A_H A_L = Warning A_LL = Trip F23-PP-651	~~5~~
PIT75303	PIC75303	2000-F23-PID-PR-40002	F23PW651/652 PRESSURE		Cascade	X	X	X	PY75303					0	2,000	kPa	-	-	-	-	PI23520.PV + 0	5	PI23520.PV + 50	5	PI23520.PV + 250	5	-	-	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	0	100	0	100	% spd	F75-PW-651 / F75-PW-652	No	Yokogawa DCS	A_H = Warning A_L = Warning A_LL = Trip F23-PP-651	4
FIT23541	FIC23541	2000-F23-PID-PR-40003	F23PP652 FLOWRATE		Automatic	X	X	~~X~~	~~FY23541~~	750	1,500	974	m³/h	0	2,000	m³/h	-	-	-	-	50	5	100	5	-	-	-	-	F23-PP-652 NOT RUN	0	100	30	80	% spd	F23-PP-652	No	Yokogawa DCS	Controller output to be limited to not exceed PIT-23540 A_H A_L = Warning A_LL = Trip F23-PP-652	~~5~~
PIT75306	PIC75306	2000-F23-PID-PR-40003	F23PW653/654 PRESSURE		Cascade	X	X	X	PY75306					0	2,000	kPa	-	-	-	-	PI23540.PV + 0	5	PI23540.PV + 50	5	PI23540.PV + 250	5	-	-	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	0	100	0	100	% spd	F23-PW-653 / F23-PW-654	No	Yokogawa DCS	A_H = Warning A_L = Warning A_LL = Trip F23-PP-652	4
LIT75300	LIC75300	2000-F75-PID-PR-40014	F75TK651 LEVEL		Automatic	X	X			70	100	90	%	0	100	%	-	-	-	-	5	5	50	5	95	5	99	5	-	0	100	0	100	% open	LCV75300	Yes	Yokogawa DCS	A_HH = Close LCV75300 A_H = Warning A_L = Warning; Trip F75-PW-665 A_LL = Trip F23-PW-651/652/653/654	4

MISCELLANEOUS CONTROLLERS

Tag Name	P&ID	Abbreviated Description	Hold	Normal State	Alarm State	Hardwired (y/n)	Alarm Delay (secs)	Alarm Masking Conditions	Control System	Comment	Rev
HS23521	2000-F23-PID-PR-40002	F23PP651 SPEED REF SEL		Operator Selection	No Alarm	N	-	-	Yokogawa DCS	HS23521 Toggle Switch to select between the two modes of control of the speed of F23-PP-651. ON = Reference speed control (with offset) from F23-PP-466 OFF = Output from FIC23521	1
HS23541	2000-F23-PID-PR-40003	F23PP652 SPEED REF SEL		Operator Selection	No Alarm	N	-	-	Yokogawa DCS	HS23541 Toggle Switch to select between the two modes of control of the speed of F23-PP-652. ON = Reference speed control (with offset) from F23-PP-467 OFF = Output from FIC23541	1
KY75665	2000-F75-PID-PR-40014	F75PW665 RUN ON TIMER		Timer PT = 600 Seconds	No Alarm	N	-	-	Yokogawa DCS	Timer to stop F75-PW-665 after a set runtime period. Run on time to be configurable from PCS with engineer access.	1
F23PP466_LOC	2000-F23-PID-PR-10002	F23PP466 DISCH LOC		Operator Selection	No Alarm	N	-	-	Yokogawa DCS	Included graphic to select discharge location for line D, only one can be selected: Discharge 1 - Line D to FIM I Discharge 2 - Line D to FIM III If Discharge 1 is SELECTED, FDI23521 is masked If Discharge 2 is SELECTED, FDI75902 is masked During Flushing: If Discharge 1 is SELECTED, FIC23417SP = 700 m^3/h If Discharge 2 is SELECTED, FIC23417SP = 1,200 m^3/h	2
F23PP467_LOC	2000-F23-PID-PR-10002	F23PP467 DISCH LOC		Operator Selection	No Alarm	N	-	-	Yokogawa DCS	Included graphic to select discharge location for line B, only one can be selected: Discharge 1 - Line B to Kaltails Discharge 2 - Line B to FIM III If Discharge 1 is SELECTED, FDI23541 is masked If Discharge 2 is SELECTED, FDI30940B is masked During Flushing: If Discharge 1 is SELECTED, FIC23430SP = 1,000 m^3/h If Discharge 2 is SELECTED, FIC23430SP = 1,200 m^3/h	2
F23PP468_LOC	2000-F23-PID-PR-10002	F23PP468 DISCH LOC		Operator Selection	No Alarm	N	-	-	Yokogawa DCS	Included graphic to select discharge location for line A, only one can be selected: Discharge 1 - Line A to Kaltails Discharge 2 - Line A to FIM II Discharge 3 - Line A to FIM II EXT If Discharge 1 is SELECTED, FDI30909 is masked If Discharge 2 is SELECTED, FDI30940A is masked If Discharge 3 is SELECTED, FDI30940A is masked During Flushing: If Discharge 1 is SELECTED, FIC23441SP = 1,000 m^3/h If Discharge 2 is SELECTED, FIC23441SP = 1,000 m^3/h If Discharge 3 is SELECTED, FIC23441SP = 1,000 m^3/h	2
F23PP469_LOC	2000-F23-PID-PR-10002	F23PP469 DISCH LOC		Operator Selection	No Alarm	N	-	-	Yokogawa DCS	Included graphic to select discharge location for line C, only one can be selected: Discharge 1 - Line C to Kaltails Discharge 2 - Line C to FIM II Discharge 3 - Line C to FIM II EXT If Discharge 1 is SELECTED, FDI30908 is masked If Discharge 2 is SELECTED, FDI30941 is masked If Discharge 3 is SELECTED, FDI30941 is masked During Flushing: If Discharge 1 is SELECTED, FIC23452SP = 1,000 m^3/h If Discharge 2 is SELECTED, FIC23452SP = 1,000 m^3/h If Discharge 3 is SELECTED, FIC23452SP = 1,000 m^3/h	2

MISCELLANEOUS CONTROLLERS

Tag Name	P&ID	Abbreviated Description	Hold	Normal State	Alarm State	Hardwired (y/n)	Alarm Delay (secs)	Alarm Masking Conditions	Control System	Comment	Rev
FDI23592A_SEL	2000-F23-PID-PR-40005	D TO FIM III LINE 1S DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23592B_SEL	2000-F23-PID-PR-40005	B TO FIM III LINE 1S DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23593A_SEL	2000-F23-PID-PR-40005	B TO FIM III LINE 1N DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23593B_SEL	2000-F23-PID-PR-40005	D TO FIM III LINE 1N DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23594A_SEL	2000-F23-PID-PR-40005	B TO FIM III LINE 3W DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23594B_SEL	2000-F23-PID-PR-40005	D TO FIM III LINE 3W DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23595A_SEL	2000-F23-PID-PR-40005	D TO FIM III LINE 3E DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23595B_SEL	2000-F23-PID-PR-40005	B TO FIM III LINE 3E DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23596A_SEL	2000-F23-PID-PR-40005	B TO FIM III LINE 2W DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23596B_SEL	2000-F23-PID-PR-40005	D TO FIM III LINE 2W DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23597A_SEL	2000-F23-PID-PR-40005	D TO FIM III LINE 2E DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5
FDI23597B_SEL	2000-F23-PID-PR-40005	B TO FIM III LINE 2E DIFF FLOW		Operator Selection	No Alarm	N	-	-	Yokogawa DCS		5

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks									Control System	Comments	Rev
																Description (Unhealthy Interlocks and Healthy Start Permissives	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissives	Process	Critical	Hardwired to Drive	Bypass	Reset Required			
F23-PP-466	2000-F23-PID-PR-10002	FTAILS PUMP 1	VSD		0	100	%	30	100	165	X	X	F23-SEQ-1001 / F23-SEQ-1003	F23-SEQ-1002											Yokogawa DCS		3
															1	F23-PP-466 Comms Failure	-	0				x						1
															2	F23-PP-466 Fail to Start / Stop	-	0				x						1
															3	F23-PP-466 Drive Fault	-	0				x						1
															4	F23-PP-466 E-Stop	-	0				x	x					1
															5	F23-PP-466 MCC Unhealthy	-	0				x						1
															6	F23-PP-466 Drive Out of Service	-	0				x						1
															7	F23-PP-466 Drive Ready	-	0				x						1
															8	F23-FN-466 NOT RUN	-	5				x			x		Gearbox Fan	0
															9	F23-FN-462 NOT RUN	-	5				x			x		Motor Cooling Fan	1
															10	F23-PP-651 NOT HEALTHY AND F30-PP-79 NOT HEALTHY	x					x		x			FIM III Booster Pump 1 and FIM I Booster Pump Bypass interlock in flush mode only	1
															11	SV23480 AUTO OR OPEN	x	5		x				x			Gland water isolation valve in auto or open	2
															12	FI23481 < A_LL AND F23-PP-466 RUNNING	x	300			x			x			Low low gland water flow. **Mask if F23-PP-466 RUN for less than 10 secs**	**5**
															13	LIC23401 > A_L	x	5		x				x			Final tails tank level low	1
															14	LIC23401 < A_LL	x	5			x			x			Final tails tank level low low	1
															15	FIC23417 < A_LL AND F23-PP-466 RUNNING	x	600			x			x			Low low discharge flow	1
															16	PI23419A > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP466_LOC = DISCHARGE 2	1
															17	PI23419B > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP466_LOC = DISCHARGE 1	1
															18	XV23411 NOT OPEN	x	30			x			x			Suction valve closed Bypass interlock in flush mode only	2
															19	XV23413 NOT CLOSED	x	30			x			x			Suction drain valve open	1
															20	XV23414 NOT OPEN	x	30			x			x			Discharge valve closed. Bypass interlock for start sequence	1
															21	**TI23470 > A_HH**	**x**	**5**			**x**			**x**			**Gearbox temperature high high**	**5**
															22	**FDI75902 < A_L**	**x**	**600**			**x**			**x**			**Line D to FIM I Differential Flow Low** **Mask if F23PP466_LOC = DISCHARGE 2**	**5**
															23	**FDI75902 > A_H**	**x**	**600**			**x**			**x**			**Line D to FIM I Differential Flow High** **Mask if F23PP466_LOC = DISCHARGE 2**	**5**
															24	**FDI23521 < A_L**	**x**	**600**			**x**			**x**			**Line D to FIM III Tailings Differential Flow Low** **Mask if F23PP466_LOC = DISCHARGE 1**	**5**
															25	**FDI23521 > A_H**	**x**	**600**			**x**			**x**			**Line D to FIM III Tailings Differential Flow High** **Mask if F23PP466_LOC = DISCHARGE 1**	**5**
F23-FN-462	2000-F23-PID-PR-10002	FTAILS PUMP 1 FAN	DOL							7.30	X	X	F23-SEQ-1001 / F23-SEQ-1003	F23-PP-466 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-462 Comms Failure	-	0				x						1
															2	F23-FN-462 Fail to Start / Stop	-	0				x						1
															3	F23-FN-462 Drive Fault	-	0				x						1
															4	F23-FN-462 E-Stop	-	0				x	x					1
															5	F23-FN-462 MCC Unhealthy	-	0				x						1
															6	F23-FN-462 Drive Out of Service	-	0				x						1
															7	F23-FN-462 Drive Ready	-	0				x						1
F23-FN-466	2000-F23-PID-PR-10002	FTAILS PUMP 1 G/BOX FAN	DOL							0.58	X	X	F23-SEQ-1001 / F23-SEQ-1003	F23-PP-466 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-466 Comms Failure	-	0				x						1
															2	F23-FN-466 Fail to Start / Stop	-	0				x						1
															3	F23-FN-466 Drive Fault	-	0				x						1
															4	F23-FN-466 E-Stop	-	0				x	x					1
															5	F23-FN-466 MCC Unhealthy	-	0				x						1
															6	F23-FN-466 Drive Out of Service	-	0				x						1
															7	F23-FN-466 Drive Ready	-	0				x						1
F23-PP-467	2000-F23-PID-PR-10002	FTAILS PUMP 2	VSD		0	100	%	30	100	165	X	X	F23-SEQ-1004 / F23-SEO-1006	F23-SEQ-1005											Yokogawa DCS		3
															1	F23-PP-467 Comms Failure	-	0				x						1
															2	F23-PP-467 Fail to Start / Stop	-	0				x						1
															3	F23-PP-467 Drive Fault	-	0				x						1
															4	F23-PP-467 E-Stop	-	0				x	x					1
															5	F23-PP-467 MCC Unhealthy	-	0				x						1
															6	F23-PP-467 Drive Out of Service	-	0				x						1
															7	F23-PP-467 Drive Ready	-	0				x						1
															8	F23-FN-467 NOT RUN	-	5				x			x		Gearbox Fan	0
															9	F23-FN-463 NOT RUN	-	5				x			x		Motor Cooling Fan	1
															10	F23-PP-652 NOT HEALTHY AND F30-PP-78A NOT HEALTHY	x					x		x			FIM III Booster Pump 2 and Kaltails Booster Pump A Bypass interlock in flush mode only	1
															11	SV23484 AUTO OR OPEN	x	5		x				x			Gland water isolation valve in auto or open	2
															12	FI23485 < A_LL AND F23-PP-467 RUNNING	x	300			x			x			Low low gland water flow. **Mask if F23-PP-466 RUN for less than 10 secs**	**5**
															13	LIC23401 > A_L	x	5		x				x			Final tails tank level low	1
															14	LIC23401 < A_LL	x	5			x			x			Final tails tank level low low	1

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks: Description (Unhealthy Interlocks and Healthy Start Permissives)	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissive	Process	Critical Hardwired to Drive	Bypass	Reset Required	Control System	Comments	Rev
															15	FIC23430 < A_LL AND F23-PP-467 RUNNING	x	600			x			x			Low low discharge flow	1
															16	PI23432A > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP467_LOC = DISCHARGE 2	1
															17	PI23432B > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP467_LOC = DISCHARGE 1	1
															18	XV23424 NOT OPEN	x	30			x			x			Suction valve closed Bypass interlock in flush mode only	2
															19	XV23426 NOT CLOSED	x	30			x			x			Suction drain valve open	1
															20	XV23427 NOT OPEN	x	30			x			x			Discharge valve closed. Bypass interlock for start sequence	1
															21	**TI23472 > A_HH**	**x**	**5**			**x**			**x**			**Gearbox temperature high high**	**5**
															22	**FDI30940B < A_L**	**x**	**600**			**x**			**x**			**Line B to Kaltails Differential Flow Low Mask if F23PP467_LOC = DISCHARGE 2**	**5**
															23	**FDI30940B > A_H**	**x**	**600**			**x**			**x**			**Line B to Kaltails Differential Flow High Mask if F23PP467_LOC = DISCHARGE 2**	**5**
															24	**FDI23541 < A_L**	**x**	**600**			**x**			**x**			**Line B to FIM II Tailings Differential Flow Low Mask if F23PP467_LOC = DISCHARGE 1**	**5**
															25	**FDI23541 > A_H**	**x**	**600**			**x**			**x**			**Line B to FIM II Tailings Differential Flow High Mask if F23PP467_LOC = DISCHARGE 1**	**5**
F23-FN-463	2000-F23-PID-PR-10002	FTAILS PUMP 2 FAN	DOL							7.30	X	X	F23-SEQ-1004 / F23-SEQ-1006	F23-PP-467 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-463 Comms Failure	-	0				x						1
															2	F23-FN-463 Fail to Start / Stop	-	0				x						1
															3	F23-FN-463 Drive Fault	-	0				x						1
															4	F23-FN-463 E-Stop	-	0				x	x					1
															5	F23-FN-463 MCC Unhealthy	-	0				x						1
															6	F23-FN-463 Drive Out of Service	-	0				x						1
															7	F23-FN-463 Drive Ready	-	0				x						1
F23-FN-467	2000-F23-PID-PR-10002	FTAILS PUMP 2 G/BOX FAN	DOL							0.58	X	X	F23-SEQ-1004 / F23-SEQ-1006	F23-PP-467 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-467 Comms Failure	-	0				x						1
															2	F23-FN-467 Fail to Start / Stop	-	0				x						1
															3	F23-FN-467 Drive Fault	-	0				x						1
															4	F23-FN-467 E-Stop	-	0				x	x					1
															5	F23-FN-467 MCC Unhealthy	-	0				x						1
															6	F23-FN-467 Drive Out of Service	-	0				x						1
															7	F23-FN-467 Drive Ready	-	0				x						1
F23-PP-468	2000-F23-PID-PR-10002	FTAILS PUMP 3	VSD		0	100	%	30	100	165	X	X	F23-SEQ-1007 / F23-SEQ-1009	F23-SEQ-1008											Yokogawa DCS		3
															1	F23-PP-468 Comms Failure	-	0				x						1
															2	F23-PP-468 Fail to Start / Stop	-	0				x						1
															3	F23-PP-468 Drive Fault	-	0				x						1
															4	F23-PP-468 E-Stop	-	0				x	x					1
															5	F23-PP-468 MCC Unhealthy	-	0				x						1
															6	F23-PP-468 Drive Out of Service	-	0				x						1
															7	F23-PP-468 Drive Ready	-	0				x						1
															8	F23-FN-468 NOT RUN	-	5				x			x		Gearbox Fan	0
															9	F23-FN-464 NOT RUN	-	5				x			x		Motor Cooling Fan	1
															10	F30-PP-77B NOT HEALTHY AND F30-PP-78A NOT HEALTHY	x					x		x			FIM II Booster Pump 2 and Kaltails Booster Pump A Bypass interlock in flush mode only	1
															11	SV23488 AUTO OR OPEN	x	5		x				x			Gland water isolation valve in auto or open	2
															12	FI23489 < A_LL AND F23-PP-468 RUNNING	x	300			x			x			Low low gland water flow. **Mask if F23-PP-466 RUN for less than 10 secs**	**5**
															13	LIC23401 > A_L	x	5		x				x			Final tails tank level low	1
															14	LIC23401 < A_LL	x	5			x			x			Final tails tank level low low	1
															15	FIC23441 < A_LL AND F23-PP-468 RUNNING	x	600			x			x			Low low discharge flow	1
															16	PI23443A > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP468_LOC = DISCHARGE 2 OR 3	3
															17	PI23443B > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP468_LOC = DISCHARGE 1	1
															18	XV23435 NOT OPEN	x	30			x			x			Suction valve closed Bypass interlock in flush mode only	2
															19	XV23437 NOT CLOSED	x	30			x			x			Suction drain valve open	1
															20	XV23438 NOT OPEN	x	30			x			x			Discharge valve closed. Bypass interlock for start sequence	1
															21	**TI23474 > A_HH**	**x**	**5**			**x**			**x**			**Gearbox temperature high high**	**5**
															22	**FDI30940A < A_L**	**x**	**600**			**x**			**x**			**Line A to Kaltails Differential Flow Low Mask if F23PP468_LOC = DISCHARGE 2 OR F23PP468_LOC = DISCHARGE 3**	**5**
															23	**FDI30940A > A_H**	**x**	**600**			**x**			**x**			**Line A to Kaltails Differential Flow High Mask if F23PP468_LOC = DISCHARGE 2 OR F23PP468_LOC = DISCHARGE 3**	**5**
															24	**FDI30909 < A_L**	**x**	**600**			**x**			**x**			**Line A to FIM II Differential Flow Low Mask if F23PP468_LOC = DISCHARGE 1**	**5**
															25	**FDI30909 > A_H**	**x**	**600**			**x**			**x**			**Line A to FIM II Differential Flow High Mask if F23PP468_LOC = DISCHARGE 1**	**5**

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks: Description (Unhealthy Interlocks and Healthy Start Permissives)	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissives	Process	Critical	Hardwired to Drive	Bypass	Reset Required	Control System	Comments	Rev
F23-FN-464	2000-F23-PID-PR-10002	FTAILS PUMP 3 FAN	DOL							7.30	X	X	F23-SEQ-1007 / F23-SEQ-1009	F23-PP-468 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-464 Comms Failure	-	0				x						1
															2	F23-FN-464 Fail to Start / Stop	-	0				x						1
															3	F23-FN-464 Drive Fault	-	0				x						1
															4	F23-FN-464 E-Stop	-	0				x	x					1
															5	F23-FN-464 MCC Unhealthy	-	0				x						1
															6	F23-FN-464 Drive Out of Service	-	0				x						1
															7	F23-FN-464 Drive Ready	-	0				x						1
F23-FN-468	2000-F23-PID-PR-10002	FTAILS PUMP 3 G/BOX FAN	DOL							0.58	X	X	F23-SEQ-1007 / F23-SEQ-1009	F23-PP-468 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-468 Comms Failure	-	0				x						1
															2	F23-FN-468 Fail to Start / Stop	-	0				x						1
															3	F23-FN-468 Drive Fault	-	0				x						1
															4	F23-FN-468 E-Stop	-	0				x	x					1
															5	F23-FN-468 MCC Unhealthy	-	0				x						1
															6	F23-FN-468 Drive Out of Service	-	0				x						1
															7	F23-FN-468 Drive Ready	-	0				x						1
F23-PP-469	2000-F23-PID-PR-10002	FTAILS PUMP 4	VSD		0	100	%	30	100	165	X	X	F23-SEQ-1010 / F23-SEQ-1012	F23-SEQ-1011											Yokogawa DCS		3
															1	F23-PP-469 Comms Failure	-	0				x						1
															2	F23-PP-469 Fail to Start / Stop	-	0				x						1
															3	F23-PP-469 Drive Fault	-	0				x						1
															4	F23-PP-469 E-Stop	-	0				x	x					1
															5	F23-PP-469 MCC Unhealthy	-	0				x						1
															6	F23-PP-469 Drive Out of Service	-	0				x						1
															7	F23-PP-469 Drive Ready	-	0				x						1
															8	F23-FN-469 NOT RUN	-	5				x			x		Gearbox Fan	0
															9	F23-FN-465 NOT RUN	-	5				x			x		Motor Cooling Fan	1
															10	F30-PP-77A NOT HEALTHY AND F30-PP-78B NOT HEALTHY	x					x		x			FIM II Booster Pump 1 and Kaltails Booster Pump B Bypass interlock in flush mode only	1
															11	SV23492 AUTO OR OPEN	x	5		x				x			Gland water isolation valve in auto or open	2
															12	FI23493 < A_LL AND F23-PP-469 RUNNING	x	300			x			x			Low low gland water flow. **Mask if F23-PP-466 RUN for less than 10 secs**	**5**
															13	LIC23401 > A_L	x	5		x				x			Final tails tank level low	1
															14	LIC23401 < A_LL	x	5			x			x			Final tails tank level low low	1
															15	FIC23452 < A_LL AND F23-PP-469 RUNNING	x	600			x			x			Low low discharge flow	1
															16	PI23454A > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP469_LOC = DISCHARGE 2 OR 3	3
															17	PI23454B > A_HH	x	5			x			x			High high discharge pressure Mask if F23PP469_LOC = DISCHARGE 1	1
															18	XV23446 NOT OPEN	x	30			x			x			Suction valve closed Bypass interlock in flush mode only	2
															19	XV23448 NOT CLOSED	x	30			x			x			Suction drain valve open	1
															20	XV23449 NOT OPEN	x	30			x			x			Discharge valve closed. Bypass interlock for start sequence	1
															21	**TI23474 > A_HH**	**x**	**5**			**x**			**x**			**Gearbox temperature high high**	**5**
															22	**FDI30941 < A_L**	**x**	**600**			**x**			**x**			**Line C to Kaltails Differential Flow Low Mask if F23PP469_LOC = DISCHARGE 2 OR F23PP469_LOC = DISCHARGE 3**	**5**
															23	**FDI30941 > A_H**	**x**	**600**			**x**			**x**			**Line C to Kaltails Differential Flow High Mask if F23PP469_LOC = DISCHARGE 2 OR F23PP469_LOC = DISCHARGE 3**	**5**
															24	**FDI30908 < A_L**	**x**	**600**			**x**			**x**			**Line C to FIM II Differential Flow Low Mask if F23PP469_LOC = DISCHARGE 1**	**5**
															25	**FDI30908 > A_H**	**x**	**600**			**x**			**x**			**Line C to FIM II Differential Flow High Mask if F23PP4698_LOC = DISCHARGE 1**	**5**
F23-FN-465	2000-F23-PID-PR-10002	FTAILS PUMP 4 FAN	DOL							7.30	X	X	F23-SEQ-1010 / F23-SEQ-1012	F23-PP-469 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-465 Comms Failure	-	0				x						1
															2	F23-FN-465 Fail to Start / Stop	-	0				x						1
															3	F23-FN-465 Drive Fault	-	0				x						1
															4	F23-FN-465 E-Stop	-	0				x	x					1
															5	F23-FN-465 MCC Unhealthy	-	0				x						1
															6	F23-FN-465 Drive Out of Service	-	0				x						1
															7	F23-FN-465 Drive Ready	-	0				x						1

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks										Control System	Comments	Rev
																Description (Unhealthy Interlocks and Healthy Start Permissives	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissives	Process	Critical	Hardwired to Drive	Bypass	Reset Required			
F23-FN-469	2000-F23-PID-PR-10002	FTAILS PUMP 4 G/BOX FAN	DOL							0.58	X	X	F23-SEQ-1010 / F23-SEQ-1012	F23-PP-469 NOT RUN FOR 30 SECS												Yokogawa DCS		3
															1	F23-FN-469 Comms Failure	-	0				x						1
															2	F23-FN-469 Fail to Start / Stop	-	0				x						1
															3	F23-FN-469 Drive Fault	-	0				x						1
															4	F23-FN-469 E-Stop	-	0				x	x					1
															5	F23-FN-469 MCC Unhealthy	-	0				x						1
															6	F23-FN-469 Drive Out of Service	-	0				x						1
															7	F23-FN-469 Drive Ready	-	0				x						1
F23-HY-420	2000-F23-PID-PR-40002	FIM 3 BOOSTER STATION HPU	FDR							13.4	X	X														Yokogawa DCS		5
															~~1~~	~~F23-HY-420 Comms Failure~~	~~-~~	~~0~~				~~x~~						~~5~~
															~~2~~	~~F23-HY-420 Fail to Start / Stop~~	~~-~~	~~0~~				~~x~~						~~5~~
															3	F23-HY-420 Drive Fault	-	0				x						1
															4	F23-HY-420 E-Stop	-	0				x	x					1
															~~5~~	~~F23-HY-420 MCC Unhealthy~~	~~-~~	~~0~~				~~x~~						~~5~~
															~~6~~	~~F23-HY-420 Drive Out of Service~~	~~-~~	~~0~~				~~x~~						~~5~~
															7	F23-HY-420 Drive Ready	-	0				x						1
F23-PP-651	2000-F23-PID-PR-40002	FIM 3 BOOSTER 1	VSD		0	100	%	0	100	171	X	X	F23-SEQ-2001 / F23-SEQ-2003	F23-SEQ-2002												Yokogawa DCS		3
															1	F23-PP-651 Comms Failure	-	0				x						1
															2	F23-PP-651 Fail to Start / Stop	-	0				x						1
															3	F23-PP-651 Drive Fault	-	0				x						1
															4	F23-PP-651 E-Stop	-	0				x	x					1
															5	F23-PP-651 MCC Unhealthy	-	0				x						1
															6	F23-PP-651 Drive Out of Service	-	0				x						1
															7	F23-PP-651 Drive Ready	-	0				x						1
															8	F23-FN-651 NOT RUN	-	5				x			x		Gearbox Fan	1
															9	F23-FN-653 NOT RUN	-	5				x			x		Motor Cooling Fan	1
															10	UA23519 ACTIVE	-					x			x		Bursting disc	1
															11	F23-PP-466 NOT RUN	-					x		x			Final Tails Pump 1 Bypass interlock in flush mode only	1
															12	PI23511 > A_L	-	5		x				x			Suction pressure	1
															13	FI23523 > A_L	-	5		x				x			Seal water flow	1
															14	FI23523 < A_LL AND F23-PP-651 RUNNING	-	300			x			x			Seal water flow low low	1
															15	PIC75303 < A_LL	-	5			x			x			Seal water pressure low low	1
															16	PI23520 > A_HH	-	5			x			x			Discharge pressure high high	0
															17	FIC23521 < A_LL AND F23-PP-651 RUNNING	-	600			x			x			Discharge flow low low	1
															18	ZIC23527 NOT ACTIVE OR ZIO23527 ACTIVE	-	30			x			x			Suction drain valve open	1
															19	XV23512 NOT OPEN AND XV23514 NOT OPEN	-	30			x			x			Suction valve closed and flushing valve closed	0
															20	XV CLOSED	-	30			x			x			Discharge valve closed. Bypass interlock for start sequence	1
															21	XV23518 OPEN		30			x			x			Discharge drain valve open	1
															22	~~LIC75300 < A_LL~~ F23-PW-651 NOT RUNNING AND F23-PW-652 NOT RUNNING	-	5				x					Seal water pumps not running ~~tank level low low~~	5
															23	F23TE651A >=A_HH OR F23TE651B >=A_HH OR F23TE651C >=A_HH	-	5				x					Drive winding temperature high high	0
															24	F23TE651D >=A_HH	-	5				x					Drive end bearing temperature high high	0
															25	F23TE651E >=A_HH	-	5				x					Non Drive end bearing temperature high high	0
															~~26~~	~~F23TE651F >=A_HH~~	~~-~~	~~5~~				~~x~~					~~Ambient temperature high high.~~	5
															27	FDI23592A < A_L	x	600			x			x			Line D to FIM III Line 1 South Differential Flow Low	5
															28	FDI23592A > A_H	x	600			x			x			Line D to FIM III Line 1 South Differential Flow High	5
															29	FDI23593B < A_L	x	600			x			x			Line D to FIM III Line 1 North Differential Flow Low	5
															30	FDI23593B > A_H	x	600			x			x			Line D to FIM III Line 1 North Differential Flow High	5
															31	FDI23594B < A_L	x	600			x			x			Line D to FIM III Line 3 West Differential Flow Low	5
															32	FDI23594B > A_H	x	600			x			x			Line D to FIM III Line 3 West Differential Flow High	5
															33	FDI23595A < A_L	x	600			x			x			Line D to FIM III Line 3 East Differential Flow Low	5
															34	FDI23595A > A_H	x	600			x			x			Line D to FIM III Line 3 East Differential Flow High	5
															35	FDI23596B < A_L	x	600			x			x			Line D to FIM III Line 2 West Differential Flow Low	5
															36	FDI23596B > A_H	x	600			x			x			Line D to FIM III Line 2 West Differential Flow High	5
															37	FDI23597A < A_L	x	600			x			x			Line D to FIM III Line 2 East Differential Flow Low	5
															38	FDI23597A > A_H	x	600			x			x			Line D to FIM III Line 2 East Differential Flow High	5

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks: Description (Unhealthy Interlocks and Healthy Start Permissives)	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissives	Process	Critical	Hardwired to Drive	Bypass	Reset Required	Control System	Comments	Rev
F23-FN-651	2000-F23-PID-PR-40002	FIM 3 BOOSTER 1 G/BOX FAN	DOL							3.16	X	X	F23-SEQ-2001 / F23-SEQ-2003	F23-PP-651 NOT RUN FOR 30 SECS											Yokogawa DCS		3
															1	F23-FN-651 Comms Failure	-	0				x						1
															2	F23-FN-651 Fail to Start / Stop	-	0				x						1
															3	F23-FN-651 Drive Fault	-	0				x						1
															4	F23-FN-651 E-Stop	-	0				x	x					1
															5	F23-FN-651 MCC Unhealthy	-	0				x						1
															6	F23-FN-651 Drive Out of Service	-	0				x						1
															7	F23-FN-651 Drive Ready	-	0				x						1
F23-FN-653	2000-F23-PID-PR-40002	FIM 3 BOOSTER 1 MTR FAN	DOL							3.16	X	X	F23-SEQ-2001 / F23-SEQ-2003	F23-PP-651 NOT RUN FOR 30 SECS													Yokogawa DCS		3
															1	F23-FN-653 Comms Failure	-	0				x						1
															2	F23-FN-653 Fail to Start / Stop	-	0				x						1
															3	F23-FN-653 Drive Fault	-	0				x						1
															4	F23-FN-653 E-Stop	-	0				x	x					1
															5	F23-FN-653 MCC Unhealthy	-	0				x						1
															6	F23-FN-653 Drive Out of Service	-	0				x						1
															7	F23-FN-653 Drive Ready	-	0				x						1
F23-PP-652	2000-F23-PID-PR-40003	FIM 3 BOOSTER 2	VSD		0	100	%	0	100	171	X	X	F23-SEQ-2004 / F23-SEQ-2006	F23-SEQ-2005													Yokogawa DCS		3
															1	F23-PP-652 Comms Failure	-	0				x						1
															2	F23-PP-652 Fail to Start / Stop	-	0				x						1
															3	F23-PP-652 Drive Fault	-	0				x						1
															4	F23-PP-652 E-Stop	-	0				x	x					1
															5	F23-PP-652 MCC Unhealthy	-	0				x						1
															6	F23-PP-652 Drive Out of Service	-	0				x						1
															7	F23-PP-652 Drive Ready	-	0				x						1
															8	F23-FN-652 NOT RUN	-	5				x			x		Gearbox Fan	1
															9	F23-FN-654 NOT RUN	-	5				x			x		Motor Cooling Fan	1
															10	UA23539 ACTIVE	-					x			x		Bursting disc	1
															11	F23-PP-467 NOT RUN	-					x		x			Final Tails Pump 2 Bypass interlock in flush mode only	1
															12	PI23531 > A_L	-	5		x				x			Suction pressure	1
															13	FI23543 > A_L	-	5		x				x			Seal water flow	1
															14	FI23543 < A_LL AND F23-PP-652 RUNNING	-	300			x			x			Seal water flow low low	0
															15	PIC75306 < A_LL	-	5			x			x			Seal water pressure low low	1
															16	PI23540 > A_HH	-	5			x			x			Discharge pressure high high	0
															17	FIC23541 < A_LL AND F23-PP-652 RUNNING	-	600			x			x			Discharge flow low low	1
															18	ZIC23545 NOT ACTIVE OR ZIO23545 ACTIVE	-	30			x			x			Suction drain valve open	1
															19	XV23532 NOT OPEN AND XV23534 NOT OPEN	-	30			x			x			Suction valve closed and flushing valve closed	0
															20	XV23537 CLOSED	-	30			x			x			Discharge valve closed. Bypass interlock for start sequence	1
															21	XV23538 OPEN	-	30			x			x			Discharge drain valve open	1
															22	~~LIC75300 < A_LL~~ F23-PW-653 NOT RUNNING AND F23-PW-654 NOT RUNNING	-	5				x					Seal water pumps not running ~~tank level low low~~	5
															23	F23TE652A >=A_HH OR F23TE652B >=A_HH OR F23TE652C >=A_HH	-	5				x					Drive winding temperature high high	0
															24	F23TE652D >=A_HH	-	5				x					Drive end bearing temperature high high	0
															25	F23TE652E >=A_HH	-	5				x					Non Drive end bearing temperature high high	0
															~~26~~	~~F23TE652F >=A_HH~~	~~-~~	~~5~~				~~x~~					~~Ambient temperature high high.~~	5
															27	FDI23592B < A_L	x	600				x		x			Line B to FIM III Line 1 South Differential Flow Low	5
															28	FDI23592B > A_H	x	600				x		x			Line B to FIM III Line 1 South Differential Flow High	5
															29	FDI23593A < A_L	x	600				x		x			Line B to FIM III Line 1 North Differential Flow Low	5
															30	FDI23593A > A_H	x	600				x		x			Line B to FIM III Line 1 North Differential Flow High	5
															31	FDI23594A < A_L	x	600				x		x			Line B to FIM III Line 3 West Differential Flow Low	5
															32	FDI23594A > A_H	x	600				x		x			Line B to FIM III Line 3 West Differential Flow High	5
															33	FDI23595B < A_L	x	600				x		x			Line B to FIM III Line 3 East Differential Flow Low	5
															34	FDI23595B > A_H	x	600				x		x			Line B to FIM III Line 3 East Differential Flow High	5
															35	FDI23596A < A_L	x	600				x		x			Line B to FIM III Line 2 West Differential Flow Low	5
															36	FDI23596A > A_H	x	600				x		x			Line B to FIM III Line 2 West Differential Flow High	5
															37	FDI23597B < A_L	x	600				x		x			Line B to FIM III Line 2 East Differential Flow Low	5
															38	FDI23597B > A_H	x	600				x		x			Line B to FIM III Line 2 East Differential Flow High	5

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks									Control System	Comments	Rev
																Description (Unhealthy Interlocks and Healthy Start Permissives	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissives	Process	Critical	Hardwired to Drive	Bypass	Reset Required			
F23-FN-652	2000-F23-PID-PR-40003	FIM3 BOOSTER 2 G/BOX FAN	DOL							3.16	X	X	F23-SEQ-2004 / F23-SEQ-2006	F23-PP-652 NOT RUN FOR 30 SECS												Yokogawa DCS		3
															1	F23-FN-652 Comms Failure	-	0				x						1
															2	F23-FN-652 Fail to Start / Stop	-	0				x						1
															3	F23-FN-652 Drive Fault	-	0				x						1
															4	F23-FN-652 E-Stop	-	0				x	x					1
															5	F23-FN-652 MCC Unhealthy	-	0				x						1
															6	F23-FN-652 Drive Out of Service	-	0				x						1
															7	F23-FN-652 Drive Ready	-	0				x						1
F23-FN-654	2000-F23-PID-PR-40003	FIM3 BOOSTER 2 MTR FAN	DOL							3.16	X	X	F23-SEQ-2004 / F23-SEQ-2006	F23-PP-652 NOT RUN FOR 30 SECS												Yokogawa DCS		3
															1	F23-FN-654 Comms Failure	-	0				x						1
															2	F23-FN-654 Fail to Start / Stop	-	0				x						1
															3	F23-FN-654 Drive Fault	-	0				x						1
															4	F23-FN-654 E-Stop	-	0				x	x					1
															5	F23-FN-654 MCC Unhealthy	-	0				x						1
															6	F23-FN-654 Drive Out of Service	-	0				x						1
															7	F23-FN-654 Drive Ready	-	0				x						1
F00-PS-464	2000-F75-PID-PR-25010	EVENT POND PUMP	DOL							190	X	X		~~LI75621 < P_L (Delay 60 seconds)~~												Yokogawa DCS	LOCAL START ONLY	**5**
															1	F00-PS-464 Comms Failure	-	0				x						1
															2	F00-PS-464 Fail to Start / Stop	-	0				x						1
															3	F00-PS-464 Drive Fault	-	0				x						1
															4	F00-PS-464 E-Stop	-	0				x	x					1
															5	F00-PS-464 MCC Unhealthy	-	0				x						1
															6	F00-PS-464 Drive Out of Service	-	0				x						1
															7	F00-PS-464 Drive Ready	-	0				x						1
															8	**LI75621 < P_L**	**-**	**60**			**x**						**Event pond level low**	**5**
F00-PS-465	2000-F75-PID-PR-25010	SILT TRAP PUMP	DOL							190	X	X		~~LI75620 < P_L (Delay 60 seconds)~~												Yokogawa DCS	LOCAL START ONLY	**5**
															1	F00-PS-465 Comms Failure	-	0				x						1
															2	F00-PS-465 Fail to Start / Stop	-	0				x						1
															3	F00-PS-465 Drive Fault	-	0				x						1
															4	F00-PS-465 E-Stop	-	0				x	x					1
															5	F00-PS-465 MCC Unhealthy	-	0				x						1
															6	F00-PS-465 Drive Out of Service	-	0				x						1
															7	F00-PS-465 Drive Ready	-	0				x						1
															8	**LI75620 < P_L**	**-**	**60**			**x**						**Event pond silt trap level low**	**5**
F75-FI-600	2000-F75-PID-PR-40014	FIM3 BOOSTER SEAL FILTER	**FDR**	002						2.21	X	X														Yokogawa DCS		**5**
															1	F75-FI-600 Comms Failure	-	0				x						1
															2	F75-FI-600 Fail to Start / Stop	-	0				x						1
															3	F75-FI-600 Drive Fault	-	0				x						1
															4	F75-FI-600 E-Stop	-	0				x	x					1
															5	F75-FI-600 MCC Unhealthy	-	0				x						1
															6	F75-FI-600 Drive Out of Service	-	0				x						1
															7	F75-FI-600 Drive Ready	-	0				x						1
F23-PW-651	2000-F75-PID-PR-40014	FIM3 BOOSTER 1 SEAL PUMP 1	VSD		0	100	%	40	100	3.16	X	X	F23-SEQ-2001 / F23-SEQ-2003	F23-SEQ-2002												Yokogawa DCS		3
													FI23523 < A_LL AND F23-PW-652 RUN					30								Duty pump run and seal water flow low low	1
															1	F23-PW-651 Comms Failure	-	0				x						1
															2	F23-PW-651 Fail to Start / Stop	-	0				x						1
															3	F23-PW-651 Drive Fault	-	0				x						1
															4	F23-PW-651 E-Stop	-	0				x	x					1
															5	F23-PW-651 MCC Unhealthy	-	0				x						1
															6	F23-PW-651 Drive Out of Service	-	0				x						1
															7	F23-PW-651 Drive Ready	-	0				x						1
															8	LIC75300 < A_LL	-	5			x			x			Seal water tank level low low	4
															9	XV75301 NOT OPEN	-	30			x			x			Isolation valve closed	1
															10	FI23523 < A_LL AND F23-PW-652 RUN	-	1,800			x			x			Seal water not healthy for F23-PP-651	1

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks										Control System	Comments	Rev
																Description (Unhealthy Interlocks and Healthy Start Permissives	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissives	Process	Critical	Hardwired to Drive	Bypass	Reset Required			
F23-PW-652	2000-F75-PID-PR-40014	FIM3 BOOSTER 1 SEAL PUMP 2	VSD		0	100	%	40	100	3.16	X	X	F23-SEQ-2001 / F23-SEQ-2003	F23-SEQ-2002												Yokogawa DCS		3
													FI23523 < A_LL AND F23-PW-651 RUN				-	30									Duty pump run and seal water flow low low	1
															1	F23-PW-652 Comms Failure	-	0				x						1
															2	F23-PW-652 Fail to Start / Stop	-	0				x						1
															3	F23-PW-652 Drive Fault	-	0				x						1
															4	F23-PW-652 E-Stop	-	0				x	x					1
															5	F23-PW-652 MCC Unhealthy	-	0				x						1
															6	F23-PW-652 Drive Out of Service	-	0				x						1
															7	F23-PW-652 Drive Ready	-	0				x						1
															8	LIC75300 < A_LL	-	5			x			x			Seal water tank level low low	4
															9	XV75301 NOT OPEN	-	30			x			x			Isolation valve closed	1
															10	FI23523 < A_LL AND F23-PW-651 RUN	-	1,800			x			x			Seal water not healthy for F23-PP-652	4
F23-PW-653	2000-F75-PID-PR-40014	FIM3 BOOSTER 2 SEAL PUMP 1	VSD		0	100	%	40	100	3.16	X	X	F23-SEQ-2004 / F23-SEQ-2006	F23-SEQ-2005												Yokogawa DCS		3
													FI23543 < A_LL AND F23-PW-654 RUN				-	30									Duty pump run and seal water flow low low	1
															1	F23-PW-653 Comms Failure	-	0				x						1
															2	F23-PW-653 Fail to Start / Stop	-	0				x						1
															3	F23-PW-653 Drive Fault	-	0				x						1
															4	F23-PW-653 E-Stop	-	0				x	x					1
															5	F23-PW-653 MCC Unhealthy	-	0				x						1
															6	F23-PW-653 Drive Out of Service	-	0				x						1
															7	F23-PW-653 Drive Ready	-	0				x						1
															8	LIC75300 < A_LL	-	5			x			x			Seal water tank level low low	4
															9	XV75302 NOT OPEN	-	30			x			x			Isolation valve closed	1
															10	FI23543 < A_LL AND F23-PW-653 RUN	-	1,800			x			x			Seal water not healthy for F23-PP-653	4
F23-PW-654	2000-F75-PID-PR-40014	FIM3 BOOSTER 2 SEAL PUMP 2	VSD		0	100	%	40	100	3.16	X	X	F23-SEQ-2004 / F23-SEQ-2006	F23-SEQ-2005												Yokogawa DCS		3
													FI23543 < A_LL AND F23-PW-653 RUN				-	30									Duty pump run and seal water flow low low	1
															1	F23-PW-654 Comms Failure	-	0				x						1
															2	F23-PW-654 Fail to Start / Stop	-	0				x						1
															3	F23-PW-654 Drive Fault	-	0				x						1
															4	F23-PW-654 E-Stop	-	0				x	x					1
															5	F23-PW-654 MCC Unhealthy	-	0				x						1
															6	F23-PW-654 Drive Out of Service	-	0				x						1
															7	F23-PW-654 Drive Ready	-	0				x						1
															8	LIC75300 < A_LL	-	5			x			x			Seal water tank level low low	4
															9	XV75302 NOT OPEN	-	30			x			x			Isolation valve closed	1
															10	FI23543 < A_LL AND F23-PW-653 RUN	-	1,800			x			x			Seal water not healthy for F23-PP-654	4
F75-PW-665	2000-F75-PID-PR-40014	FIM3 BOOSTER W/DOWN PUMP	DOL							3.16	X														Yokogawa DCS	Additional (two) local start/stop buttons HS75316 & HS75317	4
															1	F75-PW-665 Comms Failure	-	0				x						1
															2	F75-PW-665 Fail to Start / Stop	-	0				x						1
															3	F75-PW-665 Drive Fault	-	0				x						1
															4	F75-PW-665 E-Stop	-	0				x	x					1
															5	F75-PW-665 MCC Unhealthy	-	0				x						1
															6	F75-PW-665 Drive Out of Service	-	0				x						1
															7	F75-PW-665 Drive Ready	-	0				x						1
															8	LIC75300 < A_L	-	5			x			x			Seal water tank level low (suction point for washdown pump)	0
															9	KY75665 ACTIVE	-	0			x							4
F23-SE-431	2000-F75-PID-PR-40014	FIM3 BOOSTER SAFETY SHWR	FDR							4.27	X	X													Yokogawa DCS	Safety shower chiller	4
															1	F23-SE-431 Comms Failure	-	0				x						1
															2	F23-SE-431 Fail to Start / Stop	-	0				x						1
															3	F23-SE-431 Drive Fault	-	0				x						1
															4	F23-SE-431 E-Stop	-	0				x	x					1
															5	F23-SE-431 MCC Unhealthy	-	0				x						1
															6	F23-SE-431 Drive Out of Service	-	0				x						1
															7	F23-SE-431 Drive Ready	-	0				x						1

CALCULATIONS

Tag Source	Tag Name	P&ID	Description (Software Tag Only)	Unit	Hold	Inputs				Calc. Value	x = Operator Input * = Engineer Input			Calculation	Control System	Comments	Rev
						#	Tag	Unit	Description			Min	Max				
F23CALC40001	RY23401	2000-F23-PID-PR-10002	FINAL TAILS PUMP RATIO			1	F23-PP-466_RUN	-	FINAL TAILS PUMP 1 RUNNING (TRUE = 1; FALSE = 0)					IF(SUM(F23-PP-466_RUN, F23-PP-467_RUN, F23-PP-468_RUN, F23-PP-469_RUN) = 4) AND LINE D = FIM I, F23-PP-466R = 0.2; F23-PP-467R = F23-PP-468R = F23-PP-469R = ((1-F23-PP-466R) / 3)	Yokogawa DCS	Ratio Control to exclude any pump that is running in flushing duty.	3
						2	F23-PP-467_RUN	-	FINAL TAILS PUMP 2 RUNNING (TRUE = 1; FALSE = 0)								2
						3	F23-PP-468_RUN	-	FINAL TAILS PUMP 3 RUNNING (TRUE = 1; FALSE = 0)					IF(SUM(F23-PP-466_RUN, F23-PP-467_RUN, F23-PP-468_RUN, F23-PP-469_RUN) = 4) AND LINE D = FIM III, F23-PP-466R = F23-PP-467R = F23-PP-468R = F23-PP-469R = (1 / (SUM(F23-PP-466_RUN, F23-PP-467_RUN, F23-PP-468_RUN, F23-PP-469_RUN)			2
						4	F23-PP-469_RUN	-	FINAL TAILS PUMP 4 RUNNING (TRUE = 1; FALSE = 0)								2
						5	LINE D TSF SELECTION	-	Refer F23PP466_LOC		x	FIM I	FIM III				1
						6	LINE B TSF SELECTION	-	Refer F23PP467_LOC		x	KT	FIM III	IF(SUM(F23-PP-466_RUN, F23-PP-467_RUN, F23-PP-468_RUN, F23-PP-469_RUN) = 3) AND LINE D = FIM I, F23-PP-466R = 0.2; F23-PP-467R = F23-PP-468R OR F23-PP-468R = F23-PP-469R OR F23-PP-467R = F23-PP-469R = ((1-F23-PP-466R) / 2)			1
						7	LINE A TSF SELECTION	-	Refer F23PP468_LOC		x	ALL	FIM III				1
						8	LINE C TSF SELECTION	-	Refer F23PP469_LOC		x	ALL	FIM III	IF(SUM(F23-PP-466_RUN, F23-PP-467_RUN, F23-P-468_RUN, F23-PP-469_RUN) = 3) AND LINE D = FIM III, F23-PP-466R = F23-PP-467R = F23-PP-468R OR F23-PP-466R = F23-PP-468R = F23-PP-469R OR F23-PP-466R = F23-PP-467R = F23-PP-469R = (1 / (SUM(F23-PP-466_RUN, F23-PP-467_RUN, F23-PP-468_RUN OR F23-PP-466_RUN, F23-PP-468_RUN, F23-PP-469_RUN OR F23-PP-466_RUN, F23-PP-467_RUN, F23-PP-469_RUN)			1
						9	F23-PP-466R	-	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE FLOWRATE RATIO	x	x	0.15	0.25			Specify ratio for F23-PP-466 if FIM I selected	0
						10	F23-PP-467R	-	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE FLOWRATE RATIO	x							2
						11	F23-PP-468R	-	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE FLOWRATE RATIO	x							2
						12	F23-PP-469R	-	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE FLOWRATE RATIO	x				IF F23-PP-466_RUN = 0; F23-PP-467R = F23-PP-468R = F23-PP-469R = (1 / (SUM(F23-PP-467_RUN, F23-PP-468_RUN, F23-PP-			2
F23CALC40002	FY23417	2000-F23-PID-PR-10002	F23PP466_ DSCH FLOW SP	m³/h		1	LIC23401B_CV	m³/h	FINAL TAILS TANK LEVEL CONTROLLER B					FY23417 = LIC23401B_CV * F23-PP-466R	Yokogawa DCS	Minimum output of calculation = 500 m³/h when pumping to FIM I, 1,000 m³/h when pumping to FIM III	2
						2	F23-PP-466R	-	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE FLOWRATE RATIO							Maximum output of calculation = 800 m³/h when pumping to FIM I, 1,500 m³/h when pumping to FIM III	0
						3	FY23417	m³/h	F23PP466_ DSCH FLOW SP	x							2
F23CALC40003	WI23417	2000-F23-PID-PR-10002	F23PP466 DRY TONNES	t/h		1	FIT23417_PV	m³/h	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE FLOWRATE					WY23417 = ((SGSolids - (SGSolids x SGLiquid) / DIT23418_PV) / (SGSolids - SGLiquid)) * FIT23417_PV * DIT23418_PV	Yokogawa DCS		2
						2	DIT23418_PV	t/m³	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE DENSITY								0
						3	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07, typical for all calculations	2
						4	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88, typical for all calculations	2
						5	WI23417	t/h	F23PP466 DRY TONNES	x							2
F23CALC40004	DNI23418	2000-F23-PID-PR-10002	F23PP466 DENSITY	%w/w		1	DIT23418_PV	t/m³	F23-PP-466 FINAL TAILS PUMP 1 DISCHARGE DENSITY					DY23418 = 100 * ((SolidsSG - (SolidsSG x SolutionSG) / DIT23418_PV) / (SolidsSG - SolutionSG))	Yokogawa DCS		2
						2	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						3	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						4	DNI23418	%w/w	F23PP466 DENSITY	x							2
F23CALC40005	FY23430	2000-F23-PID-PR-10002	F23PP467_ DSCH FLOW SP	m³/h		1	LIC23401B_CV	m³/h	FINAL TAILS TANK LEVEL CONTROLLER B					FY23430 = LIC23401B_CV * F23-PP-467R	Yokogawa DCS	Minimum output of calculation = 750 m³/h when pumping to Kaltails, 1,000 m³/h when pumping to FIM III	2
						2	F23-PP-467R	-	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE FLOWRATE RATIO							Maximum output of calculation = 1,200 m³/h when pumping to Kaltails, 1,500 m³/h when pumping to FIM III	0
						3	FY23430	m³/h	F23PP467_ DSCH FLOW SP	x							2
F23CALC40006	WI23430	2000-F23-PID-PR-10002	F23PP467 DRY TONNES	t/h		1	FIT23430_PV	m³/h	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE FLOWRATE					WY23430 = ((SGSolids - (SGSolids x SGLiquid) / DIT23431_PV) / (SGSolids - SGLiquid)) * FIT23430_PV * DIT23431_PV	Yokogawa DCS		2
						2	DIT23431_PV	t/m³	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE DENSITY								0
						3	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						4	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						5	WI23430	t/h	F23PP467 DRY TONNES	x							2
F23CALC40007	DNI23431	2000-F23-PID-PR-10002	F23PP467 DENSITY	%w/w		1	DIT23431_PV	t/m³	F23-PP-467 FINAL TAILS PUMP 2 DISCHARGE DENSITY					DY23431 = 100 * ((SolidsSG - (SolidsSG x SolutionSG) / DIT23431_PV) / (SolidsSG - SolutionSG))	Yokogawa DCS		2
						2	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						3	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						4	DNI23431	%w/w	F23PP467 DENSITY	x							2
F23CALC40008	FY23441	2000-F23-PID-PR-10002	F23PP468_ DSCH FLOW SP	m³/h		1	LIC23401B_CV	m³/h	FINAL TAILS TANK LEVEL CONTROLLER B					FY23441 = LIC23401B_CV * F23-PP-468R	Yokogawa DCS	Minimum output of calculation = 750 m³/h when pumping to Kaltails, FIM II and FIM II EXT	2
						2	F23-PP-468R	-	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE FLOWRATE RATIO							Maximum output of calculation = 1,200 m³/h when pumping Kaltails, 1,150 m³/h when pumping to FIM II and 1,500 m³/h when pumping to FIM II EXT	0
						3	FY23441	m³/h	F23PP468_ DSCH FLOW SP	x							2
F23CALC40009	WI23441	2000-F23-PID-PR-10002	F23PP468 DRY TONNES	t/h		1	FIT23441_PV	m³/h	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE FLOWRATE					WY23441 = ((SGSolids - (SGSolids x SGLiquid) / DIT23442_PV) / (SGSolids - SGLiquid)) * FIT23441_PV * DIT23442_PV	Yokogawa DCS		2
						2	DIT23442_PV	t/m³	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE DENSITY								0
						3	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						4	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						5	WI23441	t/h	F23PP468 DRY TONNES	x							2
F23CALC40010	DNI23441	2000-F23-PID-PR-10002	F23PP468 DENSITY	%w/w		1	DIT23442_PV	t/m³	F23-PP-468 FINAL TAILS PUMP 3 DISCHARGE DENSITY					DY23441 = 100 * ((SolidsSG - (SolidsSG x SolutionSG) / DIT23442_PV) / (SolidsSG - SolutionSG))	Yokogawa DCS		2
						2	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						3	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						4	DNI23441	%w/w	F23PP468 DENSITY	x							2
F23CALC40011	FY23452	2000-F23-PID-PR-10002	F23PP469_ DSCH FLOW SP	m³/h		1	LIC23401B_CV	m³/h	FINAL TAILS TANK LEVEL CONTROLLER B					FY23452 = LIC23401B_CV * F23-PP-469R	Yokogawa DCS	Minimum output of calculation = 750 m³/h when pumping to Kaltails, FIM II and FIM II EXT	2
						2	F23-PP-469R	-	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE FLOWRATE RATIO							Maximum output of calculation = 1,200 m³/h when pumping Kaltails, 1,150 m³/h when pumping to FIM II and 1,500 m³/h when pumping to FIM II EXT	0
						3	FY23452	m³/h	F23PP469_ DSCH FLOW SP	x							2

CALCULATIONS

Tag Source	Tag Name	P&ID	Description (Software Tag Only)	Unit	Hold	Inputs				Calc. Value	x = Operator Input * = Engineer Input			Calculation	Control System	Comments	Rev
						#	Tag	Unit	Description			Min	Max				
F23CALC40012	WI23452	2000-F23-PID-PR-10002	F23PP469 DRY TONNES	t/h		1	FIT23452_PV	m³/h	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE FLOWRATE					WY23452 = ((SGSolids - (SGSolids x SGLiquid) / DIT23453_PV) / (SGSolids - SGLiquid)) * FIT23452_PV * DIT23453_PV	Yokogawa DCS		2
						2	DIT23453_PV	t/m³	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE DENSITY								0
						3	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						4	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						5	WI23452	t/h	F23PP469 DRY TONNES	x							2
F23CALC40013	DNI23453	2000-F23-PID-PR-10002	F23PP469 DENSITY	%w/w		1	DIT23453_PV	t/m³	F23-PP-469 FINAL TAILS PUMP 4 DISCHARGE DENSITY					DY23453 = 100 * ((SolidsSG - (SolidsSG x SolutionSG) / DIT23453_PV) / (SolidsSG - SolutionSG))	Yokogawa DCS		2
						2	SGLiquid	t/m³	LIQUID SG		*	0.98	1.10			Default water SG = 1.07	0
						3	SGSolids	t/m³	DRY SOLIDS SG		*	2.70	3.00			Default solids SG = 2.88	0
						4	DNI23453	%w/w	F23PP469 DENSITY	x							2
F23CALC40014	FDI23521	2000-F23-PID-PR-40002	D TO FIM3 BS DIFF FLOW	m³/h		1	FIC23417_PV	m³/h	FINAL TAILS PUMP 1 DISCHARGE FLOWRATE					FDI23521 = FIC23417_PV - FIC23521_PV	Yokogawa DCS		2
						2	FIC23521_PV	m³/h	FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE								0
						3	FDI23521	m³/h	D TO FIM3 BS DIFF FLOW	x							2
F23CALC40015	PY75303	2000-F23-PID-PR-40002	F23PP651 GSW PRESS	kPa		1	PIT23520_PV	kPa	FIM III BOOSTER PUMP 1 DISCHARGE PRESSURE TRANSMITTER					PY75303 = PIT23520_PV + SEAL PRESSURE	Yokogawa DCS		2
						2	SEAL PRESSURE - PIT23520	kPa	SEAL WATER KPA ABOVE PUMP D/C KPA REQUIRED		*	100	200				4
						3	PY75303	kPa	F23PP651 GSW PRESS	x							2
F23CALC40016	FDI23541	2000-F23-PID-PR-40003	B TO FIM3 BS DIFF FLOW	m³/h		1	FIC23430_PV	m³/h	FINAL TAILS PUMP 2 DISCHARGE FLOWRATE					FDI23541 = FIC23430_PV - FIC23541_PV	Yokogawa DCS		2
						2	FIC23541_PV	m³/h	FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE								0
						3	FDI23541	m³/h	B TO FIM3 BS DIFF FLOW	x							2
F23CALC40017	PY75306	2000-F23-PID-PR-40003	F23PP652 GSW PRESS	kPa		1	PIT23540_PV	kPa	FIM III BOOSTER PUMP 2 DISCHARGE PRESSURE TRANSMITTER					PY75306 = PIT23540_PV + SEAL PRESSURE	Yokogawa DCS		2
						2	SEAL PRESSURE - PIT23540	kPa	SEAL WATER KPA ABOVE PUMP D/C KPA REQUIRED		*	100	200				4
						3	PY75306	kPa	F23PP652 GSW PRESS	x							2
F23CALC40018	**FDI23592A**	**2000-F23-PID-PR-40005**	**D TO FIM III LINE 1S DIFF FLOW**	**m³/h**		**1**	**FIC23521_PV**	**m³/h**	**FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE**					**FDI23592A = FIT23592_PV - FIC23521_PV**	**Yokogawa DCS**		**5**
						2	**FIT23592_PV**	**m³/h**	**FIM III LINE 1 SOUTH FLOW TRANSMITTER**								**5**
						3	**FDI23592A**	**m³/h**	**D TO FIM III LINE 1S DIFF FLOW**	**x**							**5**
F23CALC40019	**FDI23592B**	**2000-F23-PID-PR-40005**	**B TO FIM III LINE 1S DIFF FLOW**	**m³/h**		**1**	**FIC23541_PV**	**m³/h**	**FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE**					**FDI23592B = FIT23592_PV - FIC23541_PV**	**Yokogawa DCS**		**5**
						2	**FIT23592_PV**	**m³/h**	**FIM III LINE 1 SOUTH FLOW TRANSMITTER**								**5**
						3	**FDI23592B**	**m³/h**	**B TO FIM III LINE 1S DIFF FLOW**	**x**							**5**
F23CALC40020	**FDI23593A**	**2000-F23-PID-PR-40005**	**B TO FIM III LINE 1N DIFF FLOW**	**m³/h**		**1**	**FIC23541_PV**	**m³/h**	**FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE**					**FDI23593A = FIT23593_PV - FIC23541_PV**	**Yokogawa DCS**		**5**
						2	**FIT23593_PV**	**m³/h**	**FIM III LINE 1 NORTH FLOW TRANSMITTER**								**5**
						3	**FDI23593A**	**m³/h**	**B TO FIM III LINE 1N DIFF FLOW**	**x**							**5**
F23CALC40021	**FDI23593B**	**2000-F23-PID-PR-40005**	**D TO FIM III LINE 1N DIFF FLOW**	**m³/h**		**1**	**FIC23521_PV**	**m³/h**	**FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE**					**FDI23593B = FIT23593_PV - FIC23521_PV**	**Yokogawa DCS**		**5**
						2	**FIT23593_PV**	**m³/h**	**FIM III LINE 1 NORTH FLOW TRANSMITTER**								**5**
						3	**FDI23593B**	**m³/h**	**D TO FIM III LINE 1N DIFF FLOW**	**x**							**5**
F23CALC40022	**FDI23594A**	**2000-F23-PID-PR-40005**	**B TO FIM III LINE 3W DIFF FLOW**	**m³/h**		**1**	**FIC23541_PV**	**m³/h**	**FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE**					**FDI23594A = FIT23594_PV - FIC23541_PV**	**Yokogawa DCS**		**5**
						2	**FIT23594_PV**	**m³/h**	**FIM III LINE 3 WEST FLOW TRANSMITTER**								**5**
						3	**FDI23594A**	**m³/h**	**B TO FIM III LINE 3W DIFF FLOW**	**x**							**5**
F23CALC40023	**FDI23594B**	**2000-F23-PID-PR-40005**	**D TO FIM III LINE 3W DIFF FLOW**	**m³/h**		**1**	**FIC23521_PV**	**m³/h**	**FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE**					**FDI23594B = FIT23594_PV - FIC23521_PV**	**Yokogawa DCS**		**5**
						2	**FIT23594_PV**	**m³/h**	**FIM III LINE 3 WEST FLOW TRANSMITTER**								**5**
						3	**FDI23594B**	**m³/h**	**D TO FIM III LINE 3W DIFF FLOW**	**x**							**5**
F23CALC40024	**FDI23595A**	**2000-F23-PID-PR-40005**	**D TO FIM III LINE 3E DIFF FLOW**	**m³/h**		**1**	**FIC23521_PV**	**m³/h**	**FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE**					**FDI23595A = FIT23595_PV - FIC23521_PV**	**Yokogawa DCS**		**5**
						2	**FIT23595_PV**	**m³/h**	**FIM III LINE 3 EAST FLOW TRANSMITTER**								**5**
						3	**FDI23595A**	**m³/h**	**D TO FIM III LINE 3E DIFF FLOW**	**x**							**5**
F23CALC40025	**FDI23595B**	**2000-F23-PID-PR-40005**	**B TO FIM III LINE 3E DIFF FLOW**	**m³/h**		**1**	**FIC23541_PV**	**m³/h**	**FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE**					**FDI23595B = FIT23595_PV - FIC23541_PV**	**Yokogawa DCS**		**5**
						2	**FIT23595_PV**	**m³/h**	**FIM III LINE 3 EAST FLOW TRANSMITTER**								**5**
						3	**FDI23595B**	**m³/h**	**B TO FIM III LINE 3E DIFF FLOW**	**x**							**5**
F23CALC40026	**FDI23596A**	**2000-F23-PID-PR-40005**	**B TO FIM III LINE 2W DIFF FLOW**	**m³/h**		**1**	**FIC23541_PV**	**m³/h**	**FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE**					**FDI23596A = FIT23596_PV - FIC23541_PV**	**Yokogawa DCS**		**5**
						2	**FIT23596_PV**	**m³/h**	**FIM III LINE 2 WEST FLOW TRANSMITTER**								**5**
						3	**FDI23596A**	**m³/h**	**B TO FIM III LINE 2W DIFF FLOW**	**x**							**5**
F23CALC40027	**FDI23596B**	**2000-F23-PID-PR-40005**	**D TO FIM III LINE 2W DIFF FLOW**	**m³/h**		**1**	**FIC23521_PV**	**m³/h**	**FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE**					**FDI23596B = FIT23596_PV - FIC23521_PV**	**Yokogawa DCS**		**5**
						2	**FIT23596_PV**	**m³/h**	**FIM III LINE 2 WEST FLOW TRANSMITTER**								**5**
						3	**FDI23596B**	**m³/h**	**D TO FIM III LINE 2W DIFF FLOW**	**x**							**5**
F23CALC40028	**FDI23597A**	**2000-F23-PID-PR-40005**	**D TO FIM III LINE 2E DIFF FLOW**	**m³/h**		**1**	**FIC23521_PV**	**m³/h**	**FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE**					**FDI23597A = FIT23597_PV - FIC23521_PV**	**Yokogawa DCS**		**5**
						2	**FIT23597_PV**	**m³/h**	**FIM III LINE 2 EAST FLOW TRANSMITTER**								**5**
						3	**FDI23597A**	**m³/h**	**D TO FIM III LINE 2E DIFF FLOW**	**x**							**5**

CALCULATIONS

Tag Source	Tag Name	P&ID	Description (Software Tag Only)	Unit	Hold	Inputs				Calc. Value	x = Operator Input * = Engineer Input			Calculation	Control System	Comments	Rev
						#	Tag	Unit	Description			Min	Max				
F23CALC40029	FDI23597B	2000-F23-PID-PR-40005	B TO FIM III LINE 2E DIFF FLOW	m³/h		1	FIC23541_PV	m³/h	FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE					FDI23597B = FIT23597_PV - FIC23541_PV	Yokogawa DCS		5
						2	FIT23597_PV	m³/h	FIM III LINE 2 EAST FLOW TRANSMITTER								5
						3	FDI23597B	m³/h	B TO FIM III LINE 2E DIFF FLOW	x							5
~~F75CALC40001~~	~~FDI75611~~	~~2000-F75-PID-PR-25008~~	~~DD3&4 RETURN DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIT75333_PV~~	~~m³/h~~	~~DECANT DAM 4 RETURN FLOW TRANSMITTER~~								~~5~~
						~~2~~	~~FIT75682_PV~~	~~m³/h~~	~~DECANT DAM 3 RETURN FLOW TRANSMITTER~~					~~FDI75611 = FIT75611_PV - FIT75333_PV - FIT75682_PV~~	~~Yokogawa DCS~~		~~5~~
						~~3~~	~~FIT75611_PV~~	~~m³/h~~	~~DECANT DAM 3 & 4 RETURN FLOW TRANSMITTER~~								~~5~~
						~~4~~	~~FDI75611~~	~~m³/h~~	~~DD3&4 RETURN DIFF FLOW~~	~~x~~							~~5~~
~~F75CALC40005~~	~~FDI75321~~	~~2000-F75-PID-PR-40014~~	~~BW TO F75TK651 DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIT17503_PV~~	~~m³/h~~	~~BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER~~								~~5~~
						~~2~~	~~FIT75321_PV~~	~~m³/h~~	~~BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER 2~~					~~FDI75321 = FIT17503_PV - FIT75321_PV~~	~~Yokogawa DCS~~		~~5~~
						~~3~~	~~FDI75321~~	~~m³/h~~	~~BW TO F75TK651 DIFF FLOW~~	~~x~~							~~5~~
F75CALC40006	FDI75902	2000-F75-PID-PR-40020	D TO FIM1 DIFF FLOW	m³/h		1	FIC23417_PV	m³/h	FINAL TAILS PUMP 1 DISCHARGE FLOWRATE								3
						2	FIT75902_PV	m³/h	FIM I BOOSTER PUMP DISCHARGE FLOWRATE					FDI75902 = FIC23417_PV - FIT75902_PV	Yokogawa DCS	Existing PLC	4
						3	FDI75902	m³/h	D TO FIM1 DIFF FLOW	x							2
F75CALC40007	FDI30909	2000-F75-PID-PR-40022	A TO FIM2 DIFF FLOW	m³/h		1	FIC23441_PV	m³/h	FINAL TAILS PUMP 3 DISCHARGE FLOWRATE								3
						2	FIT30909_PV	m³/h	FIM II BOOSTER PUMP 2 DISCHARGE FLOWRATE					FDI30909 = FIC23441_PV - FIT30909_PV	Yokogawa DCS	Existing PLC	0
						3	FDI30909	m³/h	A TO FIM2 DIFF FLOW	x							2
F75CALC40008	FDI30908	2000-F75-PID-PR-40022	C TO FIM2 DIFF FLOW	m³/h		1	FIC23452_PV	m³/h	FINAL TAILS PUMP 4 DISCHARGE FLOWRATE								3
						2	FIT30908_PV	m³/h	FIM II BOOSTER PUMP 1 DISCHARGE FLOWRATE					FDI30908 = FIC23452_PV - FIT30908_PV	Yokogawa DCS	Existing PLC	0
						3	FDI30908	m³/h	C TO FIM2 DIFF FLOW	x							2
F75CALC40009	FDI30940A	2000-F75-PID-PR-40023	A TO KALTAILS DIFF FLOW	m³/h		1	FIC23441_PV	m³/h	FINAL TAILS PUMP 3 DISCHARGE FLOWRATE								3
						2	FIT30940_PV	m³/h	KALTAILS BOOSTER PUMP 1 DISCHARGE FLOWRATE					FDI30940A = FIC23441_PV - FIT30940_PV	Yokogawa DCS	Existing PLC	0
						3	FDI30940A	m³/h	A TO KALTAILS DIFF FLOW	x							2
F75CALC40010	FDI30940B	2000-F75-PID-PR-40023	B TO KALTAILS DIFF FLOW	m³/h		1	FIC23430_PV	m³/h	FINAL TAILS PUMP 2 DISCHARGE FLOWRATE								3
						2	FIT30940_PV	m³/h	KALTAILS BOOSTER PUMP 1 DISCHARGE FLOWRATE					FDI30940B = FIC23430_PV - FIT30940_PV	Yokogawa DCS	Existing PLC	0
						3	FDI30940B	m³/h	B TO KALTAILS DIFF FLOW	x							2
F75CALC40011	FDI30941	2000-F75-PID-PR-40023	C TO KALTAILS DIFF FLOW	m³/h		1	FIC23452_PV	m³/h	FINAL TAILS PUMP 4 DISCHARGE FLOWRATE								3
						2	FIT30941_PV	m³/h	KALTAILS BOOSTER PUMP 2 DISCHARGE FLOWRATE					FDI30941 = FIC23452_PV - FIT30941_PV	Yokogawa DCS	Existing PLC	0
						3	FDI30941	m³/h	C TO KALTAILS DIFF FLOW	x							2
F75CALC40012	FDI30115	2000-F75-PR-PID-40025	LINE C TO FIM II F DIFF FLOW	m³/h		1	FIC23452_PV	m³/h	FINAL TAILS PUMP 4 DISCHARGE FLOWRATE								5
						2	FIT30115_PV	m³/h	LINE C TO FIM II CELL F FLOWRATE					FDI30115 = FIC23452_PV - FIT30115_PV	Yokogawa DCS	Existing PLC	5
						3	FDI30115	m³/h	LINE C TO FIM II F DIFF FLOW	x							5
F75CALC40013	FDI30116	2000-F75-PR-PID-40025	LINE A TO FIM II F DIFF FLOW	m³/h		1	FIC23441_PV	m³/h	FINAL TAILS PUMP 3 DISCHARGE FLOWRATE								5
						2	FIT30116_PV	m³/h	LINE A TO FIM II CELL F FLOWRATE					FDI30116 = FIC23441_PV - FIT30116_PV	Yokogawa DCS	Existing PLC	5
						3	FDI30116	m³/h	LINE A TO FIM II F DIFF FLOW	x							5
F75CALC40014	FDI30117	2000-F75-PR-PID-40026	LINE C TO FIM II E DIFF FLOW	m³/h		1	FIC23452_PV	m³/h	FINAL TAILS PUMP 4 DISCHARGE FLOWRATE								5
						2	FIT30117_PV	m³/h	LINE C TO FIM II CELL E FLOWRATE					FDI30117 = FIC23452_PV - FIT30117_PV	Yokogawa DCS	Existing PLC	5
						3	FDI30117	m³/h	LINE C TO FIM II E DIFF FLOW	x							5
F75CALC40015	FDI30118	2000-F75-PR-PID-40026	LINE C TO FIM II G DIFF FLOW	m³/h		1	FIC23452_PV	m³/h	FINAL TAILS PUMP 4 DISCHARGE FLOWRATE								5
						2	FIT30118_PV	m³/h	LINE C TO FIM II CELL G FLOWRATE					FDI30118 = FIC23452_PV - FIT30118_PV	Yokogawa DCS	Existing PLC	5
						3	FDI30118	m³/h	LINE C TO FIM II G DIFF FLOW	x							5
F75CALC40016	FDI30119	2000-F75-PR-PID-40026	LINE A TO FIM II E DIFF FLOW	m³/h		1	FIC23441_PV	m³/h	FINAL TAILS PUMP 3 DISCHARGE FLOWRATE								5
						2	FIT30119_PV	m³/h	LINE A TO FIM II CELL E FLOWRATE					FDI30119= FIC23441_PV - FIT30119_PV	Yokogawa DCS	Existing PLC	5
						3	FDI30119	m³/h	LINE A TO FIM II E DIFF FLOW	x							5
F75CALC40017	FDI30120	2000-F75-PR-PID-40026	LINE A TO FIM II G DIFF FLOW	m³/h		1	FIC23441_PV	m³/h	FINAL TAILS PUMP 3 DISCHARGE FLOWRATE								5
						2	FIT30120_PV	m³/h	LINE A TO FIM II CELL G FLOWRATE					FDI30120 = FIC23441_PV - FIT30120_PV	Yokogawa DCS	Existing PLC	5
						3	FDI30120	m³/h	LINE A TO FIM II G DIFF FLOW	x							5

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
FIC23417	2000-F23-PID-PR-10002	F23PP466 FLOWRATE		Analogue PID	A_L	100	m³/h	5	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
FIC23417	2000-F23-PID-PR-10002	F23PP466 FLOWRATE		Analogue PID	A_LL	50	m³/h	5	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
DNI23418	2000-F23-PID-PR-10002	F23PP466 DISCH DENSITY		Analogue	A_L	50	%	5	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
DNI23418	2000-F23-PID-PR-10002	F23PP466 DISCH DENSITY		Analogue	A_H	60	%	5	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
PI23419A	2000-F23-PID-PR-10002	F23PP466 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-466 NOT RUN OR XV75609 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23419A	2000-F23-PID-PR-10002	F23PP466 DISCH PRESSURE		Analogue	A_H	850	kPa	5	F23-PP-466 NOT RUN OR XV75609 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23419A	2000-F23-PID-PR-10002	F23PP466 DISCH PRESSURE		Analogue	A_HH	900	kPa	5	F23-PP-466 NOT RUN OR XV75609 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI23419B	2000-F23-PID-PR-10002	F23PP466 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-466 NOT RUN OR XV75608 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23419B	2000-F23-PID-PR-10002	F23PP466 DISCH PRESSURE		Analogue	A_H	950	kPa	5	F23-PP-466 NOT RUN OR XV75608 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23419B	2000-F23-PID-PR-10002	F23PP466 DISCH PRESSURE		Analogue	A_HH	1,000	kPa	5	F23-PP-466 NOT RUN OR XV75608 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FIC23430	2000-F23-PID-PR-10002	F23PP467 FLOWRATE		Analogue PID	A_L	100	m³/h	5	F23-PP-467 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
FIC23430	2000-F23-PID-PR-10002	F23PP467 FLOWRATE		Analogue PID	A_LL	50	m³/h	5	F23-PP-467 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
DNI23431	2000-F23-PID-PR-10002	F23PP467 DISCH DENSITY		Analogue	A_L	50	%	5	F23-PP-467 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
DNI23431	2000-F23-PID-PR-10002	F23PP467 DISCH DENSITY		Analogue	A_H	60	%	5	F23-PP-467 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
PI23432A	2000-F23-PID-PR-10002	F23PP467 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-467 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23432A	2000-F23-PID-PR-10002	F23PP467 DISCH PRESSURE		Analogue	A_H	650	kPa	5	F23-PP-467 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23432A	2000-F23-PID-PR-10002	F23PP467 DISCH PRESSURE		Analogue	A_HH	690	kPa	5	F23-PP-467 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI23432B	2000-F23-PID-PR-10002	F23PP467 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-467 NOT RUN OR XV23532 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23432B	2000-F23-PID-PR-10002	F23PP467 DISCH PRESSURE		Analogue	A_H	950	kPa	5	F23-PP-467 NOT RUN OR XV23532 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23432B	2000-F23-PID-PR-10002	F23PP467 DISCH PRESSURE		Analogue	A_HH	1,000	kPa	5	F23-PP-467 NOT RUN OR XV23532 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FIC23441	2000-F23-PID-PR-10002	F23PP468 FLOWRATE		Analogue PID	A_L	100	m³/h	5	F23-PP-468 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
FIC23441	2000-F23-PID-PR-10002	F23PP468 FLOWRATE		Analogue PID	A_LL	50	m³/h	5	F23-PP-468 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
DNI23442	2000-F23-PID-PR-10002	F23PP468 DISCH DENSITY		Analogue	A_L	50	%	5	F23-PP-468 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
DNI23442	2000-F23-PID-PR-10002	F23PP468 DISCH DENSITY		Analogue	A_H	60	%	5	F23-PP-468 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
PI23443A	2000-F23-PID-PR-10002	F23PP468 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-468 NOT RUN OR KV30933 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23443A	2000-F23-PID-PR-10002	F23PP468 DISCH PRESSURE		Analogue	A_H	650	kPa	5	F23-PP-468 NOT RUN OR KV30933 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23443A	2000-F23-PID-PR-10002	F23PP468 DISCH PRESSURE		Analogue	A_HH	690	kPa	5	F23-PP-468 NOT RUN OR KV30933 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI23443B	2000-F23-PID-PR-10002	F23PP468 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-468 NOT RUN OR KV30901 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23443B	2000-F23-PID-PR-10002	F23PP468 DISCH PRESSURE		Analogue	A_H	1,150	kPa	5	F23-PP-468 NOT RUN OR KV30901 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23443B	2000-F23-PID-PR-10002	F23PP468 DISCH PRESSURE		Analogue	A_HH	1,200	kPa	5	F23-PP-468 NOT RUN OR KV30901 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FIC23452	2000-F23-PID-PR-10002	F23PP469 FLOWRATE		Analogue PID	A_L	100	m³/h	5	F23-PP-469 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
FIC23452	2000-F23-PID-PR-10002	F23PP469 FLOWRATE		Analogue PID	A_LL	50	m³/h	5	F23-PP-469 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
DNI23453	2000-F23-PID-PR-10002	F23PP469 DISCH DENSITY		Analogue	A_L	50	%	5	F23-PP-469 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
DNI23453	2000-F23-PID-PR-10002	F23PP469 DISCH DENSITY		Analogue	A_H	60	%	5	F23-PP-469 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Density outside of normal operating range. Operator to investiage in field	3
PI23454A	2000-F23-PID-PR-10002	F23PP469 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-469 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23454A	2000-F23-PID-PR-10002	F23PP469 DISCH PRESSURE		Analogue	A_H	650	kPa	5	F23-PP-469 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23454A	2000-F23-PID-PR-10002	F23PP469 DISCH PRESSURE		Analogue	A_HH	690	kPa	5	F23-PP-469 NOT RUN OR KV30932 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI23454B	2000-F23-PID-PR-10002	F23PP469 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-469 NOT RUN OR KV30900 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23454B	2000-F23-PID-PR-10002	F23PP469 DISCH PRESSURE		Analogue	A_H	1,150	kPa	5	F23-PP-469 NOT RUN OR KV30900 CLOSED	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23454B	2000-F23-PID-PR-10002	F23PP469 DISCH PRESSURE		Analogue	A_HH	1,200	kPa	5	F23-PP-469 NOT RUN OR KV30900 CLOSED	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
TI23470	2000-F23-PID-PR-10002	F23PP466 GEARBOX TEMP		Analogue	A_H	85	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	5
TI23470	2000-F23-PID-PR-10002	F23PP466 GEARBOX TEMP		Analogue	A_HH	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
TI23472	2000-F23-PID-PR-10002	F23PP467 GEARBOX TEMP		Analogue	A_H	85	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	5
TI23472	2000-F23-PID-PR-10002	F23PP467 GEARBOX TEMP		Analogue	A_HH	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
TI23474	2000-F23-PID-PR-10002	F23PP468 GEARBOX TEMP		Analogue	A_H	85	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	5
TI23474	2000-F23-PID-PR-10002	F23PP468 GEARBOX TEMP		Analogue	A_HH	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
TI23476	2000-F23-PID-PR-10002	F23PP469 GEARBOX TEMP		Analogue	A_H	85	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	5
TI23476	2000-F23-PID-PR-10002	F23PP469 GEARBOX TEMP		Analogue	A_HH	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
PAL23471	2000-F23-PID-PR-10002	F23PP466 GEARBOX PRES LO		Digital	Alarm			30	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
PAL23473	2000-F23-PID-PR-10002	F23PP467 GEARBOX PRES LO		Digital	Alarm			30	F23-PP-467 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
PAL23475	2000-F23-PID-PR-10002	F23PP468 GEARBOX PRES LO		Digital	Alarm			30	F23-PP-468 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
PAL23477	2000-F23-PID-PR-10002	F23PP469 GEARBOX PRES LO		Digital	Alarm			30	F23-PP-469 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	5
F23SW602_HTY	2000-F23-PID-PR-10002	F23SW602 ACB SHUNT OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23SW602_AF	2000-F23-PID-PR-10002	F23SW602 ARC FLASH ALM		Digital	Alarm				-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Rapid (>5 to <15 minutes)	Low	Electrician to investigate	4
F23SW602_TRP	2000-F23-PID-PR-10002	F23SW602 ACB TRIP		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23SW602_SD	2000-F23-PID-PR-10002	F23SW602 INC SRG OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23SR601_ESTP	2000-F23-PID-PR-10002	F23SW601 EMERG SD1		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23COM601_PS1	2000-F23-PID-PR-10002	F23COM601 UPS PS1 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23COM601_PS2	2000-F23-PID-PR-10002	F23COM601 UPS PS2 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23COM601_PS3	2000-F23-PID-PR-10002	F23COM601 48V PS3 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23BC601_FL1	2000-F23-PID-PR-10002	F23BC601 MCR BATT CHR FL		Digital	Alarm				-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to investigate	4
F23BC601_FL2	2000-F23-PID-PR-10002	F23BC601 BATT CHR ALM		Digital	Alarm				-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to investigate	4
F23BC601_FL3	2000-F23-PID-PR-10002	F23BC601 MCR BATT CHR EF		Digital	Alarm				-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to investigate	4
F23SW602_PS1	2000-F23-PID-PR-10002	F23SW602 24V UPS PS1 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23SW602_PS2	2000-F23-PID-PR-10002	F23SW602 24V UPS PS2 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Switchroom tripped. Electrician to investigate	4
F23DCS601_PS1	2000-F23-PID-PR-10002	F23DCS601 24V PS1 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Electrician to investigate	4
F23DCS601_PS2	2000-F23-PID-PR-10002	F23DCS601 24V PS2 OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Electrician to investigate	4
F23TR602_ALM	2000-F23-PID-PR-10002	F23TR602 TRF TEMP OK		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Prompt (>15 to <30 minutes)	Low	Operator to investigate air conditioning and cause of high temperature	4
F23TR602_TRP	2000-F23-PID-PR-10002	F23TR602 TRF TRIP		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Operator to investigate air conditioning and cause of high temperature	4
F23TR602_FLT	2000-F23-PID-PR-10002	F23TR602 TRF FAULT		Digital	Alarm				-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Prompt (>15 to <30 minutes)	Low	Operator to investigate air conditioning and cause of high temperature	4
PI23511	2000-F23-PID-PR-40002	F23PP651 SUCT PRESSURE		Analogue	A_H	800	kPa	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump suction pressure. Advise Operator / Maintenance to investigate	3
PI23511	2000-F23-PID-PR-40002	F23PP651 SUCT PRESSURE		Analogue	A_L	20	kPa	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump suction pressure. Advise Operator / Maintenance to investigate	3
PI23520	2000-F23-PID-PR-40002	F23PP651 DISCH PRESSURE		Analogue	A_HH	1,100	kPa	5	F23-PP-651 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI23520	2000-F23-PID-PR-40002	F23PP651 DISCH PRESSURE		Analogue	A_H	1,000	kPa	5	F23-PP-651 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23520	2000-F23-PID-PR-40002	F23PP651 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-651 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
FIC23521	2000-F23-PID-PR-40002	F23PP651 FLOWRATE		Analogue PID	A_L	100	m³/h	5	F23-PP-651 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
FIC23521	2000-F23-PID-PR-40002	F23PP651 FLOWRATE		Analogue PID	A_LL	50	m³/h	5	F23-PP-651 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
FDI23521	2000-F23-PID-PR-40002	LINE D FIM III DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI23521	2000-F23-PID-PR-40002	LINE D FIM III DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-466 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FI23523	2000-F23-PID-PR-40002	F23PP651 SEAL WATER FLOW		Analogue	A_H	20	L/min	5	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High seal water flow, could indicate seal failure. Advise Operator / Maintenance to investigate	3
FI23523	2000-F23-PID-PR-40002	F23PP651 SEAL WATER FLOW		Analogue	A_L	13	L/min	5	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Check gland water system is healthy and inline auto valves are in the correct position. Else, advise Field Operator / Maintenance to investigate immediately	3

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
FI23523	2000-F23-PID-PR-40002	F23PP651 SEAL WATER FLOW		Analogue	A_LL	6.5	L/min	25	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
PIC75303	2000-F23-PID-PR-40002	F23PW651/652 PRESSURE		Analogue PID	A_H	SP + 250	kPa	5	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High seal water pressure. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	4
PIC75303	2000-F23-PID-PR-40002	F23PW651/652 PRESSURE		Analogue PID	A_L	SP + 50	kPa	5	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Low seal water pressure. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	4
PIC75303	2000-F23-PID-PR-40002	F23PW651/652 PRESSURE		Analogue PID	A_LL	SP + 0	kPa	5	F23-PW-651 NOT RUN AND F23-PW-652 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
TI23651A	2000-F23-PID-PR-40002	F23PP651 WINDING 1 RTD		Analogue	A_HH	130	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23651A	2000-F23-PID-PR-40002	F23PP651 WINDING 1 RTD		Analogue	A_H	120	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23651B	2000-F23-PID-PR-40002	F23PP651 WINDING 2 RTD		Analogue	A_HH	130	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23651B	2000-F23-PID-PR-40002	F23PP651 WINDING 2 RTD		Analogue	A_H	120	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23651C	2000-F23-PID-PR-40002	F23PP651 WINDING 3 RTD		Analogue	A_HH	130	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23651C	2000-F23-PID-PR-40002	F23PP651 WINDING 3 RTD		Analogue	A_H	120	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23651D	2000-F23-PID-PR-40002	F23PP651 DE BEARING RTD		Analogue	A_HH	95	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23651D	2000-F23-PID-PR-40002	F23PP651 DE BEARING RTD		Analogue	A_H	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23651E	2000-F23-PID-PR-40002	F23PP651 NDE BEARING RTD		Analogue	A_HH	95	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23651E	2000-F23-PID-PR-40002	F23PP651 NDE BEARING RTD		Analogue	A_H	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23651F	2000-F23-PID-PR-40002	F23PP651 AMBIENT RTD		Analogue	A_HH	50	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23651F	2000-F23-PID-PR-40002	F23PP651 AMBIENT RTD		Analogue	A_H	45	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
UA23519	2000-F23-PID-PR-40002	F23PP651 BRST DISC		Digital	Alarm			3	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Immediate (<5 minutes)	High	Potential tails line leak. Operator to investigate in field	3
XI23571	2000-F23-PID-PR-40002	F23HY420 FAULT		Digital	Fault			3	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Rapid (>5 to <15 minutes)	High	Hydraulic power pack fault. Operator to investigate in field	4
PI23531	2000-F23-PID-PR-40003	F23PP652 SUCT PRESSURE		Analogue	A_H	800	kPa	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump suction pressure. Advise Operator / Maintenance to investigate	3
PI23531	2000-F23-PID-PR-40003	F23PP652 SUCT PRESSURE		Analogue	A_L	20	kPa	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump suction pressure. Advise Operator / Maintenance to investigate	3
PI23540	2000-F23-PID-PR-40003	F23PP652 DISCH PRESSURE		Analogue	A_HH	1,100	kPa	5	F23-PP-652 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI23540	2000-F23-PID-PR-40003	F23PP652 DISCH PRESSURE		Analogue	A_H	1,000	kPa	5	F23-PP-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI23540	2000-F23-PID-PR-40003	F23PP652 DISCH PRESSURE		Analogue	A_L	100	kPa	5	F23-PP-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low pump discharge pressure. Advise Operator / Maintenance to investigate	3
FIC23541	2000-F23-PID-PR-40003	F23PP652 FLOWRATE		Analogue PID	A_L	100	m³/h	5	F23-PP-652 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
FIC23541	2000-F23-PID-PR-40003	F23PP652 FLOWRATE		Analogue PID	A_LL	50	m³/h	5	F23-PP-652 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Tails Line Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
FDI23541	2000-F23-PID-PR-40003	LINE B FIM III DIFF FLOW		Analogue	A_H	50	m³/h	**600**	**F23-PP-467 NOT RUN**	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	**5**
FDI23541	2000-F23-PID-PR-40003	LINE B FIM III DIFF FLOW		Analogue	A_L	-50	m³/h	**600**	**F23-PP-467 NOT RUN**	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	**5**
FI23543	2000-F23-PID-PR-40003	F23PP652 SEAL WATER FLOW		Analogue	A_H	20	L/min	5	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High seal water flow, could indicate seal failure. Advise Operator / Maintenance to investigate	3
FI23543	2000-F23-PID-PR-40003	F23PP652 SEAL WATER FLOW		Analogue	A_L	13	L/min	5	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Check gland water system is healthy and inline auto valves are in the correct position. Else, advise Field Operator / Maintenance to investigate immediately	3
FI23543	2000-F23-PID-PR-40003	F23PP652 SEAL WATER FLOW		Analogue	A_LL	6.5	L/min	25	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
PIC75306	2000-F23-PID-PR-40003	F23PW653/654 PRESSURE		Analogue PID	A_H	SP + 250	kPa	5	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High seal water pressure. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	4
PIC75306	2000-F23-PID-PR-40003	F23PW653/654 PRESSURE		Analogue PID	A_L	SP + 50	kPa	5	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Low seal water pressure. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	4
PIC75306	2000-F23-PID-PR-40003	F23PW653/654 PRESSURE		Analogue PID	A_LL	SP + 0	kPa	5	F23-PW-653 NOT RUN AND F23-PW-654 NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
TI23652A	2000-F23-PID-PR-40003	F23PP652 WINDING 1 RTD		Analogue	A_HH	130	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23652A	2000-F23-PID-PR-40003	F23PP652 WINDING 1 RTD		Analogue	A_H	120	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23652B	2000-F23-PID-PR-40003	F23PP652 WINDING 2 RTD		Analogue	A_HH	130	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23652B	2000-F23-PID-PR-40003	F23PP652 WINDING 2 RTD		Analogue	A_H	120	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23652C	2000-F23-PID-PR-40003	F23PP652 WINDING 3 RTD		Analogue	A_HH	130	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23652C	2000-F23-PID-PR-40003	F23PP652 WINDING 3 RTD		Analogue	A_H	120	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
TI23652D	2000-F23-PID-PR-40003	F23PP652 DE BEARING RTD		Analogue	A_HH	95	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23652D	2000-F23-PID-PR-40003	F23PP652 DE BEARING RTD		Analogue	A_H	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23652E	2000-F23-PID-PR-40003	F23PP652 NDE BEARING RTD		Analogue	A_HH	95	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23652E	2000-F23-PID-PR-40003	F23PP652 NDE BEARING RTD		Analogue	A_H	90	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
TI23652F	2000-F23-PID-PR-40003	F23PP652 AMBIENT RTD		Analogue	A_HH	50	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Plant Tripped	Logging	Equipment tripped. Operator to investigate & escalate to maintenance	4
TI23652F	2000-F23-PID-PR-40003	F23PP652 AMBIENT RTD		Analogue	A_H	45	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to report and investigate high temperature from pump motor	4
UA23539	2000-F23-PID-PR-40003	F23PP652 BRST DISC		Digital	Alarm			3	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Immediate (<5 minutes)	High	Potential tails line leak. Operator to investigate in field	3
FI23592	**2000-F23-PID-PR-40005**	**FIM III LINE 1 SOUTH FLOW**		**Analogue**	**A_L**	**100**	**m³/h**	**5**	**-**	**Yokogawa DCS**	**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**	**Prompt (>15 to <30 minutes)**	**Low**	**Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately**	**5**
FDI23592A	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 1S DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23592A	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 1S DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23592B	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 1S DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23592B	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 1S DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FI23593	**2000-F23-PID-PR-40005**	**FIM III LINE 1 NORTH FLOW**		**Analogue**	**A_L**	**100**	**m³/h**	**5**	**-**	**Yokogawa DCS**	**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**	**Prompt (>15 to <30 minutes)**	**Low**	**Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately**	**5**
FDI23593A	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 1N DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23593A	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 1N DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23593B	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 1N DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23593B	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 1N DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FI23594	**2000-F23-PID-PR-40005**	**FIM III LINE 3 WEST FLOW**		**Analogue**	**A_L**	**100**	**m³/h**	**5**	**-**	**Yokogawa DCS**	**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**	**Prompt (>15 to <30 minutes)**	**Low**	**Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately**	**5**
FDI23594A	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 3W DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23594A	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 3W DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23594B	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 3W DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23594B	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 3W DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FI23595	**2000-F23-PID-PR-40005**	**FIM III LINE 3 EAST FLOW**		**Analogue**	**A_L**	**100**	**m³/h**	**5**	**-**	**Yokogawa DCS**	**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**	**Prompt (>15 to <30 minutes)**	**Low**	**Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately**	**5**
FDI23595A	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 3E DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23595A	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 3E DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23595B	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 3E DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23595B	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 3E DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FI23596	**2000-F23-PID-PR-40005**	**FIM III LINE 2 WEST FLOW**		**Analogue**	**A_L**	**100**	**m³/h**	**5**	**-**	**Yokogawa DCS**	**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**	**Prompt (>15 to <30 minutes)**	**Low**	**Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately**	**5**
FDI23596A	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 2W DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23596A	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 2W DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23596B	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 2W DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23596B	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 2W DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FI23597	**2000-F23-PID-PR-40005**	**FIM III LINE 2 EAST FLOW**		**Analogue**	**A_L**	**100**	**m³/h**	**5**	**-**	**Yokogawa DCS**	**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**	**Prompt (>15 to <30 minutes)**	**Low**	**Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately**	**5**
FDI23597A	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 2E DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23597A	**2000-F23-PID-PR-40005**	**LINE D TO FIM III 2E DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PP-651 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23597B	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 2E DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
FDI23597B	**2000-F23-PID-PR-40005**	**LINE B TO FIM III 2E DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F23-PW-652 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential tails line leak. Operator to investigate in field**	**5**
UI23575A	2000-F23-TER-IC-40008	F23SR601 TRF ROOM TEMP		Analogue	A_HH	38	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to investigate air conditioning and cause of high room temperature.	4
UI23575A	2000-F23-TER-IC-40008	F23SR601 TRF ROOM TEMP		Analogue	A_H	35	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Prompt (>15 to <30 minutes)	Low	Operator to investigate air conditioning and cause of high room temperature.	4
UI23576A	2000-F23-TER-IC-40008	F23SR601 HV ROOM TEMP		Analogue	A_HH	38	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to investigate air conditioning and cause of high room temperature.	4
UI23576A	2000-F23-TER-IC-40008	F23SR601 HV ROOM TEMP		Analogue	A_H	35	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Prompt (>15 to <30 minutes)	Low	Operator to investigate air conditioning and cause of high room temperature.	4

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
UI23577A	2000-F23-TER-IC-40008	F23SR601 LV ROOM TEMP		Analogue	A_HH	38	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Operator to investigate air conditioning and cause of high room temperature.	4
UI23577A	2000-F23-TER-IC-40008	F23SR601 LV ROOM TEMP		Analogue	A_H	35	℃	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Prompt (>15 to <30 minutes)	Low	Operator to investigate air conditioning and cause of high room temperature.	4
F23FIP601_ALM	2000-F23-TER-IC-40009	23FIP601 FIRE ALARM		Digital	Alarm				-	Yokogawa DCS	Financial 3: Loss equivalent greater than 6 hours production. Repairs costing >$500k	Rapid (>5 to <15 minutes)	Urgent	Fire alarm. Follow site specific Fire Emergency Plan.	4
F23FIP601_FLT	2000-F23-TER-IC-40009	23FIP601 FIRE FAULT		Digital	Fault				-	Yokogawa DCS	Financial 3: Loss equivalent greater than 6 hours production. Repairs costing >$500k	Rapid (>5 to <15 minutes)	Urgent	Fire alarm fault. Follow site specific Fire Emergency Plan.	4
F23FIP601I_SL	2000-F23-TER-IC-40009	23FIP601 FIRE ISO		Digital	Alarm				-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Fire alarm. Follow site specific Fire Emergency Plan.	4
FI23481	2000-F24-PID-PR-10004	F23PP466 GW FLOW		Analogue	A_L	1.9	m³/h	5	F23-PP-466 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Check gland water system is healthy and inline auto valves are in the correct position. Else, advise Field Operator / Maintenance to investigate immediately	3
FI23481	2000-F24-PID-PR-10004	F23PP466 GW FLOW		Analogue	A_LL	1.6	m³/h	25	F23-PP-466 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI23485	2000-F24-PID-PR-10004	F23PP467 GW FLOW		Analogue	A_L	1.9	m³/h	5	F23-PP-467 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Check gland water system is healthy and inline auto valves are in the correct position. Else, advise Field Operator / Maintenance to investigate immediately	3
FI23485	2000-F24-PID-PR-10004	F23PP467 GW FLOW		Analogue	A_LL	1.6	m³/h	25	F23-PP-467 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI23489	2000-F24-PID-PR-10004	F23PP468 GW FLOW		Analogue	A_L	1.9	m³/h	5	F23-PP-468 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Check gland water system is healthy and inline auto valves are in the correct position. Else, advise Field Operator / Maintenance to investigate immediately	3
FI23489	2000-F24-PID-PR-10004	F23PP468 GW FLOW		Analogue	A_LL	1.6	m³/h	25	F23-PP-468 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI23493	2000-F24-PID-PR-10004	F23PP469 GW FLOW		Analogue	A_L	1.9	m³/h	5	F23-PP-469 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	Urgent	Check gland water system is healthy and inline auto valves are in the correct position. Else, advise Field Operator / Maintenance to investigate immediately	3
FI23493	2000-F24-PID-PR-10004	F23PP469 GW FLOW		Analogue	A_LL	1.6	m³/h	25	F23-PP-469 RUN FOR LESS THAN 10 SECS	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75610	2000-F75-PID-PR-25008	KALTAILS DECANT FLOW		Analogue	A_L	50	m³/h	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75611	2000-F75-PID-PR-25008	DD3&4 RETURN FLOW		Analogue	A_L	50	m³/h	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
~~FDI75611~~	~~2000-F75-PID-PR-25008~~	~~DD3&4 RETURN DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~F75-PW-74 NOT RUN AND F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN~~	~~Yokogawa DCS~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential decant water line leak. Operator to investigate in field~~	~~5~~
~~FDI75611~~	~~2000-F75-PID-PR-25008~~	~~DD3&4 RETURN DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~F75-PW-74 NOT RUN AND F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN~~	~~Yokogawa DCS~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential decant water line leak. Operator to investigate in field~~	~~5~~
FI75612	2000-F75-PID-PR-25008	FIM I DEC&OFFSITE ROASTER SAL FLOW		Analogue	A_L	50	m³/h	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
LI75620	2000-F75-PID-PR-25010	F00PS465 LEVEL		Analogue	A_H	95	%	5	-	Yokogawa DCS	Environmental 1: Localised environmental impact, contained, with no regulatory reporting	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75621	2000-F75-PID-PR-25010	F00PN401 LEVEL		Analogue	A_H	95	%	5	-	Yokogawa DCS	Environmental 1: Localised environmental impact, contained, with no regulatory reporting	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
~~FDI75321~~	~~2000-F75-PID-PR-40014~~	~~F75TK651 INFLOW DIFF~~		~~Analogue~~	~~A_H~~	~~10~~	~~m³/h~~	~~30~~	~~SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN~~	~~Yokogawa DCS~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential bore water line leak. Operator to investigate in field~~	~~5~~
~~FDI75321~~	~~2000-F75-PID-PR-40014~~	~~F75TK651 INFLOW DIFF~~		~~Analogue~~	~~A_L~~	~~-10~~	~~m³/h~~	~~30~~	~~SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN~~	~~Yokogawa DCS~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential bore water line leak. Operator to investigate in field~~	~~5~~
LIC75300	2000-F75-PID-PR-40014	F75TK651 LEVEL		Analogue PID	A_HH	99	%	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High tank level. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	4
LIC75300	2000-F75-PID-PR-40014	F75TK651 LEVEL		Analogue PID	A_H	95	%	5	-	Yokogawa DCS	Environmental 1: Localised environmental impact, contained, with no regulatory reporting	Prompt (>15 to <30 minutes)	Logging	Feed valve tripped to prevent overflowing tank	4
LIC75300	2000-F75-PID-PR-40014	F75TK651 LEVEL		Analogue PID	A_L	50	%	5	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
LIC75300	2000-F75-PID-PR-40014	F75TK651 LEVEL		Analogue PID	A_LL	5	%	5	-	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
XI75366	2000-F75-PID-PR-40014	F75FI600 FAULT	~~002~~	Digital	Fault			3	-	Yokogawa DCS	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Rapid (>5 to <15 minutes)	High	Seal water elf-cleaning filter fault. Operator to investigate in field	5
FDI75902	2000-F75-PID-PR-40020	D TO FIM1 DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-466 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI75902	2000-F75-PID-PR-40020	D TO FIM1 DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-466 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30909	2000-F75-PID-PR-40022	A TO FIM2 DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-468 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30909	2000-F75-PID-PR-40022	A TO FIM2 DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-468 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30908	2000-F75-PID-PR-40022	C TO FIM2 DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-469 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30908	2000-F75-PID-PR-40022	C TO FIM2 DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-469 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30940A	2000-F75-PID-PR-40023	A TO KALTAILS DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-468 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30940A	2000-F75-PID-PR-40023	A TO KALTAILS DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-468 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30940B	2000-F75-PID-PR-40023	B TO KALTAILS DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-467 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30940B	2000-F75-PID-PR-40023	B TO KALTAILS DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-467 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
FDI30941	2000-F75-PID-PR-40023	C TO KALTAILS DIFF FLOW		Analogue	A_H	50	m³/h	600	F23-PP-469 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30941	2000-F75-PID-PR-40023	C TO KALTAILS DIFF FLOW		Analogue	A_L	-50	m³/h	600	F23-PP-469 NOT RUN	Yokogawa PLC	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30115	2000-F75-PR-PID-40025	LINE C TO FIM II F DIFF FLOW		Analogue	A_H	50	m³/h	600	F30-PP-77A NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30115	2000-F75-PR-PID-40025	LINE C TO FIM II F DIFF FLOW		Analogue	A_L	-50	m³/h	600	F30-PP-77A NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30116	2000-F75-PR-PID-40025	LINE A TO FIM II F DIFF FLOW		Analogue	A_H	50	m³/h	600	F30-PP-77B NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30116	2000-F75-PR-PID-40025	LINE A TO FIM II F DIFF FLOW		Analogue	A_L	-50	m³/h	600	F30-PP-77B NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30117	2000-F75-PR-PID-40026	LINE C TO FIM II E DIFF FLOW		Analogue	A_H	50	m³/h	600	F30-PP-77A NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30117	2000-F75-PR-PID-40026	LINE C TO FIM II E DIFF FLOW		Analogue	A_L	-50	m³/h	600	F30-PP-77A NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30118	2000-F75-PR-PID-40026	LINE C TO FIM II G DIFF FLOW		Analogue	A_H	50	m³/h	600	F30-PP-77A NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30118	2000-F75-PR-PID-40026	LINE C TO FIM II G DIFF FLOW		Analogue	A_L	-50	m³/h	600	F30-PP-77A NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30119	2000-F75-PR-PID-40026	LINE A TO FIM II E DIFF FLOW		Analogue	A_H	50	m³/h	600	F30-PP-77B NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30119	2000-F75-PR-PID-40026	LINE A TO FIM II E DIFF FLOW		Analogue	A_L	-50	m³/h	600	F30-PP-77B NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30120	2000-F75-PR-PID-40026	LINE A TO FIM II G DIFF FLOW		Analogue	A_H	50	m³/h	600	F30-PP-77B NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI30120	2000-F75-PR-PID-40026	LINE A TO FIM II G DIFF FLOW		Analogue	A_L	-50	m³/h	600	F30-PP-77B NOT RUN	Yokogawa DCS	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5

Appendix B MOSCAD RTU Process Control Tables

Owner **FIM TSF PROJECT**

PROCESS CONTROL TABLES - MOSCAD RTU

2000-F00-PHL-PR-40001 - APPENDIX B

Table of Contents		Total	On Hold	Complete (%)	FAT Complete (%)	SAT Complete (%)
1	**Analogue**	73	0	100.0	0.0	0.0
2	**Digital**	25	0	100.0	0.0	0.0
3	**Totalisers**	27	0	100.0	0.0	0.0
4	**Open/Close Valves**	7	0	100.0	0.0	0.0
5	**Control Valves**	1	0	100.0	0.0	0.0
6	**PID Controllers**	4	0	100.0	0.0	0.0
7	**Miscellaneous Controllers**	5	0	100.0	0.0	0.0
8	**Interlocks**	239	0	100.0	0.0	0.0
9	**Calculations**	25	0	100.0	0.0	0.0
	Total	406	0	100.0	0.0	0.0

Alarms	Priority	Count	%
1	Urgent (Critical Failure)	0	0
2	High (Fault/Error)	56	37.3
3	Low (Warning)	66	44.0
4	Logging	28	18.7
	Total	150	100

Abbreviations	
FC	Fail closed
FO	Fail open
FL	Fail last
LCP	Local control panel
LCS	Local control station
A_L	Low alarm
A_H	High alarm
A_LL	Low-Low alarm
A_HH	High-High alarm
P_L	Low process switchpoint
P_H	High process switchpoint
P_LL	Low-Low process switchpoint
P_HH	High-High process switchpoint
OIT	Operator interface terminal
Range	Instrument range
SEQ	Sequence
TBA	To be advised
v	Vendor controlled
VTA	Vendor to advise
x	PCS controlled
*	Denotes a sequence stop in place of an interlock

ANALOGUE INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	Trend (Sec)	Range			Process Switchpoints				Alarms								Alarm Masking	Control System	Comments	Rev
							Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)				
IT23586	II23586	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK CURRENT STATUS SIGNAL	F23HY421 CURRENT		1	0	13.4	A	-	-	-	-	-	-	-	-	-	-	-	-	-	MOSCAD RTU	Display only	4
FIT23592	FI23592	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH FLOW TRANSMITTER	FIM III LINE 1 SOUTH FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	100	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	4
~~F23CALC40018~~	~~FDI23592A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III LINE 1 SOUTH DIFFERENTIAL FLOW CALCULATION~~	~~LINE D TO FIM III 1S DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F23CALC40019~~	~~FDI23592B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III LINE 1 SOUTH DIFFERENTIAL FLOW CALCULATION~~	~~LINE B TO FIM III 1S DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
FIT23593	FI23593	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH FLOW TRANSMITTER	FIM III LINE 1 NORTH FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	100	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	4
~~F23CALC40020~~	~~FDI23593A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III LINE 1 NORTH DIFFERENTIAL FLOW CALCULATION~~	~~LINE B TO FIM III 1N DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F23CALC40021~~	~~FDI23593B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III LINE 1 NORTH DIFFERENTIAL FLOW CALCULATION~~	~~LINE D TO FIM III 1N DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
FIT23594	FI23594	2000-F23-PID-PR-40005	FIM III LINE 3 WEST FLOW TRANSMITTER	FIM III LINE 3 WEST FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	100	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	4
~~F23CALC40022~~	~~FDI23594A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III LINE 3 WEST DIFFERENTIAL FLOW CALCULATION~~	~~LINE B TO FIM III 3W DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F23CALC40023~~	~~FDI23594B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III LINE 3 WEST DIFFERENTIAL FLOW CALCULATION~~	~~LINE D TO FIM III 3W DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
FIT23595	FI23595	2000-F23-PID-PR-40005	FIM III LINE 3 EAST FLOW TRANSMITTER	FIM III LINE 3 EAST FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	100	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	4
~~F23CALC40024~~	~~FDI23595A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III LINE 3 EAST DIFFERENTIAL FLOW CALCULATION~~	~~LINE D TO FIM III 3E DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F23CALC40025~~	~~FDI23595B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III LINE 3 EAST DIFFERENTIAL FLOW CALCULATION~~	~~LINE B TO FIM III 3E DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
FIT23596	FI23596	2000-F23-PID-PR-40005	FIM III LINE 2 WEST FLOW TRANSMITTER	FIM III LINE 2 WEST FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	100	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	4
~~F23CALC40026~~	~~FDI23596A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III LINE 2 WEST DIFFERENTIAL FLOW CALCULATION~~	~~LINE B TO FIM III 2W DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F23CALC40027~~	~~FDI23596B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III LINE 2 WEST DIFFERENTIAL FLOW CALCULATION~~	~~LINE D TO FIM III 2W DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
FIT23597	FI23597	2000-F23-PID-PR-40005	FIM III LINE 2 EAST FLOW TRANSMITTER	FIM III LINE 2 EAST FLOW		1	0	2,000	m³/h	-	-	-	-	-	-	100	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	4
~~F23CALC40028~~	~~FDI23597A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III LINE 2 EAST DIFFERENTIAL FLOW CALCULATION~~	~~LINE D TO FIM III 2E DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F23CALC40029~~	~~FDI23597B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III LINE 2 EAST DIFFERENTIAL FLOW CALCULATION~~	~~LINE B TO FIM III 2E DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	-	-	-	-	-	-	~~-50~~	~~30~~	~~50~~	~~30~~	-	-	-	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
LIT75268	LI75268	2000-F75-PID-PR-40001	FIM II CELL G SUPERNATANT LEVEL TRANSMITTER	FIM II CELL G LEVEL		1	0	100	%	-	-	-	-	-	-	5	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning	1
FIT75269	FI75269	2000-F75-PID-PR-40001	FIM II EXTENSION CELL G DECANT FLOW TRANSMITTER	F75PW664 DISCH FLOW		1	0	1,000	m³/h	-	-	-	-	10	5	20	5	-	-	-	-	F75-PW-664 NOT RUN	MOSCAD RTU	A_L = Warning A_LL = Trip F75-PW-664	1
LIT75267	LI75267	2000-F75-PID-PR-40001	FIM II CELL G ANTISCALANT IBC LEVEL TRANSMITTER	FIM II CELL G AS LEVEL		1	0	100	%	-	-	-	-	5	5	10	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning A_LL = Replace IBC	1
FIT75270	FI75270	2000-F75-PID-PR-40031	FIM II EXTENSION CELL G DECANT FLOW TRANSMITTER 2	FIM II EXT CELL G DECANT FLOW2		1	0	1,000	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	F75-PW-664 NOT RUN	MOSCAD RTU	A_L = Warning	2
F75CALC40018	FDI75270	2000-F75-PID-PR-40031	FIM II EXTENSION CELL G DECANT RETURN DIFFERENTIAL FLOW	FIM II EXT DECANT DIFF FLOW		1	-50	50	m³/h	-	-	-	-	-	-	-25	30	25	30	-	-	**F75-PS-593 NOT RUN AND F75-PW-664 NOT RUN**	MOSCAD RTU	**Alarm masked for 600 seconds on startup of pumps** A_H = Warning A_L = Warning	**5**
LIT75318	LI75318	2000-F75-PID-PR-40002	FIM III CELL 1 SUPERNATANT LEVEL TRANSMITTER	FIM III CELL 1 LEVEL		1	0	100	%	-	-	-	-	-	-	5	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning	1
FIT75319	FI75319	2000-F75-PID-PR-40002	FIM III CELL 1 DECANT PUMP FLOW TRANSMITTER	F75PW600 DISCH FLOW		1	0	1,000	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	F75-PW-600 NOT RUN	MOSCAD RTU	A_L = Warning	1
FIT75329	FI75329	2000-F75-PID-PR-40002	FIM III DECANT FLOW TRANSMITTER	FIM III DECANT FLOW		1	0	1,000	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	1
F75CALC40002	FDI75329	2000-F75-PID-PR-40002	FIM III DECANT RETURN DIFFERENTIAL FLOW	FIM II DECANT DIFF FLOW		1	-50	50	m³/h	-	-	-	-	-	-	-25	30	25	30	-	-	**F75-PW-600 NOT RUN AND F75-PW-601 NOT RUN AND F75-PW-602 NOT RUN**	MOSCAD RTU	**Alarm masked for 600 seconds on startup of decant pumps** A_H = Warning A_L = Warning	**5**
LIT75324	LI75324	2000-F75-PID-PR-40002	FIM III CELL 1 ANTISCALANT IBC LEVEL TRANSMITTER	FIM III CELL 1 AS LEVEL		1	0	100	%	-	-	-	-	5	5	10	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning A_LL = Replace IBC	1
LIT75325	LI75325	2000-F75-PID-PR-40003	FIM III CELL 2 SUPERNATANT LEVEL TRANSMITTER	FIM III CELL 2 LEVEL		1	0	100	%	-	-	-	-	-	-	5	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning	1
FIT75326	FI75326	2000-F75-PID-PR-40003	FIM III CELL 2 DECANT PUMP FLOW TRANSMITTER	F75PW601 DISCH FLOW		1	0	1,000	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	F75-PW-601 NOT RUN	MOSCAD RTU	A_L = Warning	1
LIT75327	LI75327	2000-F75-PID-PR-40003	FIM III CELL 3 SUPERNATANT LEVEL TRANSMITTER	FIM III CELL 3 LEVEL		1	0	100	%	-	-	-	-	-	-	5	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning	1
FIT75328	FI75328	2000-F75-PID-PR-40003	FIM III CELL 3 DECANT PUMP FLOW TRANSMITTER	F75PW602 DISCH FLOW		1	0	1,000	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	F75-PW-602 NOT RUN	MOSCAD RTU	A_L = Warning	1
LIT75342	LI75342	2000-F75-PID-PR-40003	FIM III CELL 2 ANTISCALANT IBC LEVEL TRANSMITTER	FIM III CELL 2 AS LEVEL		1	0	100	%	-	-	-	-	5	5	10	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning A_LL = Replace IBC	1
LIT75343	LI75343	2000-F75-PID-PR-40003	FIM III CELL 3 ANTISCALANT IBC LEVEL TRANSMITTER	FIM III CELL 3 AS LEVEL		1	0	100	%	-	-	-	-	5	5	10	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning A_LL = Replace IBC	1
LIT75362	LI75362	2000-F75-PID-PR-40004	DECANT DAM 4 OVERFLOW POND LEVEL TRANSMITTER	DD4 O/F POND LEVEL		1	0	100	%	-	-	-	-	5	5	10	5	95	5	99	5	-	MOSCAD RTU	A_HH = Trip F75-PS-556/557 AND F75-PW-600/601/602 A_H = Warning A_L = Warning A_LL = Trip F75-PW-606/607/608	1
LIT75364	LI75364	2000-F75-PID-PR-40004	DECANT DAM 4 ANTISCALANT IBC LEVEL TRANSMITTER	DD4 AS LEVEL		1	0	100	%	-	-	-	-	5	5	10	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning A_LL = Replace IBC	1
PIT75338	PI75338	2000-F75-PID-PR-40004	DECANT DAM 4 FLUSHING PRESSURE TRANSMITTER	F75PW606-608 DISCH PRESSURE		1	0	1,600	kPa	-	-	-	-	-	-	-	-	1,350	5	1,400	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	A_HH = Trip F75-PW-606/607/608 A_H = Warning	3
PIT75339	PI75339	2000-F75-PID-PR-40004	DECANT DAM 4 RETURN PRESSURE TRANSMITTER	DD4 RETURN PRESSURE		1	0	1,600	kPa	-	-	-	-	-	-	-	-	850	5	900	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	A_HH = Trip F75-PW-606/607/608 A_H = Warning	3

ANALOGUE INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	Trend (Sec)	Range			Process Switchpoints				Alarms								Alarm Masking	Control System	Comments	Rev
							Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)				
LIT75264	LI75264	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 1 LEVEL TRANSMITTER	F75PS559 LEVEL		1	0	100	%	-	25	75	-	5	5	10	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PS-559	1
FIT75264	FI75264	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 1 FLOW TRANSMITTER	F75PS559 DISCH FLOW		1	0	200	m³/h	-	-	-	-	5	5	10	5	-	-	-	-	F75-PS-559 NOT RUN	MOSCAD RTU	A_L = Warning A_LL = Trip F75-PS-559	1
LIT75265	LI75265	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP LEVEL TRANSMITTER	F75PS593 LEVEL		1	0	100	%	-	25	75	-	5	5	10	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PS-593	1
FIT75265	FI75265	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP FLOW TRANSMITTER	F75PS593 DISCH FLOW		1	0	200	m³/h	-	-	-	-	5	5	10	5	-	-	-	-	F75-PS-593 NOT RUN	MOSCAD RTU	A_L = Warning A_LL = Trip F75-PS-593	1
LIT75365	LI75365	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 2 LEVEL TRANSMITTER	F75PS560 LEVEL		1	0	100	%	-	25	75	-	5	5	10	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PS-560	1
FIT75365	FI75365	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 2 FLOW TRANSMITTER	F75PS560 DISCH FLOW		1	0	200	m³/h	-	-	-	-	5	5	10	5	-	-	-	-	F75-PS-560 NOT RUN	MOSCAD RTU	A_L = Warning A_LL = Trip F75-PS-560	1
PIT75730	PI75730	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 1 PRESSURE TRANSMITTER	F75PS801 DISCH PRESSURE		1	0	1,000	kPa	-	-	-	-	-	-	-	-	650	5	700	5	-	MOSCAD RTU	A_HH = Trip F75-PS-801 A_H = Warning	1
FIT75730	FI75730	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 1 FLOW TRANSMITTER	F75PS801 DISCH FLOW		1	0	50	m³/h	-	-	-	-	2	5	5	5	-	-	-	-	F75-PS-801 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-801	1
PIT75731	PI75731	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 2 PRESSURE TRANSMITTER	F75PS802 DISCH PRESSURE		1	0	1,000	kPa	-	-	-	-	-	-	-	-	650	5	700	5	-	MOSCAD RTU	A_HH = Trip F75-PS-802 A_H = Warning	1
FIT75731	FI75731	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 2 FLOW TRANSMITTER	F75PS802 DISCH FLOW		1	0	50	m³/h	-	-	-	-	2	5	5	5	-	-	-	-	F75-PS-802 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-802	1
PIT75732	PI75732	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 3 PRESSURE TRANSMITTER	F75PS803 DISCH PRESSURE		1	0	1,000	kPa	-	-	-	-	-	-	-	-	650	5	700	5	-	MOSCAD RTU	A_HH = Trip F75-PS-803 A_H = Warning	1
FIT75732	FI75732	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 3 FLOW TRANSMITTER	F75PS803 DISCH FLOW		1	0	50	m³/h	-	-	-	-	2	5	5	5	-	-	-	-	F75-PS-803 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-803	1
PIT75733	PI75733	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 4 PRESSURE TRANSMITTER	F75PS804 DISCH PRESSURE		1	0	1,000	kPa	-	-	-	-	-	-	-	-	650	5	700	5	-	MOSCAD RTU	A_HH = Trip F75-PS-804 A_H = Warning	1
FIT75733	FI75733	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 4 FLOW TRANSMITTER	F75PS804 DISCH FLOW		1	0	50	m³/h	-	-	-	-	2	5	5	5	-	-	-	-	F75-PS-804 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-804	1
FIT75793	FI75793	2000-F75-PID-PR-40009	FIM III SEEPAGE WATER FLOW TRANSMITTER	FIM III SEEPAGE FLOW		1	0	600	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	1
F75CALC40003	FDI75793	2000-F75-PID-PR-40009	FIM III SEEPAGE DIFFERENTIAL FLOW	FIM III SEEPAGE DIFF FLOW		1	-50	50	m³/h	-	-	-	-	-	-	-25	30	25	30	-	-	**F75-PS-558 NOT RUN AND F75-PS-801 NOT RUN AND F75-PS-802 NOT RUN AND F75-PS-803 NOT RUN AND F75-PS-804 NOT RUN**	MOSCAD RTU	**Alarm masked for 600 seconds on startup of pumps** A_H = Warning A_L = Warning	**5**
LIT75785	LI75785	2000-F75-PID-PR-40011	FIM III TOE DRAIN PUMP LEVEL TRANSMITTER	F75PS558 LEVEL		1	0	100	%	-	25	75	-	5	5	10	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PS-558	1
FIT75785	FI75785	2000-F75-PID-PR-40011	FIM III TOE DRAIN PUMP FLOW TRANSMITTER	F75PS558 DISCH FLOW		1	0	200	m³/h	-	-	-	-	5	5	10	5	-	-	-	-	F75-PS-558 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-558	1
LIT75789	LI75789	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 LEVEL TRANSMITTER	F75PS556 LEVEL		1	0	100	%	-	25	75	-	5	5	10	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PS-556	1
FIT75789	FI75789	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 FLOW TRANSMITTER	F75PS556 DISCH FLOW		1	0	200	m³/h	-	-	-	-	5	5	10	5	-	-	-	-	F75-PS-556 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-556	1
LIT75790	LI75790	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 LEVEL TRANSMITTER	F75PS557 LEVEL		1	0	100	%	-	25	75	-	5	5	10	5	99	5	-	-	-	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PS-557	1
FIT75790	FI75790	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 FLOW TRANSMITTER	F75PS557 DISCH FLOW		1	0	200	m³/h	-	-	-	-	5	5	10	5	-	-	-	-	F75-PS-557 NOT RUN	MOSCAD RTU	A_L = Warning A_LL - Trip F75-PS-557	1
FIT75792	FI75792	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE FLOW TRANSMITTER	FIM III U/DRAIN FLOW		1	0	400	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	1
F75CALC40004	FDI75792	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE DIFFERENTIAL FLOW	FIM III U/DRAIN DIFF FLOW		1	-50	50	m³/h	-	-	-	-	-	-	-25	30	25	30	-	-	**F75-PS-556 NOT RUN AND F75-PS-557 NOT RUN**	MOSCAD RTU	**Alarm masked for 600 seconds on startup of pumps** A_H = Warning A_L = Warning	**5**
~~F75CALC40005~~	~~FDI75321~~	~~2000-F75-PID-PR-40014~~	~~BORE WATER TO FIM III SEAL WATER TANK DIFFERENTIAL FLOW~~	~~F75TK651 INFLOW DIFF~~		~~1~~	~~-25~~	~~25~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-10~~	~~600~~	~~10~~	~~600~~	~~-~~	~~-~~	**SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN**	~~MOSCAD RTU~~	**Alarm masked for 600 seconds on startup of pumps** ~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F75CALC40012~~	~~FDI30115~~	~~2000-F75-PR-PID-40025~~	~~LINE C TO FIM II CELL F DIFFERENTIAL FLOW CALCULATION~~	~~LINE C TO FIM II F DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-50~~	~~30~~	~~50~~	~~30~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F75CALC40013~~	~~FDI30116~~	~~2000-F75-PR-PID-40025~~	~~LINE A TO FIM II CELL F DIFFERENTIAL FLOW CALCULATION~~	~~LINE A TO FIM II F DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-50~~	~~30~~	~~50~~	~~30~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F75CALC40014~~	~~FDI30117~~	~~2000-F75-PR-PID-40026~~	~~LINE C TO FIM II CELL E DIFFERENTIAL FLOW CALCULATION~~	~~LINE C TO FIM II E DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-50~~	~~30~~	~~50~~	~~30~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F75CALC40015~~	~~FDI30118~~	~~2000-F75-PR-PID-40026~~	~~LINE C TO FIM II CELL G DIFFERENTIAL FLOW CALCULATION~~	~~LINE C TO FIM II G DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-50~~	~~30~~	~~50~~	~~30~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F75CALC40016~~	~~FDI30119~~	~~2000-F75-PR-PID-40026~~	~~LINE A TO FIM II CELL E DIFFERENTIAL FLOW CALCULATION~~	~~LINE A TO FIM II E DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-50~~	~~30~~	~~50~~	~~30~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
~~F75CALC40017~~	~~FDI30120~~	~~2000-F75-PR-PID-40026~~	~~LINE A TO FIM II CELL G DIFFERENTIAL FLOW CALCULATION~~	~~LINE A TO FIM II G DIFF FLOW~~		~~1~~	~~-100~~	~~100~~	~~m³/h~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-~~	~~-50~~	~~30~~	~~50~~	~~30~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~A_H = Warning~~ ~~A_L = Warning~~	~~5~~
F75CALC40001	FDI75611	2000-F75-PID-PR-25008	DECANT DAM 3 AND 4 RETURN DIFFERENTIAL FLOW CALCULATION	DD3&4 RETURN DIFF FLOW		1	-100	100	m³/h	-	-	-	-	-	-	-50	600	50	600	-	-	**F75-PW-74 NOT RUN AND F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN**	MOSCAD RTU	**Alarm masked for 600 seconds on startup of decant pumps** A_H = Warning A_L = Warning	**5**
FIT17503	FI17503	2000-F175-PID-PR-40002	BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER	F75TK651 INFLOW		1	0	25	m³/h	-	-	-	-	-	-	-	-	-	-	-	-	-	MOSCAD RTU		1
FIT75794	FI75794	2000-F175-PID-PR-40002	FIM III SEEPAGE WATER FLOW TRANSMITTER 2	FIM III SEEPAGE FLOW 2		1	0	1,000	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	-	MOSCAD RTU	A_L = Warning	1
F75CALC40020	FDI75794	2000-F175-PID-PR-40002	FIM III SEEPAGE DIFFERENTIAL FLOW 2	FIM III SEEPAGE DIFF FLOW 2		1	-50	50	m³/h	-	-	-	-	-	-	-25	30	25	30	-	-	**F75-PS-558 NOT RUN AND F75-PS-801 NOT RUN AND F75-PS-802 NOT RUN AND F75-PS-803 NOT RUN AND F75-PS-804 NOT RUN**	MOSCAD RTU	**Alarm masked for 600 seconds on startup of pumps** A_H = Warning A_L = Warning	**5**

DIGITAL INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	State 0	State 1	Alarm State	Source	Alarm Delay	Alarm Priority	Alarm Masking	Control System	Comments	Rev
XS23580	XI23580	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK RUN	F23HY421 RUNNING		Inactive (N_Running)	Active (Running)	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
XS23581	XI23581	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK FAULT	F23HY421 FAULT		Fault	Normal	Fault	Hardwired to PCS	3	Med	-	MOSCAD RTU		4
XS23582	XY23582	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK START SIGNAL	F23HY421 START CMD		Off	On	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
XS23583	XI23583	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK READY	F23HY421 READY		Inactive (N_Ready)	Active (Ready)	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
XS23584	XI23584	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK FIELD E-STOP SIGNAL	F23HY421 PB STOP		Pressed	Not Pressed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
XS23585	XI23585	2000-F23-PID-PR-40005	FIM III VALVE STATION HYDRAULIC POWER PACK FIELD START SIGNAL	F23HY421 PB START		Not Pressed	Pressed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23700	ZIO23700	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH ISOLATION VALVE 1 OPEN	FIM III LINE 1N VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23700	ZIC23700	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH ISOLATION VALVE 1 CLOSED	FIM III LINE 1N VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23701	ZIO23701	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH ISOLATION VALVE 2 OPEN	FIM III LINE 1N VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23701	ZIC23701	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH ISOLATION VALVE 2 CLOSED	FIM III LINE 1N VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23702	ZIO23702	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH & SOUTH ISOLATION VALVE OPEN	FIM III LINE 1N & 1S VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23702	ZIC23702	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH & SOUTH ISOLATION VALVE CLOSED	FIM III LINE 1N & 1S VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23703	ZIO23703	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH ISOLATION VALVE 1 OPEN	FIM III LINE 1S VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23703	ZIC23703	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH ISOLATION VALVE 1 CLOSED	FIM III LINE 1S VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23704	ZIO23704	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH ISOLATION VALVE 2 OPEN	FIM III LINE 1S VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23704	ZIC23704	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH ISOLATION VALVE 2 CLOSED	FIM III LINE 1S VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23705	ZIO23705	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH FLUSHING WATER VALVE OPEN	FIM III LINE 1S FLUSH OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23705	ZIC23705	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH FLUSHING WATER VALVE CLOSED	FIM III LINE 1S FLUSH CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23706	ZIO23706	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH FLUSHING WATER VALVE OPEN	FIM III LINE 1N FLUSH OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23706	ZIC23706	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH FLUSHING WATER VALVE CLOSED	FIM III LINE 1N FLUSH CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23707	ZIO23707	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 1 OPEN	FIM III LINE 2W VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23707	ZIC23707	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 1 CLOSED	FIM III LINE 2W VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23708	ZIO23708	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 1 OPEN	FIM III LINE 2E VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23708	ZIC23708	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 1 CLOSED	FIM III LINE 2E VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23709	ZIO23709	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 2 OPEN	FIM III LINE 2W VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23709	ZIC23709	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 2 CLOSED	FIM III LINE 2W VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23710	ZIO23710	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 3 OPEN	FIM III LINE 2W VLV 3 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23710	ZIC23710	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 3 CLOSED	FIM III LINE 2W VLV 3 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23711	ZIO23711	2000-F23-PID-PR-40005	FIM III LINE 2 EAST & WEST ISOLATION VALVE OPEN	FIM III LINE 2E & 2W VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23711	ZIC23711	2000-F23-PID-PR-40005	FIM III LINE 2 EAST & WEST ISOLATION VALVE CLOSED	FIM III LINE 2E & 2W VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23712	ZIO23712	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 2 OPEN	FIM III LINE 2E VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23712	ZIC23712	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 2 CLOSED	FIM III LINE 2E VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23713	ZIO23713	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 3 OPEN	FIM III LINE 2E VLV 3 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23713	ZIC23713	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 3 CLOSED	FIM III LINE 2E VLV 3 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23714	ZIO23714	2000-F23-PID-PR-40005	FIM III LINE 2 EAST FLUSHING WATER VALVE OPEN	FIM III LINE 2E FLUSH OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23714	ZIC23714	2000-F23-PID-PR-40005	FIM III LINE 2 EAST FLUSHING WATER VALVE CLOSED	FIM III LINE 2E FLUSH CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23715	ZIO23715	2000-F23-PID-PR-40005	FIM III LINE 2 WEST FLUSHING WATER VALVE OPEN	FIM III LINE 2W FLUSH OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23715	ZIC23715	2000-F23-PID-PR-40005	FIM III LINE 2 WEST FLUSHING WATER VALVE CLOSED	FIM III LINE 2W FLUSH CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23716	ZIO23716	2000-F23-PID-PR-40005	FIM III LINE 3 WEST ISOLATION VALVE 1 OPEN	FIM III LINE 3W VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23716	ZIC23716	2000-F23-PID-PR-40005	FIM III LINE 3 WEST ISOLATION VALVE 1 CLOSED	FIM III LINE 3W VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23717	ZIO23717	2000-F23-PID-PR-40005	FIM III LINE 3 WEST ISOLATION VALVE 2 OPEN	FIM III LINE 3W VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23717	ZIC23717	2000-F23-PID-PR-40005	FIM III LINE 3 WEST ISOLATION VALVE 2 CLOSED	FIM III LINE 3W VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23718	ZIO23718	2000-F23-PID-PR-40005	FIM III LINE 3 EAST & WEST ISOLATION VALVE OPEN	FIM III LINE 3E & 3W VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23718	ZIC23718	2000-F23-PID-PR-40005	FIM III LINE 3 EAST & WEST ISOLATION VALVE CLOSED	FIM III LINE 3E & 3W VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23719	ZIO23719	2000-F23-PID-PR-40005	FIM III LINE 3 EAST ISOLATION VALVE 1 OPEN	FIM III LINE 3E VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23719	ZIC23719	2000-F23-PID-PR-40005	FIM III LINE 3 EAST ISOLATION VALVE 1 CLOSED	FIM III LINE 3E VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23720	ZIO23720	2000-F23-PID-PR-40005	FIM III LINE 3 EAST ISOLATION VALVE 2 OPEN	FIM III LINE 3E VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4

DIGITAL INSTRUMENTS

Tag Source	Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Hold	State 0	State 1	Alarm State	Source	Alarm Delay	Alarm Priority	Alarm Masking	Control System	Comments	Rev
ZSC23720	ZIC23720	2000-F23-PID-PR-40005	FIM III LINE 3 EAST ISOLATION VALVE 2 CLOSED	FIM III LINE 3E VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23721	ZIO23721	2000-F23-PID-PR-40005	FIM III LINE 3 EAST FLUSHING WATER VALVE OPEN	FIM III LINE 3E FLUSH OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23721	ZIC23721	2000-F23-PID-PR-40005	FIM III LINE 3 EAST FLUSHING WATER VALVE CLOSED	FIM III LINE 3E FLUSH CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO23722	ZIO23722	2000-F23-PID-PR-40005	FIM III LINE 3 WEST FLUSHING WATER VALVE OPEN	FIM III LINE 3W FLUSH OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC23722	ZIC23722	2000-F23-PID-PR-40005	FIM III LINE 3 WEST FLUSHING WATER VALVE CLOSED	FIM III LINE 3W FLUSH CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
~~LSH75271~~	~~LAH75271~~	~~2000-F75-PID-PR-40001~~	~~FIM II DECANT PUMP CELL G SUCTION LINE LEVEL SWITCH HIGH~~	~~F30PW664 SUC PRM~~		~~Inactive~~	~~Active~~	~~-~~	~~Hardwired to PCS~~	~~-~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~No alarm. Indication only~~	~~5~~
LSH75363	LAH75363	2000-F75-PID-PR-40004	DECANT DAM 4 SUCTION LINE LEVEL SWITCH HIGH	DD4 PP SUC PRM		Inactive	Active	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO75340	ZIO75340	2000-F75-PID-PR-40004	DECANT DAM 4 ~~FLUSHING WATER~~ DECANT RETURN ISOLATION VALVE OPEN	DD4 RET VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	5
ZSC75340	ZIC75340	2000-F75-PID-PR-40004	DECANT DAM 4 ~~FLUSHING WATER~~ DECANT RETURN ISOLATION VALVE CLOSED	DD4 RET VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	5
ZSO75330	ZIO75330	2000-F75-PID-PR-40004	DECANT DAM 4 TURRET 1 ISOLATION VALVE OPEN	DD4 TRT 1 ISO VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75330	ZIC75330	2000-F75-PID-PR-40004	DECANT DAM 4 TURRET 1 ISOLATION VALVE CLOSED	DD4 TRT 1 ISO VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO75331	ZIO75331	2000-F75-PID-PR-40004	DECANT DAM 4 TURRET 2 ISOLATION VALVE OPEN	DD4 TRT 2 ISO VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75331	ZIC75331	2000-F75-PID-PR-40004	DECANT DAM 4 TURRET 2 ISOLATION VALVE CLOSED	DD4 TRT 2 ISO VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO75341	ZIO75341	2000-F75-PID-PR-40004	DECANT DAM 4 OVERFLOW TURRET ISOLATION VALVE OPEN	DD4 O/F TRT ISO VLV OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75341	ZIC75341	2000-F75-PID-PR-40004	DECANT DAM 4 OVERFLOW TURRET ISOLATION VALVE CLOSED	DD4 O/F TRT ISO VLV CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
FSL75335	FAL75335	2000-F75-PID-PR-40004	DECANT DAM 4 PUMP 1 FLOW SWITCH LOW	F75PW606 FLOW SWITCH		Alarm	Normal	Alarm	Hardwired to PCS	3	Med	F75-PW-606 NOT RUN	MOSCAD RTU		1
FSL75336	FAL75336	2000-F75-PID-PR-40004	DECANT DAM 4 PUMP 2 FLOW SWITCH LOW	F75PW607 FLOW SWITCH		Alarm	Normal	Alarm	Hardwired to PCS	3	Med	F75-PW-607 NOT RUN	MOSCAD RTU		1
FSL75337	FAL75337	2000-F75-PID-PR-40004	DECANT DAM 4 PUMP 3 FLOW SWITCH LOW	F75PW608 FLOW SWITCH		Alarm	Normal	Alarm	Hardwired to PCS	3	Med	F75-PW-608 NOT RUN	MOSCAD RTU		1
ZSO75283	ZIO75283	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 1 OPEN	F75PS593 SUC VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	1
ZSC75283	ZIC75283	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 1 CLOSED	F75PS593 SUC VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	1
ZSO75284	ZIO75284	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 2 OPEN	F75PS593 SUC VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75284	ZIC75284	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 2 CLOSED	F75PS593 SUC VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO75284	ZIO75284	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 2 OPEN	F75PS593 SUC VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75284	ZIC75284	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 2 CLOSED	F75PS593 SUC VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO75795	ZIO75795	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 ISOLATION VALVE 1 OPEN	F75PS556 SUC VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	1
ZSC75795	ZIC75795	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 ISOLATION VALVE 1 CLOSED	F75PS556 SUC VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	1
ZSO75796	ZIO75796	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 ISOLATION VALVE 1 OPEN	F75PS557 SUC VLV 1 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	1
ZSC75796	ZIC75796	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 ISOLATION VALVE 1 CLOSED	F75PS557 SUC VLV 1 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	1
ZSO75797	ZIO75797	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 ISOLATION VALVE 2 OPEN	F75PS556 SUC VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75797	ZIC75797	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 ISOLATION VALVE 2 CLOSED	F75PS556 SUC VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSO75798	ZIO75798	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 ISOLATION VALVE 2 OPEN	F75PS557 SUC VLV 2 OP		N_Open	Open	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4
ZSC75798	ZIC75798	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 ISOLATION VALVE 2 CLOSED	F75PS557 SUC VLV 2 CLS		N_Closed	Closed	-	Hardwired to PCS	-	-	-	MOSCAD RTU	No alarm. Indication only	4

TOTALISERS

Tag Source	Tag Name	Type	P&ID	Source Description	Abbreviated Description (Faceplate)	Hold	Units	Reset Period			Control System	Comments	Rev
								Shift	Day	Cont			
FIT23592	FQI23592	FQI	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH FLOW TRANSMITTER	FIM III LINE 1 SOUTH TOTAL FLOW		m³	x	x	x	MOSCAD RTU		4
FIT23593	FQI23593	FQI	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH FLOW TRANSMITTER	FIM III LINE 1 NORTH TOTAL FLOW		m³	x	x	x	MOSCAD RTU		4
FIT23594	FQI23594	FQI	2000-F23-PID-PR-40005	FIM III LINE 3 WEST FLOW TRANSMITTER	FIM III LINE 3 WEST TOTAL FLOW		m³	x	x	x	MOSCAD RTU		4
FIT23595	FQI23595	FQI	2000-F23-PID-PR-40005	FIM III LINE 3 EAST FLOW TRANSMITTER	FIM III LINE 3 EAST TOTAL FLOW		m³	x	x	x	MOSCAD RTU		4
FIT23596	FQI23596	FQI	2000-F23-PID-PR-40005	FIM III LINE 2 WEST FLOW TRANSMITTER	FIM III LINE 2 WEST TOTAL FLOW		m³	x	x	x	MOSCAD RTU		4
FIT23597	FQI23597	FQI	2000-F23-PID-PR-40005	FIM III LINE 2 EAST FLOW TRANSMITTER	FIM III LINE 2 EAST TOTAL FLOW		m³	x	x	x	MOSCAD RTU		4
FIT75269	FQI75269	FQI	2000-F75-PID-PR-40001	FIM II EXTENSION CELL G DECANT FLOW TRANSMITTER	F75PW664 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75270	FQI75270	FQI	2000-F75-PID-PR-40031	FIM II EXTENSION CELL G DECANT FLOW TRANSMITTER 2	F75PW664 TOTAL FLOW 2		m³	x	x	x	MOSCAD RTU		2
FIT75319	FQI75319	FQI	2000-F75-PID-PR-40002	FIM III CELL 1 DECANT PUMP FLOW TRANSMITTER	F75PW600 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75329	FQI75329	FQI	2000-F75-PID-PR-40002	FIM III DECANT FLOW TRANSMITTER	FIM III TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75326	FQI75326	FQI	2000-F75-PID-PR-40003	FIM III CELL 2 DECANT PUMP FLOW TRANSMITTER	F75PW601 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75328	FQI75328	FQI	2000-F75-PID-PR-40003	FIM III CELL 3 DECANT PUMP FLOW TRANSMITTER	F75PW602 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75333	FQI75333	FQI	2000-F75-PID-PR-40004	DECANT DAM 4 RETURN FLOW TRANSMITTER	F75PW606/607/608 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75264	FQI75264	FQI	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 1 FLOW TRANSMITTER	F75PS559 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75265	FQI75265	FQI	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP FLOW TRANSMITTER	F75PS593 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75365	FQI75365	FQI	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 2 FLOW TRANSMITTER	F75PS560 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75730	FQI75730	FQI	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 1 FLOW TRANSMITTER	F75PS801 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75731	FQI75731	FQI	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 2 FLOW TRANSMITTER	F75PS802 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75732	FQI75732	FQI	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 3 FLOW TRANSMITTER	F75PS803 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75733	FQI75733	FQI	2000-F75-PID-PR-40008	FIM III SEEPAGE PUMP 4 FLOW TRANSMITTER	F75PS804 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75793	FQI75793	FQI	2000-F75-PID-PR-40009	FIM III SEEPAGE WATER FLOW TRANSMITTER	FIM III SEEPAGE TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75785	FQI75785	FQI	2000-F75-PID-PR-40011	FIM III TOE DRAIN PUMP FLOW TRANSMITTER	F75PS558 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75789	FQI75789	FQI	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 FLOW TRANSMITTER	F75PS556 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75790	FQI75790	FQI	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 FLOW TRANSMITTER	F75PS557 TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75792	FQI75792	FQI	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE FLOW TRANSMITTER	FIM III U/DRAIN TOTAL FLOW		m³	x	x	x	MOSCAD RTU		1
FIT75794	FQI75794	FQI	2000-F175-PID-PR-40002	FIM III SEEPAGE WATER FLOW TRANSMITTER 2	FIM III SEEPAGE TOTAL FLOW 2		m³	x	x	x	MOSCAD RTU		1
FIT17503	FQI17503	FQI	2000-F175-PID-PR-40002	BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER	F75TK651 TOTAL INFLOW		m³	x	x	x	MOSCAD RTU		1

OPEN-CLOSE VALVES

Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Valve Detail	Fail Position	Manual	Auto	Hold	Open Condition (Auto Mode)	Close Condition (Auto Mode)	Condition	Interlocks								Control System	Comments	Rev
												Description	Delay On	Delay Off	Open Permissive	Close Permissive	Open Trip	Close Trip	Bypass	Reset Required			
XV23700	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH ISOLATION VALVE 1	FIM III LINE 1N VLV 1	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23701	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH ISOLATION VALVE 2	FIM III LINE 1N VLV 2	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23702	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH & SOUTH ISOLATION VALVE	FIM III LINE 1N & 1S VLV	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23703	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH ISOLATION VALVE 1	FIM III LINE 1S VLV 1	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23704	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH ISOLATION VALVE 2	FIM III LINE 1S VLV 2	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23705	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH FLUSHING WATER VALVE	FIM III LINE 1S FLUSH	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23706	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH FLUSHING WATER VALVE	FIM III LINE 1N FLUSH	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23707	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 1	FIM III LINE 2W VLV 1	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23708	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 1	FIM III LINE 2E VLV 1	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23709	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 2	FIM III LINE 2W VLV 2	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23710	2000-F23-PID-PR-40005	FIM III LINE 2 WEST ISOLATION VALVE 3	FIM III LINE 2W VLV 3	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23711	2000-F23-PID-PR-40005	FIM III LINE 2 EAST & WEST ISOLATION VALVE	FIM III LINE 2E & 2W VLV	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23712	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 2	FIM III LINE 2E VLV 2	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23713	2000-F23-PID-PR-40005	FIM III LINE 2 EAST ISOLATION VALVE 3	FIM III LINE 2E VLV 3	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23714	2000-F23-PID-PR-40005	FIM III LINE 2 EAST FLUSHING WATER VALVE	FIM III LINE 2E FLUSH	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23715	2000-F23-PID-PR-40005	FIM III LINE 2 WEST FLUSHING WATER VALVE	FIM III LINE 2W FLUSH	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23716	2000-F23-PID-PR-40005	FIM III LINE 3 WEST ISOLATION VALVE 1	FIM III LINE 3W VLV 1	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23717	2000-F23-PID-PR-40005	FIM III LINE 3 WEST ISOLATION VALVE 2	FIM III LINE 3W VLV 2	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23718	2000-F23-PID-PR-40005	FIM III LINE 3 EAST & WEST ISOLATION VALVE	FIM III LINE 3E & 3W VLV	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23719	2000-F23-PID-PR-40005	FIM III LINE 3 EAST ISOLATION VALVE 1	FIM III LINE 3E VLV 1	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23720	2000-F23-PID-PR-40005	FIM III LINE 3 EAST ISOLATION VALVE 2	FIM III LINE 3E VLV 2	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23721	2000-F23-PID-PR-40005	FIM III LINE 3 EAST FLUSHING WATER VALVE	FIM III LINE 3E FLUSH	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV23722	2000-F23-PID-PR-40005	FIM III LINE 3 WEST FLUSHING WATER VALVE	FIM III LINE 3W FLUSH	Detail 5	FL	x	x		As per F23-SEQ-4001 to F23-SEQ- 4060	As per F23-SEQ-4001 to F23-SEQ- 4060			5								MOSCAD RTU		4
XV75340	2000-F75-PID-PR-40004	DECANT DAM 4 ~~FLUSHING WATER~~ DECANT RETURN ISOLATION VALVE	DD4 ~~FLUSHING~~ DECANT RETURN	Detail 14	FL	x	x			As per F23-SEQ-2003 As per F23-SEQ-2006			5								MOSCAD RTU		5
XV75283	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 1	F75PS593 SUC VLV 1	Detail 14	FC	x	x				F75-PS-593 RUN	F75-PS-593 RUN	5				x		x		MOSCAD RTU		4
XV75284	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP SUCTION VALVE 2	F75PS593 SUC VLV 2	Detail 14	FC	x	x				F75-PS-593 RUN	F75-PS-593 RUN	5				x		x		MOSCAD RTU		4
XV75795	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 SUCTION VALVE 1	F75PS556 SUC VLV 1	Detail 14	FC	x	x				F75-PS-556 RUN	F75-PS-556 RUN	5				x		x		MOSCAD RTU		4
XV75797	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1 SUCTION VALVE 2	F75PS556 SUC VLV 2	Detail 14	FC	x	x				F75-PS-556 RUN	F75-PS-556 RUN	5				x		x		MOSCAD RTU		4
XV75796	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 SUCTION VALVE 1	F75PS557 SUC VLV 1	Detail 14	FC	x	x				F75-PS-557 RUN	F75-PS-557 RUN	5				x		x		MOSCAD RTU		4
XV75798	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2 SUCTION VALVE 2	F75PS557 SUC VLV 2	Detail 14	FC	x	x				F75-PS-557 RUN	F75-PS-557 RUN	5				x		x		MOSCAD RTU		4

CONTROL VALVES

Tag Name	P&ID	Description	Abbreviated Description (Faceplate)	Valve Type	Fail Position	Manual	Auto	Hold	Open Condition (Auto Mode)	Close Condition (Auto Mode)	Condition	Interlocks									Control System	Comments	Rev
												Description	Delay On	Delay Off	Open Permissive	Close Permissive	Open Trip	Close Trip	Bypass	Reset Required			
PCV17504	2000-F175-PID-PR-40002	EASTERN BOREFIELDS SALINE DAM PUMP PRESSURE CONTROL VALVE	SEGDT01PP1/PP2/PP3 PRESSURE CONTROL	Detail 13	N/A	x	x														MOSCAD RTU		1

PID CONTROLLERS

Tag Source	Tag Name	P&ID	Abbreviated Description (Faceplate)	Hold	Normal Operating Mode	Modes			Process Value Tag	Setpoint Variable (SP)				Range			Process Switchpoints				Alarms								Alarm Masking	Manipulated Variable (MV)							Control System	Comment	Rev
						Manual	Auto	Cascade		Min	Max	Nom / Peak	Unit	Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_LL Delay (sec)	A_L	A_Lo Delay (sec)	A_H	A_Hi Delay (sec)	A_HH	A_HH Delay (sec)		Range Min	Range Max	Lower Limit	Uper Limit	Unit	Destination	Rev Acting			
~~LIT75268~~	~~LIC75268~~	~~2000-F75-PID-PR-40001~~	~~FIM 8 CELL G LEVEL~~		~~Automatic~~	~~X~~	~~X~~			~~70~~	~~100~~	~~90~~	~~%~~	~~0~~	~~100~~	~~%~~	~~-~~	~~-~~	~~-~~	~~-~~	~~5~~	~~5~~	~~10~~	~~5~~	~~99~~	~~5~~	~~-~~	~~-~~	~~-~~	~~0~~	~~100~~	~~0~~	~~100~~	~~% spd~~	~~F75-PW-664~~	~~Yes~~	~~MOSCAD RTU~~	~~A_H = Warning A_L = Warning A_LL = Trip F75-PW-664~~	~~5~~
LIT75332	LIC75332	2000-F75-PID-PR-40004	DECANT DAM 4 LEVEL		Automatic	X	X			50	100	75	%	0	100	%	-	-	-	-	5	5	10	5	99	5	-	-	-	0	100	0	100	% spd	F75-PW-606 / F75-PW-607 / F75-PW-608	No	MOSCAD RTU	A_H = Warning A_L = Warning A_LL = Trip F75-PW-606/607/608, Trip F75-PV-414	1
FIT75333	FIC75333	2000-F75-PID-PR-40004	F75PW606/607/608 FLOWRATE		Automatic	X	X			900	1,100	1,000	m³/h	0	1,500	m³/h	-	-	-	-	-	-	20	5	-	-	-	-	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	0	100	0	100	% spd	F75-PW-606 / F75-PW-607 / F75-PW-608	No	MOSCAD RTU	2 out of 3 pumps required for decant return above 1,100 m^3/h, below this 1 pump required. 2 out of 3 pumps required for flushing Controller output to be limited to not exceed PIT75338 A_H in flushing mode. Controller output to be limited to not exceed PIT75339 A_H in decant return mode A_L = Warning	1
PIT17504	PIC17504	2000-F175-PID-PR-40002	SEGDT01PP1/PP2/PP3 PRESSURE		Automatic	X	X			300	500	400	kPa	0	1,000	kPa	-	-	-	-	-	-	200	5	600	5	800	5	SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN	0	100	0	100	% open	PCV17504	No	MOSCAD RTU	A_HH = Trip SEGDT01PP1/PP2/PP3 A_H = Warning A_L = Warning	1

MISCELLANEOUS CONTROLLERS

Tag Name	P&ID	Abbreviated Description	Hold	Normal State	Alarm State	Hardwired (y/n)	Alarm Delay (secs)	Alarm Masking Conditions	Control System	Comment	Rev
~~HS75334~~	~~2000-F75-PID-PR-40004~~	~~F75PW606/607/608 SPEED REF SEL~~		~~Operator Selection~~	~~No Alarm~~	~~N~~	~~-~~	~~-~~	~~MOSCAD RTU~~	~~HS75333 Toggle Switch to select between the two modes of control of the speed of F75-PW-606/607/608.~~ ~~ON = Output from LIC75332~~ ~~OFF = Output from FIC75333~~	~~5~~
HS75333	2000-F75-PID-PR-40004	F75PW606/607/608 SPEED REF SEL		Operator Selection	No Alarm	N	-	-	MOSCAD RTU	HS75333 Toggle Switch to select between the two modes of control of the speed of F75-PW-606/607/608. ON = Output from LIC75332 OFF = Output from FIC75333	4
KY75730	2000-F75-PID-PR-40008	F75PS801 RESTART TIMER		Timer PT = 1800 Seconds	No Alarm	N	-	-	MOSCAD RTU	Timer to delay the automatic restart of F75-PS-801 Start 1,800 second timer when F75-PS-801 = NOT RUNNING when tripped by FIT75730 < A_LL If on restart FIT75730 < A_LL condition remains, restart timer. Delay time to be configurable from PCS with engineer access.	1
KY75731	2000-F75-PID-PR-40008	F75PS802 RESTART TIMER		Timer PT = 1800 Seconds	No Alarm	N	-	-	MOSCAD RTU	Timer to delay the automatic restart of F75-PS-802 Start 1,800 second timer when F75-PS-802 = NOT RUNNING when tripped by FIT75731 < A_LL If on restart FIT75731 < A_LL condition remains, restart timer. Delay time to be configurable from PCS with engineer access.	1
KY75732	2000-F75-PID-PR-40008	F75PS803 RESTART TIMER		Timer PT = 1800 Seconds	No Alarm	N	-	-	MOSCAD RTU	Timer to delay the automatic restart of F75-PS-803 Start 1,800 second timer when F75-PS-803 = NOT RUNNING when tripped by FIT75732 < A_LL If on restart FIT75732 < A_LL condition remains, restart timer. Delay time to be configurable from PCS with engineer access.	1
KY75733	2000-F75-PID-PR-40008	F75PS804 RESTART TIMER		Timer PT = 1800 Seconds	No Alarm	N	-	-	MOSCAD RTU	Timer to delay the automatic restart of F75-PS-804 Start 1,800 second timer when F75-PS-804 = NOT RUNNING when tripped by FIT75733 < A_LL If on restart FIT75733 < A_LL condition remains, restart timer. Delay time to be configurable from PCS with engineer access.	1

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks: Description (Unhealthy Interlocks and Healthy Start Permissives)	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissive	Process	Critical	Hardwired to Drive	Bypass	Reset Required	Control System	Comments	Rev
F23-HY-421	2000-F23-PID-PR-40005	FIM3 VALVE STATION HPU	DOL							13.4	X	X														MOSCAD RTU		4
															1	F23-HY-421 Comms Failure	-	0				x						4
															2	F23-HY-421 Fail to Start / Stop	-	0				x						4
															3	F23-HY-421 Drive Fault	-	0				x						4
															4	F23-HY-421 E-Stop	-	0				x	x					4
															5	F23-HY-421 MCC Unhealthy	-	0				x						4
															6	F23-HY-421 Drive Out of Service	-	0				x						4
															7	F23-HY-421 Drive Ready	-	0				x						4
F75-PW-664	2000-F75-PID-PR-40001	FIM II CELL G DECANT PUMP	D		0	100	%	40	100	N/A	X	X														MOSCAD RTU		5
															1	LIT75675 > A_HH	x	5			x			x			Decant Dam 3 Overflow Pond level high high	5
															~~1~~	~~F75-PW-664 Comms Failure~~	~~-~~	~~0~~				~~x~~						5
															~~2~~	~~F75-PW-664 Fail to Start / Stop~~	~~-~~	~~0~~				~~x~~						5
															~~3~~	~~F75-PW-664 Drive Fault~~	~~-~~	~~0~~				~~x~~						5
															~~4~~	~~F75-PW-664 E-Stop~~	~~-~~	~~0~~				~~x~~	~~x~~					5
															~~5~~	~~F75-PW-664 MCC Unhealthy~~	~~-~~	~~0~~				~~x~~						5
															~~6~~	~~F75-PW-664 Drive Out of Service~~	~~-~~	~~0~~				~~x~~						5
															~~7~~	~~F75-PW-664 Drive Ready~~	~~-~~	~~0~~				~~x~~						5
															~~8~~	~~LAH75271 ACTIVE~~	~~x~~	~~5~~		~~x~~				~~x~~			~~Suction line primed~~	5
															9	LIT75268 < A_L	x	5			x			x			FIM II Cell G level low ~~low~~	5
															10	FI75269 < A_L AND F75-PW-664 RUNNING	x	30			x			x			Decant pump discharge flow low ~~low~~	5
															11	FDI75270 < A_L	x	600			x			x			FIM II Cell G decant/underdrainage to Decant Dam 3 differential flow low	5
															12	FDI75270 > A_H	x	600			x			x			FIM II Cell G decant/underdrainage to Decant Dam 3 differential flow high	5
~~F75-PV-413~~	~~2000-F75-PID-PR-40001~~	~~FIM II CELL G DECANT VACUUM PUMP~~	~~DOL~~							~~4.27~~	~~X~~	~~X~~	~~F75-PW-664 RUN~~	~~LAH75271 ACTIVE~~													~~MOSCAD RTU~~		5
															~~1~~	~~F75-PV-413 Comms Failure~~	~~-~~	~~0~~				~~x~~						5
															~~2~~	~~F75-PV-413 Fail to Start / Stop~~	~~-~~	~~0~~				~~x~~						5
															~~3~~	~~F75-PV-413 Drive Fault~~	~~-~~	~~0~~				~~x~~						5
															~~4~~	~~F75-PV-413 E-Stop~~	~~-~~	~~0~~				~~x~~	~~x~~					5
															~~5~~	~~F75-PV-413 MCC Unhealthy~~	~~-~~	~~0~~				~~x~~						5
															~~6~~	~~F75-PV-413 Drive Out of Service~~	~~-~~	~~0~~				~~x~~						5
															~~7~~	~~F75-PV-413 Drive Ready~~	~~-~~	~~0~~				~~x~~						5
F75-PW-600	2000-F75-PID-PR-40002	FIM III CELL 1 DECANT PUMP	D							N/A																MOSCAD RTU		4
															1	LIT75362 > A_HH	x	5			x			x			Decant Dam 4 Overflow Pond level high high	4
															2	LIT75318 < A_L	x	5			x			x			FIM III Cell 1 level low	5
															3	FIT75319 < A_L AND F75-PW-600 RUNNING	x	30			x			x			FIM III Cell 1 decant pump discharge flow low	5
															4	FDI75329 < A_L	x	600			x			x			FIM III decant to Decant Dam 4 differential flow low	5
															5	FDI75329 > A_H	x	600			x			x			FIM III decant to Decant Dam 4 differential flow high	5
F75-PW-601	2000-F75-PID-PR-40003	FIM III CELL 2 DECANT PUMP	D							N/A																MOSCAD RTU		4
															1	LIT75362 > A_HH	x	5			x			x			Decant Dam 4 Overflow Pond level high high	4
															2	LIT75325 < A_L	x	5			x			x			FIM III Cell 2 level low	5
															3	FIT75326 < A_L AND F75-PW-601 RUNNING	x	30			x			x			FIM III Cell 2 decant pump discharge flow low	5
															4	FDI75329 < A_L	x	600			x			x			FIM III decant to Decant Dam 4 differential flow low	5
															5	FDI75329 > A_H	x	600			x			x			FIM III decant to Decant Dam 4 differential flow high	5
F75-PW-602	2000-F75-PID-PR-40003	FIM III CELL 3 DECANT PUMP	D							N/A																MOSCAD RTU		4
															1	LIT75362 > A_HH	x	5			x			x			Decant Dam 4 Overflow Pond level high high	4
															2	LIT75327 < A_L	x	5			x			x			FIM III Cell 3 level low	5
															3	FIT75328 < A_L AND F75-PW-602 RUNNING	x	30			x			x			FIM III Cell 3 decant pump discharge flow low	5
															4	FDI75329 < A_L	x	600			x			x			FIM III decant to Decant Dam 4 differential flow low	5
															5	FDI75329 > A_H	x	600			x			x			FIM III decant to Decant Dam 4 differential flow high	5

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks										Control System	Comments	Rev
																Description (Unhealthy Interlocks and Healthy Start Permissives	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissive	Process	Critical	Hardwired to Drive	Bypass	Reset Required			
F75-PW-606	2000-F75-PID-PR-40004	DECANT DAM 4 PUMP 1	VSD		0	100	%	40	100	721	X	X	MASTER TO 1 DUTY/2 STANDBY LOGIC AUTO START IF NOT RUNNING/AVAILBLE WHEN HS75333/4 is active.	MASTER TO 1 DUTY/2 STANDBY LOGIC REQUEST TO STOP FROM PLANT												MOSCAD RTU	Pump max speed to be limited to 1,400 rpm	1
															1	F75-PW-606 Comms Failure	-	0				x					1
															2	F75-PW-606 Fail to Start / Stop	-	0				x					1
															3	F75-PW-606 Drive Fault	-	0				x					1
															4	F75-PW-606 E-Stop	-	0				x	x				1
															5	F75-PW-606 MCC Unhealthy	-	0				x					1
															6	F75-PW-606 Drive Out of Service	-	0				x					1
															7	F75-PW-606 Drive Ready	-	0				x					1
															8	LAH75363 ACTIVE	x	5		x				x			Suction line primed	4
															9	F75-PW-607 RUN AND F75-PW-608 RUN	x	5			x			x			Standby pump	4
															10	ZIC75330 ACTIVE OR ZIO75330 INACTIVE AND ZIC75331 ACTIVE OR ZIO75331 INACTIVE AND ZIC75341 ACTIVE OR ZIO75341 INACTIVE	x	30			x			x			Suction valves closed	4
															11	ZIO75341 ACTIVE AND ZIC75330 ACTIVE AND ZIC75331 ACTIVE AND F75-PW-607 RUN OR F75-PW-608 RUN	x	30			x			x			Only 1 pump can operate when pumping from overflow pond only	4
															12	FAL75335 ACTIVE AND F75-PW-606 RUNNING	x	30			x			x			Decant pump flow switch	4
															13	PI75338 > A_HH	x	30			x			x			Decant flushing pressure high high	4
															14	PI75339 > A_HH	x	30			x			x			Decant return pressure high high	4
															15	LI7525 > A_HH AND LI7526 > A_HH	x	5			x			x			North Dam level high high and South Dam level high high. **Bypass when HS75333 is OFF.**	**5**
															16	LIT75332 < A_LL **AND LIT75362 < A_LL**	x	5			x			x			Decant Dam 4 level low low **and Overflow Pond level low low**	**5**
															17	LI75362 < A_LL AND ZIO75341 ACTIVE	x	5			x			x			Overflow Pond level low low	4
															18	**FDI75611 < A_L**	**x**	**600**			**x**			**x**			**Decant Dam 4 water return differential flow low**	**5**
															19	**FDI75611 > A_H**	**x**	**600**			**x**			**x**			**Decant Dam 4 water return differential flow high**	**5**
F75-PW-607	2000-F75-PID-PR-40004	DECANT DAM 4 PUMP 2	VSD		0	100	%	40	100	721	X	X														MOSCAD RTU	Pump max speed to be limited to 1,400 rpm	1
															1	F75-PW-607 Comms Failure	-	0				x					1
															2	F75-PW-607 Fail to Start / Stop	-	0				x					1
															3	F75-PW-607 Drive Fault	-	0				x					1
															4	F75-PW-607 E-Stop	-	0				x	x				1
															5	F75-PW-607 MCC Unhealthy	-	0				x					1
															6	F75-PW-607 Drive Out of Service	-	0				x					1
															7	F75-PW-607 Drive Ready	-	0				x					1
															8	LAH75363 ACTIVE	x	5		x				x			Suction line primed	4
															9	F75-PW-606 RUN AND F75-PW-608 RUN	x	5			x			x			Standby pump	4
															10	ZIC75330 ACTIVE OR ZIO75330 INACTIVE AND ZIC75331 ACTIVE OR ZIO75331 INACTIVE AND ZIC75341 ACTIVE OR ZIO75341 INACTIVE	x	30			x			x			Suction valves closed	4
															11	ZIO75341 ACTIVE AND ZIC75330 ACTIVE AND ZIC75331 ACTIVE AND F75-PW-606 RUN OR F75-PW-608 RUN	x	30			x			x			Only 1 pump can operate when pumping from overflow pond only	4
															12	FAL75336 ACTIVE AND F75-PW-607 RUNNING	x	30			x			x			Decant pump flow switch	4
															13	PI75338 > A_HH	x	30			x			x			Decant flushing pressure high high	4
															14	PI75339 > A_HH	x	30			x			x			Decant return pressure high high	4
															15	LI7525 > A_HH AND LI7526 > A_HH	x	5			x			x			North Dam level high high and South Dam level high high. **Bypass when HS75333 is OFF.**	**5**
															16	LIT75332 < A_LL **AND LIT75362 < A_LL**	x	5			x			x			Decant Dam 4 level low low **and Overflow Pond level low low**	**5**
															17	LI75362 < A_LL AND ZIO75341 ACTIVE	x	5			x			x			Overflow Pond level low low	4
															18	**FDI75611 < A_L**	**x**	**600**			**x**			**x**			**Decant Dam 4 water return differential flow low**	**5**
															19	**FDI75611 > A_H**	**x**	**600**			**x**			**x**			**Decant Dam 4 water return differential flow high**	**5**
F75-PW-608	2000-F75-PID-PR-40004	DECANT DAM 4 PUMP 3	VSD		0	100	%	40	100	721	X	X														MOSCAD RTU	Pump max speed to be limited to 1,400 rpm	1
															1	F75-PW-608 Comms Failure	-	0				x					1
															2	F75-PW-608 Fail to Start / Stop	-	0				x					1
															3	F75-PW-608 Drive Fault	-	0				x					1
															4	F75-PW-608 E-Stop	-	0				x	x				1
															5	F75-PW-608 MCC Unhealthy	-	0				x					1
															6	F75-PW-608 Drive Out of Service	-	0				x					1
															7	F75-PW-608 Drive Ready	-	0				x					1
															8	LAH75363 ACTIVE	x	5		x				x			Suction line primed	4
															9	F75-PW-606 RUN AND F75-PW-607 RUN	x	5			x			x			Standby pump	4
															10	ZIC75330 ACTIVE OR ZIO75330 INACTIVE AND ZIC75331 ACTIVE OR ZIO75331 INACTIVE AND ZIC75341 ACTIVE OR ZIO75341 INACTIVE	x	30			x			x			Suction valves closed	4
															11	ZIO75341 ACTIVE AND ZIC75330 ACTIVE AND ZIC75331 ACTIVE AND F75-PW-606 RUN OR F75-PW-607 RUN	x	30			x			x			Only 1 pump can operate when pumping from overflow pond only	4
															12	FAL75337 ACTIVE AND F75-PW-608 RUNNING	x	30			x			x			Decant pump flow switch	4
															13	PI75338 > A_HH	x	30			x			x			Decant flushing pressure high high	4

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks: Description (Unhealthy Interlocks and Healthy Start Permissives)	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissive	Process	Critical	Hardwired to Drive	Bypass	Reset Required	Control System	Comments	Rev
															14	PI75339 > A_HH	x	30			x			x			Decant return pressure high high	4
															15	LI7525 > A_HH AND LI7526 > A_HH	x	5			x			x			North Dam level high high and South Dam level high high. **Bypass when HS75333 is OFF.**	**5**
															16	LIT75332 < A_LL **AND LIT75362 < A_LL**	x	5			x			x			Decant Dam 4 level low low **and Overflow Pond level low low**	**5**
															17	LI75362 < A_LL AND ZIO75341 ACTIVE	x	5			x			x			Overflow Pond level low low	4
															18	**FDI75611 < A_L**	**x**	**600**			**x**			**x**			**Decant Dam 4 water return differential flow low**	**5**
															19	**FDI75611 > A_H**	**x**	**600**			**x**			**x**			**Decant Dam 4 water return differential flow high**	**5**
F75-PV-414	2000-F75-PID-PR-40004	DECANT DAM 4 VACUUM PUMP	DOL							4.27	X	X	F75-PW-606 RUN OR F75-PW-607 RUN OR F75-PW-608 RUN	LAH75363 ACTIVE												MOSCAD RTU		1
															1	F75-PV-414 Comms Failure	-	0				x							1
															2	F75-PV-414 Fail to Start / Stop	-	0				x							1
															3	F75-PV-414 Drive Fault	-	0				x							1
															4	F75-PV-414 E-Stop	-	0				x	x						1
															5	F75-PV-414 MCC Unhealthy	-	0				x							1
															6	F75-PV-414 Drive Out of Service	-	0				x							1
															7	F75-PV-414 Drive Ready	-	0				x							1
															8	LIT75332 < A_LL **AND LIT75362 < A_LL**	x	5			x			x			Decant Dam 4 level low low **and Overflow Pond level low low**	**5**
															9	LIT75362 < A_LL AND ZIO75341 ACTIVE	x	5			x			x			Overflow Pond level low low	4
															10	**ZIO75330 ACTIVE OR ZIO75331 ACTIVE OR ZIO75341 ACTIVE**	**x**	**5**			**x**			**x**			**Minimum of one suction line open**	**5**
F75-PS-593	2000-F75-PID-PR-40007	FIM II CELL G UNDERDRAINAGE PUMP	DOL							39	X	X	LI75265 > P_H	LI75265 < P_L												MOSCAD RTU		1
															1	F75-PS-593 Comms Failure	-	0				x							1
															2	F75-PS-593 Fail to Start / Stop	-	0				x							1
															3	F75-PS-593 Drive Fault	-	0				x							1
															4	F75-PS-593 E-Stop	-	0				x	x						1
															5	F75-PS-593 MCC Unhealthy	-	0				x							1
															6	F75-PS-593 Drive Out of Service	-	0				x							1
															7	F75-PS-593 Drive Ready	-	0				x							1
															8	FI75265 < A_LL AND F75-PS-593 RUNNING	x	30			x			x			Underdrainage pump flow low low	4
															9	LI75265 < A_LL	x	5			x			x			Underdrainage pit level low low	4
															10	**FDI75270 < A_L**	**x**	**600**			**x**			**x**			**FIM II Cell G decant/underdrainage to Decant Dam 3 differential flow low**	**5**
															11	**FDI75270 > A_H**	**x**	**600**			**x**			**x**			**FIM II Cell G decant/underdrainage to Decant Dam 3 differential flow high**	**5**
F75-PS-559	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 1	DOL							13.4	X	X	LI75264 > P_H	LI75264 < P_L												MOSCAD RTU		1
															1	F75-PS-559 Comms Failure	-	0				x							1
															2	F75-PS-559 Fail to Start / Stop	-	0				x							1
															3	F75-PS-559 Drive Fault	-	0				x							1
															4	F75-PS-559 E-Stop	-	0				x	x						1
															5	F75-PS-559 MCC Unhealthy	-	0				x							1
															6	F75-PS-559 Drive Out of Service	-	0				x							1
															7	F75-PS-559 Drive Ready	-	0				x							1
															8	FI75264 < A_LL AND F75-PS-559 RUNNING	x	30			x			x			Toe drain pump 1 flow low low	4
															9	LI75264 < A_LL	x	5			x			x			Toe drain pit level low low	4
F75-PS-560	2000-F75-PID-PR-40007	FIM II CELL G TOE DRAIN PUMP 2	DOL							13.4	X	X	LI75365 > P_H	LI75365 < P_L												MOSCAD RTU		1
															1	F75-PS-560 Comms Failure	-	0				x							1
															2	F75-PS-560 Fail to Start / Stop	-	0				x							1
															3	F75-PS-560 Drive Fault	-	0				x							1
															4	F75-PS-560 E-Stop	-	0				x	x						1
															5	F75-PS-560 MCC Unhealthy	-	0				x							1
															6	F75-PS-560 Drive Out of Service	-	0				x							1
															7	F75-PS-560 Drive Ready	-	0				x							1
															8	FI75365 < A_LL AND F75-PS-560 RUNNING	x	30			x			x			Toe drain pump 2 flow low low	4
															9	LI75365 < A_LL	x	5			x			x			Toe drain pit level low low	4
F75-PS-801	2000-F75-PID-PR-40009	FIM III SEEPAGE PUMP 1	DOL							4.27	X	X	KY75730													MOSCAD RTU		1
															1	F75-PS-801 Comms Failure	-	0				x							1
															2	F75-PS-801 Fail to Start / Stop	-	0				x							1
															3	F75-PS-801 Drive Fault	-	0				x							1
															4	F75-PS-801 E-Stop	-	0				x	x						1
															5	F75-PS-801 MCC Unhealthy	-	0				x							1
															6	F75-PS-801 Drive Out of Service	-	0				x							1
															7	F75-PS-801 Drive Ready	-	0				x							1

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks: Description (Unhealthy Interlocks and Healthy Start Permissives)	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissive	Process	Critical	Hardwired to Drive	Bypass	Reset Required	Control System	Comments	Rev
															8	FI75730 < A_LL AND F75-PS-801 RUNNING	x	30			x			x			Seepage pump 1 flow low low	4
															9	PI75730 > A_HH	x	5			x			x			Seepage pump 1 pressure high high	4
															10	**FDI75793 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															11	**FDI75793 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
															12	**FDI75794 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															13	**FDI75794 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
F75-PS-802	2000-F75-PID-PR-40009	FIM III SEEPAGE PUMP 2	DOL							4.27	X	X	KY75731												MOSCAD RTU		1
															1	F75-PS-802 Comms Failure	-	0				x						1
															2	F75-PS-802 Fail to Start / Stop	-	0				x						1
															3	F75-PS-802 Drive Fault	-	0				x						1
															4	F75-PS-802 E-Stop	-	0				x	x					1
															5	F75-PS-802 MCC Unhealthy	-	0				x						1
															6	F75-PS-802 Drive Out of Service	-	0				x						1
															7	F75-PS-802 Drive Ready	-	0				x						1
															8	FI75731 < A_LL AND F75-PS-802 RUNNING	x	30			x			x			Seepage pump 2 flow low low	4
															9	PI75731 > A_HH	x	5			x			x			Seepage pump 2 pressure high high	4
															10	**FDI75793 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															11	**FDI75793 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
															12	**FDI75794 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															13	**FDI75794 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
F75-PS-803	2000-F75-PID-PR-40009	FIM III SEEPAGE PUMP 3	DOL							4.27	X	X	KY75732												MOSCAD RTU		1
															1	F75-PS-803 Comms Failure	-	0				x						1
															2	F75-PS-803 Fail to Start / Stop	-	0				x						1
															3	F75-PS-803 Drive Fault	-	0				x						1
															4	F75-PS-803 E-Stop	-	0				x	x					1
															5	F75-PS-803 MCC Unhealthy	-	0				x						1
															6	F75-PS-803 Drive Out of Service	-	0				x						1
															7	F75-PS-803 Drive Ready	-	0				x						1
															8	FI75732 < A_LL AND F75-PS-803 RUNNING	x	30			x			x			Seepage pump 3 flow low low	4
															9	PI75732 > A_HH	x	5			x			x			Seepage pump 3 pressure high high	4
															10	**FDI75793 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															11	**FDI75793 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
															12	**FDI75794 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															13	**FDI75794 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
F75-PS-804	2000-F75-PID-PR-40009	FIM III SEEPAGE PUMP 4	DOL							4.27	X	X	KY75733												MOSCAD RTU		1
															1	F75-PS-804 Comms Failure	-	0				x						1
															2	F75-PS-804 Fail to Start / Stop	-	0				x						1
															3	F75-PS-804 Drive Fault	-	0				x						1
															4	F75-PS-804 E-Stop	-	0				x	x					1
															5	F75-PS-804 MCC Unhealthy	-	0				x						1
															6	F75-PS-804 Drive Out of Service	-	0				x						1
															7	F75-PS-804 Drive Ready	-	0				x						1
															8	FI75733 < A_LL AND F75-PS-804 RUNNING	x	30			x			x			Seepage pump 4 flow low low	4
															9	PI75733 > A_HH	x	5			x			x			Seepage pump 4 pressure high high	4
															10	**FDI75793 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															11	**FDI75793 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
															12	**FDI75794 < A_L**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow low**	5
															13	**FDI75794 > A_H**	x	600			x			x			**FIM III seepage to Eastern Borefields Saline Dam differential flow high**	5
F75-PS-558	2000-F75-PID-PR-40011	FIM III TOE DRAIN PUMP	DOL							13.4	X	X	LI75785 > P_H	LI75785 < P_L												MOSCAD RTU		1
															1	F75-PS-558 Comms Failure	-	0				x						1
															2	F75-PS-558 Fail to Start / Stop	-	0				x						1
															3	F75-PS-558 Drive Fault	-	0				x						1
															4	F75-PS-558 E-Stop	-	0				x	x					1
															5	F75-PS-558 MCC Unhealthy	-	0				x						1
															6	F75-PS-558 Drive Out of Service	-	0				x						1
															7	F75-PS-558 Drive Ready	-	0				x						1
															8	FI75785 < A_LL AND F75-PS-558 RUNNING	x	30			x			x			Toe drain pump flow low low	4
															9	LI75785 < A_LL	x	5			x			x			Toe drain pit level low low	4

DRIVE INTERLOCKS AND CONTROL

Tag Name	P&ID	Name	Drive Type	Hold	Speed Min	Speed Max	Speed Units	SP Lower Limit	SP Upper Limit	Current Max	Manual	Auto	Start Condition (Auto Mode)	Stop Condition (Auto Mode)	#	Interlocks									Control System	Comments	Rev
																Description (Unhealthy Interlocks and Healthy Start Permissives	Soft Tag Inter FCS comms	Delay On	Delay Off	Start Permissive	Process	Critical	Hardwired to Drive	Bypass	Reset Required			
															10	FDI75793 < A_L	x	600			x			x			FIM III seepage to Eastern Borefields Saline Dam differential flow low	5
															11	FDI75793 > A_H	x	600			x			x			FIM III seepage to Eastern Borefields Saline Dam differential flow high	5
															12	FDI75794 < A_L	x	600			x			x			FIM III seepage to Eastern Borefields Saline Dam differential flow low	5
															13	FDI75794 > A_H	x	600			x			x			FIM III seepage to Eastern Borefields Saline Dam differential flow high	5
F75-PS-556	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 1	DOL							39	X	X	LI75789 > P_H	LI75789 < P_L											MOSCAD RTU		1
															1	F75-PS-556 Comms Failure	-	0				x						1
															2	F75-PS-556 Fail to Start / Stop	-	0				x						1
															3	F75-PS-556 Drive Fault	-	0				x						1
															4	F75-PS-556 E-Stop	-	0				x	x					1
															5	F75-PS-556 MCC Unhealthy	-	0				x						1
															6	F75-PS-556 Drive Out of Service	-	0				x						1
															7	F75-PS-556 Drive Ready	-	0				x						1
															8	LI75362 > A_HH	x	5			x			x			Decant Dam 4 Overflow Pond level high high	4
															9	FI75789 < A_LL AND F75-PS-556 RUNNING	x	30			x			x			Underdrainage pump 1 flow low low	4
															10	LI75789 < A_LL	x	5			x			x			Underdrainage pit level low low	4
															11	FDI75792 < A_L	x	600			x			x			FIM III underdrainage to Decant Dam 4 differential flow low	5
															12	FDI75792 > A_H	x	600			x			x			FIM III underdrainage to Decant Dam 4 differential flow high	5
F75-PS-557	2000-F75-PID-PR-40012	FIM III UNDERDRAINAGE PUMP 2	DOL							39	X	X	LI75790 > P_H	LI75790 < P_L											MOSCAD RTU		1
															1	F75-PS-557 Comms Failure	-	0				x						1
															2	F75-PS-557 Fail to Start / Stop	-	0				x						1
															3	F75-PS-557 Drive Fault	-	0				x						1
															4	F75-PS-557 E-Stop	-	0				x	x					1
															5	F75-PS-557 MCC Unhealthy	-	0				x						1
															6	F75-PS-557 Drive Out of Service	-	0				x						1
															7	F75-PS-557 Drive Ready	-	0				x						1
															8	LI75362 > A_HH	x	5			x			x			Decant Dam 4 Overflow Pond level high high	4
															9	FI75790 < A_LL AND F75-PS-557 RUNNING	x	30			x			x			Underdrainage pump 2 flow low low	4
															10	LI75790 < A_LL	x	5			x			x			Underdrainage pit level low low	4
															11	FDI75792 < A_L	x	600			x			x			FIM III underdrainage to Decant Dam 4 differential flow low	5
															12	FDI75792 > A_H	x	600			x			x			FIM III underdrainage to Decant Dam 4 differential flow high	5

CALCULATIONS

Tag Source	Tag Name	P&ID	Description (Software Tag Only)	Unit	Hold	Inputs #	Inputs Tag	Inputs Unit	Inputs Description	Calc. Value	x = Operator Input * = Engineer Input	Min	Max	Calculation	Control System	Comments	Rev
~~F23CALC40018~~	~~FDI23592A~~	~~2000-F23-PID-PR-40005~~	~~D TO FIM III LINE 1S DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23521_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE~~					~~FDI23592A = FIT23592_PV - FIC23521_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23592_PV~~	~~m³/h~~	~~FIM III LINE 1 SOUTH FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23592A~~	~~m³/h~~	~~D TO FIM III LINE 1S DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40019~~	~~FDI23592B~~	~~2000-F23-PID-PR-40005~~	~~B TO FIM III LINE 1S DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23541_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE~~					~~FDI23592B = FIT23592_PV - FIC23541_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23592_PV~~	~~m³/h~~	~~FIM III LINE 1 SOUTH FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23592B~~	~~m³/h~~	~~B TO FIM III LINE 1S DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40020~~	~~FDI23593A~~	~~2000-F23-PID-PR-40005~~	~~B TO FIM III LINE 1N DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23541_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE~~					~~FDI23593A = FIT23593_PV - FIC23541_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23593_PV~~	~~m³/h~~	~~FIM III LINE 1 NORTH FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23593A~~	~~m³/h~~	~~B TO FIM III LINE 1N DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40021~~	~~FDI23593B~~	~~2000-F23-PID-PR-40005~~	~~D TO FIM III LINE 1N DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23521_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE~~					~~FDI23593B = FIT23593_PV - FIC23521_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23593_PV~~	~~m³/h~~	~~FIM III LINE 1 NORTH FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23593B~~	~~m³/h~~	~~D TO FIM III LINE 1N DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40022~~	~~FDI23594A~~	~~2000-F23-PID-PR-40005~~	~~B TO FIM III LINE 3W DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23541_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE~~					~~FDI23594A = FIT23594_PV - FIC23541_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23594_PV~~	~~m³/h~~	~~FIM III LINE 3 WEST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23594A~~	~~m³/h~~	~~B TO FIM III LINE 3W DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40023~~	~~FDI23594B~~	~~2000-F23-PID-PR-40005~~	~~D TO FIM III LINE 3W DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23521_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE~~					~~FDI23594B = FIT23594_PV - FIC23521_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23594_PV~~	~~m³/h~~	~~FIM III LINE 3 WEST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23594B~~	~~m³/h~~	~~D TO FIM III LINE 3W DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40024~~	~~FDI23595A~~	~~2000-F23-PID-PR-40005~~	~~D TO FIM III LINE 3E DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23521_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE~~					~~FDI23595A = FIT23595_PV - FIC23521_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23595_PV~~	~~m³/h~~	~~FIM III LINE 3 EAST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23595A~~	~~m³/h~~	~~D TO FIM III LINE 3E DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40025~~	~~FDI23595B~~	~~2000-F23-PID-PR-40005~~	~~B TO FIM III LINE 3E DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23541_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE~~					~~FDI23595B = FIT23595_PV - FIC23541_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23595_PV~~	~~m³/h~~	~~FIM III LINE 3 EAST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23595B~~	~~m³/h~~	~~B TO FIM III LINE 3E DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40026~~	~~FDI23596A~~	~~2000-F23-PID-PR-40005~~	~~B TO FIM III LINE 2W DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23541_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE~~					~~FDI23596A = FIT23596_PV - FIC23541_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23596_PV~~	~~m³/h~~	~~FIM III LINE 2 WEST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23596A~~	~~m³/h~~	~~B TO FIM III LINE 2W DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40027~~	~~FDI23596B~~	~~2000-F23-PID-PR-40005~~	~~D TO FIM III LINE 2W DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23521_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE~~					~~FDI23596B = FIT23596_PV - FIC23521_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23596_PV~~	~~m³/h~~	~~FIM III LINE 2 WEST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23596B~~	~~m³/h~~	~~D TO FIM III LINE 2W DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40028~~	~~FDI23597A~~	~~2000-F23-PID-PR-40005~~	~~D TO FIM III LINE 2E DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23521_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 1 DISCHARGE FLOWRATE~~					~~FDI23597A = FIT23597_PV - FIC23521_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23597_PV~~	~~m³/h~~	~~FIM III LINE 2 EAST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23597A~~	~~m³/h~~	~~D TO FIM III LINE 2E DIFF FLOW~~	~~x~~							~~5~~
~~F23CALC40029~~	~~FDI23597B~~	~~2000-F23-PID-PR-40005~~	~~B TO FIM III LINE 2E DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23541_PV~~	~~m³/h~~	~~FIM III BOOSTER PUMP 2 DISCHARGE FLOWRATE~~					~~FDI23597B = FIT23597_PV - FIC23541_PV~~	~~MOSCAD RTU~~		~~5~~
						~~2~~	~~FIT23597_PV~~	~~m³/h~~	~~FIM III LINE 2 EAST FLOW TRANSMITTER~~								~~5~~
						~~3~~	~~FDI23597B~~	~~m³/h~~	~~B TO FIM III LINE 2E DIFF FLOW~~	~~x~~							~~5~~
F75CALC40001	FDI75611	2000-F75-PID-PR-25008	DD3&4 RETURN DIFF FLOW	m³/h		1	FIT75333_PV	m³/h	DECANT DAM 4 RETURN FLOW TRANSMITTER					FDI75611 = FIT75611_PV - FIT75333_PV - FIT75682_PV	MOSCAD RTU		5
						2	FIT75682_PV	m³/h	DECANT DAM 3 RETURN FLOW TRANSMITTER								5
						3	FIT75611_PV	m³/h	DECANT DAM 3 & 4 RETURN FLOW TRANSMITTER								5
						4	FDI75611	m³/h	DD3&4 RETURN DIFF FLOW	x							5
F75CALC40002	FDI75329	2000-F75-PID-PR-40002	FIM III DEC RET DIFF FLOW	m³/h		1	FIT75319_PV	m³/h	FIM III CELL 1 DECANT PUMP FLOW TRANSMITTER					FDI75329 = FIT75329_PV - FIT75328_PV - FIT75326_PV - FIT75319_PV	MOSCAD RTU		2
						2	FIT75326_PV	m³/h	FIM III CELL 2 DECANT PUMP FLOW TRANSMITTER								0
						3	FIT75328_PV	m³/h	FIM III CELL 3 DECANT POND **FLOW** TRANSMITTER								5
						4	FIT75329_PV	m³/h	FIM III DECANT FLOW TRANSMITTER								0
						5	FDI75329	m³/h	FIM III DEC RET DIFF FLOW	x							2
F75CALC40003	FDI75793	2000-F75-PID-PR-40009	FIM III SEEP DIFF FLOW	m³/h		1	FIT75730_PV	m³/h	FIM III SEEPAGE PUMP 1 FLOW TRANSMITTER					FDI75793 = FIT75793_PV - FIT75730_PV - FIT75731_PV - FIT75732_PV - FIT75733_PV - FIT75785_PV	MOSCAD RTU		2
						2	FIT75731_PV	m³/h	FIM III SEEPAGE PUMP 2 FLOW TRANSMITTER								0
						3	FIT75732_PV	m³/h	FIM III SEEPAGE PUMP 3 FLOW TRANSMITTER								0
						4	FIT75733_PV	m³/h	FIM III SEEPAGE PUMP 4 FLOW TRANSMITTER								0
						5	FIT75785_PV	m³/h	FIM III TOE DRAIN PUMP FLOW TRANSMITTER								0

CALCULATIONS

Tag Source	Tag Name	P&ID	Description (Software Tag Only)	Unit	Hold	Inputs #	Inputs Tag	Inputs Unit	Inputs Description	Calc. Value	x = Operator Input * = Engineer Input	Min	Max	Calculation	Control System	Comments	Rev
						6	FIT75793_PV	m³/h	FIM III SEEPAGE WATER FLOW TRANSMITTER								0
						7	FDI75793	m³/h	FIM III SEEP DIFF FLOW	x							2
F75CALC40004	FDI75792	2000-F75-PID-PR-40012	FIM III UD DIFF FLOW	m³/h		1	FIT75789_PV	m³/h	FIM III UNDERDRAINAGE PUMP 1 FLOW TRANSMITTER								2
						2	FIT75790_PV	m³/h	FIM III UNDERDRAINAGE PUMP 2 FLOW TRANSMITTER					FDI75792 = FIT75792_PV - FIT75789_PV - FIT75790_PV	MOSCAD RTU		0
						3	FIT75792_PV	m³/h	FIM III UNDERDRAINAGE FLOW TRANSMITTER								2
						4	FDI75792	m³/h	FIM III UD DIFF FLOW	x							2
F75CALC40005	FDI75321	2000-F75-PID-PR-40014	BW TO F75TK651 DIFF FLOW	m³/h		1	FIT17503_PV	m³/h	BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER								5
						2	FIT75321_PV	m³/h	BORE WATER TO FIM III SEAL WATER TANK FLOW TRANSMITTER 2					FDI75321 = FIT17503_PV - FIT75321_PV	MOSCAD RTU		5
						3	FDI75321	m³/h	BW TO F75TK651 DIFF FLOW	x							5
~~F75CALC40012~~	~~FDI30115~~	~~2000-F75-PR-PID-40025~~	~~LINE C TO FIM II F DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23452_PV~~	~~m³/h~~	~~FINAL TAILS PUMP 4 DISCHARGE FLOWRATE~~							~~Existing PLC~~	5
						~~2~~	~~FIT30115_PV~~	~~m³/h~~	~~LINE C TO FIM II CELL F FLOWRATE~~					~~FDI30115 = FIC23452_PV - FIT30115_PV~~	~~MOSCAD RTU~~		5
						~~3~~	~~FDI30115~~	~~m³/h~~	~~LINE C TO FIM II F DIFF FLOW~~	~~x~~							5
~~F75CALC40013~~	~~FDI30116~~	~~2000-F75-PR-PID-40025~~	~~LINE A TO FIM II F DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23441_PV~~	~~m³/h~~	~~FINAL TAILS PUMP 3 DISCHARGE FLOWRATE~~							~~Existing PLC~~	5
						~~2~~	~~FIT30116_PV~~	~~m³/h~~	~~LINE A TO FIM II CELL F FLOWRATE~~					~~FDI30116 = FIC23441_PV - FIT30116_PV~~	~~MOSCAD RTU~~		5
						~~3~~	~~FDI30116~~	~~m³/h~~	~~LINE A TO FIM II F DIFF FLOW~~	~~x~~							5
~~F75CALC40014~~	~~FDI30117~~	~~2000-F75-PR-PID-40026~~	~~LINE C TO FIM II E DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23452_PV~~	~~m³/h~~	~~FINAL TAILS PUMP 4 DISCHARGE FLOWRATE~~							~~Existing PLC~~	5
						~~2~~	~~FIT30117_PV~~	~~m³/h~~	~~LINE C TO FIM II CELL E FLOWRATE~~					~~FDI30117 = FIC23452_PV - FIT30117_PV~~	~~MOSCAD RTU~~		5
						~~3~~	~~FDI30117~~	~~m³/h~~	~~LINE C TO FIM II E DIFF FLOW~~	~~x~~							5
~~F75CALC40015~~	~~FDI30118~~	~~2000-F75-PR-PID-40026~~	~~LINE C TO FIM II G DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23452_PV~~	~~m³/h~~	~~FINAL TAILS PUMP 4 DISCHARGE FLOWRATE~~							~~Existing PLC~~	5
						~~2~~	~~FIT30118_PV~~	~~m³/h~~	~~LINE C TO FIM II CELL G FLOWRATE~~					~~FDI30118 = FIC23452_PV - FIT30118_PV~~	~~MOSCAD RTU~~		5
						~~3~~	~~FDI30118~~	~~m³/h~~	~~LINE C TO FIM II G DIFF FLOW~~	~~x~~							5
~~F75CALC40016~~	~~FDI30119~~	~~2000-F75-PR-PID-40026~~	~~LINE A TO FIM II E DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23441_PV~~	~~m³/h~~	~~FINAL TAILS PUMP 3 DISCHARGE FLOWRATE~~							~~Existing PLC~~	5
						~~2~~	~~FIT30119_PV~~	~~m³/h~~	~~LINE A TO FIM II CELL E FLOWRATE~~					~~FDI30119= FIC23441_PV - FIT30119_PV~~	~~MOSCAD RTU~~		5
						~~3~~	~~FDI30119~~	~~m³/h~~	~~LINE A TO FIM II E DIFF FLOW~~	~~x~~							5
~~F75CALC40017~~	~~FDI30120~~	~~2000-F75-PR-PID-40026~~	~~LINE A TO FIM II G DIFF FLOW~~	~~m³/h~~		~~1~~	~~FIC23441_PV~~	~~m³/h~~	~~FINAL TAILS PUMP 3 DISCHARGE FLOWRATE~~							~~Existing PLC~~	5
						~~2~~	~~FIT30120_PV~~	~~m³/h~~	~~LINE A TO FIM II CELL G FLOWRATE~~					~~FDI30120 = FIC23441_PV - FIT30120_PV~~	~~MOSCAD RTU~~		5
						~~3~~	~~FDI30120~~	~~m³/h~~	~~LINE A TO FIM II G DIFF FLOW~~	~~x~~							5
F75CALC40018	FDI75270	2000-F75-PID-PR-40031	FIM II EXT DEC RET DIFF FLOW	m³/h		1	FIT75269_PV	m³/h	FIM II EXTENSION CELL G DECANT FLOW TRANSMITTER								4
						2	FIT75265_PV	m³/h	FIM II CELL G UNDERDRAINAGE PUMP FLOW TRANSMITTER					FDI75270 = FIT75270_PV - FIT75269_PV - FIT75265_PV	MOSCAD RTU		5
						3	FIT75270_PV	m³/h	FIM II EXTENSION CELL G DECANT FLOW TRANSMITTER 2								4
						4	FDI75270	m³/h	FIM II EXT DEC RET DIFF FLOW	x							2
F75CALC40020	FDI75794	2000-F175-PID-PR-40002	FIM III SEEP RET DIFF FLOW	m³/h		1	FIT75793_PV	m³/h	FIM III SEEPAGE WATER FLOW TRANSMITTER								0
						2	FIT75794_PV	m³/h	FIM III SEEPAGE WATER FLOW TRANSMITTER 2					FDI75794 = FIT75794_PV - FIT75793_PV	MOSCAD RTU		3
						3	FDI75794	m³/h	FIM III SEEP RET DIFF FLOW	x							2

ALARM DATABASE

Tag Name	P&ID	Abbreviated Description (Faceplate)	HOLD	Type	Alarm State	Alarm Value	Alarm Units	On Delay (sec)	Alarm Masking Condition	Control System	Impact	Urgency	Priority	Operator Response	Rev
FI23592	2000-F23-PID-PR-40005	FIM III LINE 1 SOUTH FLOW		Analogue	A_L	100	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
~~FDI23592A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 1S DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23592A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 1S DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23592B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 1S DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23592B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 1S DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
FI23593	2000-F23-PID-PR-40005	FIM III LINE 1 NORTH FLOW		Analogue	A_L	100	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
~~FDI23593A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 1N DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23593A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 1N DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23593B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 1N DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23593B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 1N DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
FI23594	2000-F23-PID-PR-40005	FIM III LINE 3 WEST FLOW		Analogue	A_L	100	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
~~FDI23594A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 3W DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23594A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 3W DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23594B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 3W DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23594B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 3W DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
FI23595	2000-F23-PID-PR-40005	FIM III LINE 3 EAST FLOW		Analogue	A_L	100	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
~~FDI23595A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 3E DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23595A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 3E DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23595B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 3E DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23595B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 3E DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
FI23596	2000-F23-PID-PR-40005	FIM III LINE 2 WEST FLOW		Analogue	A_L	100	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
~~FDI23596A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 2W DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23596A~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 2W DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23596B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 2W DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23596B~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 2W DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
FI23597	2000-F23-PID-PR-40005	FIM III LINE 2 EAST FLOW		Analogue	A_L	100	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	4
~~FDI23597A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 2E DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23597A~~	~~2000-F23-PID-PR-40005~~	~~LINE D TO FIM III 2E DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23597B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 2E DIFF FLOW~~		~~Analogue~~	~~A_H~~	~~50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
~~FDI23597B~~	~~2000-F23-PID-PR-40005~~	~~LINE B TO FIM III 2E DIFF FLOW~~		~~Analogue~~	~~A_L~~	~~-50~~	~~m³/h~~	~~30~~	~~-~~	~~MOSCAD RTU~~	~~Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.~~	~~Rapid (>5 to <15 minutes)~~	~~High~~	~~Potential tails line leak. Operator to investigate in field~~	~~5~~
XI23581	2000-F23-PID-PR-40005	F23HY421 FAULT	~~006~~	Digital	Fault			3	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Rapid (>5 to <15 minutes)	High	Hydraulic power pack fault. Operator to investigate in field	5
FDI75611	**2000-F75-PID-PR-25008**	**DD3&4 RETURN DIFF FLOW**		**Analogue**	**A_H**	**50**	**m³/h**	**600**	**F75-PW-74 NOT RUN AND F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential decant water line leak. Operator to investigate in field**	**5**
FDI75611	**2000-F75-PID-PR-25008**	**DD3&4 RETURN DIFF FLOW**		**Analogue**	**A_L**	**-50**	**m³/h**	**600**	**F75-PW-74 NOT RUN AND F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN**	**Yokogawa DCS**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential decant water line leak. Operator to investigate in field**	**5**
LI75268	2000-F75-PID-PR-40001	FIM II CELL G LEVEL		**Analogue**	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	5
LI75268	2000-F75-PID-PR-40001	FIM II CELL G LEVEL		**Analogue**	A_L	**5**	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	5
~~**LIC75268**~~	~~**2000-F75-PID-PR-40001**~~	~~**FIM II CELL G LEVEL**~~		~~**Analogue PID**~~	~~**A_LL**~~	~~**5**~~	~~**%**~~	~~**5**~~	~~-~~	~~**MOSCAD RTU**~~	~~**Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000**~~	~~**Equipment Tripped**~~	~~**Logging**~~	~~**Equipment tripped. Operator to investigate & restart if safe to do so.**~~	~~**5**~~
FI75269	2000-F75-PID-PR-40001	F75PW664 DISCH FLOW		Analogue	A_L	20	m³/h	5	F75-PW-664 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75269	2000-F75-PID-PR-40001	F75PW664 DISCH FLOW		Analogue	A_LL	10	m³/h	5	F75-PW-664 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
LI75267	2000-F75-PID-PR-40001	FIM II CELL G AS LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Antiscalant IBC will require replacement shortly. Advise Field Operator to investigate	3

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LI75267	2000-F75-PID-PR-40001	FIM II CELL G AS LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	High	Antiscalant IBC to be replaced immediately. Advise Field Operator to take replacement IBC to FIM II Cell G.	3
LI75318	2000-F75-PID-PR-40002	FIM III CELL 1 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
LI75318	2000-F75-PID-PR-40002	FIM III CELL 1 LEVEL		Analogue	A_L	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
FI75319	2000-F75-PID-PR-40002	F75PW600 DISCH FLOW		Analogue	A_L	20	m³/h	5	F75-PW-600 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75329	2000-F75-PID-PR-40002	FIM III DECANT FLOW		Analogue	A_L	20	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FDI75329	2000-F75-PID-PR-40002	FIM II DECANT DIFF FLOW		Analogue	A_H	25	m³/h	600	F75-PW-600 NOT RUN AND F75-PW-601 NOT RUN AND F75-PW-602 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
FDI75329	2000-F75-PID-PR-40002	FIM II DECANT DIFF FLOW		Analogue	A_L	-25	m³/h	600	F75-PW-600 NOT RUN AND F75-PW-601 NOT RUN AND F75-PW-602 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential tails line leak. Operator to investigate in field	5
LI75324	2000-F75-PID-PR-40002	FIM III CELL 1 AS LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Antiscalant IBC will require replacement shortly. Advise Field Operator to investigate	3
LI75324	2000-F75-PID-PR-40002	FIM III CELL 1 AS LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	High	Antiscalant IBC to be replaced immediately. Advise Field Operator to take replacement IBC to FIM III Cell 1.	3
LI75325	2000-F75-PID-PR-40003	FIM III CELL 2 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
LI75325	2000-F75-PID-PR-40003	FIM III CELL 2 LEVEL		Analogue	A_L	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
FI75326	2000-F75-PID-PR-40003	F75PW601 DISCH FLOW		Analogue	A_L	20	m³/h	5	F75-PW-601 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
LI75327	2000-F75-PID-PR-40003	FIM III CELL 3 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
LI75327	2000-F75-PID-PR-40003	FIM III CELL 3 LEVEL		Analogue	A_L	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
FI75328	2000-F75-PID-PR-40003	F75PW602 DISCH FLOW		Analogue	A_L	20	m³/h	5	F75-PW-602 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
LI75342	2000-F75-PID-PR-40003	FIM III CELL 2 AS LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Antiscalant IBC will require replacement shortly. Advise Field Operator to investigate	3
LI75342	2000-F75-PID-PR-40003	FIM III CELL 2 AS LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	High	Antiscalant IBC to be replaced immediately. Advise Field Operator to take replacement IBC to FIM III Cell 2.	3
LI75343	2000-F75-PID-PR-40003	FIM III CELL 3 AS LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Antiscalant IBC will require replacement shortly. Advise Field Operator to investigate	3
LI75343	2000-F75-PID-PR-40003	FIM III CELL 3 AS LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	High	Antiscalant IBC to be replaced immediately. Advise Field Operator to take replacement IBC to FIM III Cell 3.	3
LI75362	2000-F75-PID-PR-40004	DD4 O/F POND LEVEL		Analogue	A_HH	99	%	5	-	MOSCAD RTU	Environmental 1: Localised environmental impact, contained, with no regulatory reporting	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	4
LI75362	2000-F75-PID-PR-40004	DD4 O/F POND LEVEL		Analogue	A_H	95	%	5	-	MOSCAD RTU	Environmental 1: Localised environmental impact, contained, with no regulatory reporting	Rapid (>5 to <15 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
LI75362	2000-F75-PID-PR-40004	DD4 O/F POND LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate turret area	3
LI75362	2000-F75-PID-PR-40004	DD4 O/F POND LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75338	2000-F75-PID-PR-40004	F75PW606-608 DISCH PRESSURE		Analogue	A_HH	1400	kPa	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75338	2000-F75-PID-PR-40004	F75PW606-608 DISCH PRESSURE		Analogue	A_H	1350	kPa	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
PI75339	2000-F75-PID-PR-40004	DD4 RETURN PRESSURE		Analogue	A_HH	900	kPa	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75339	2000-F75-PID-PR-40004	DD4 RETURN PRESSURE		Analogue	A_H	850	kPa	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
LIC75332	2000-F75-PID-PR-40004	DECANT DAM 4 LEVEL		Analogue PID	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High dam level. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	3
LIC75332	2000-F75-PID-PR-40004	DECANT DAM 4 LEVEL		Analogue PID	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Low dam level. Check controllers are operating as intended. Advise Field Operator / Maintenance to investigate.	3
LIC75332	2000-F75-PID-PR-40004	DECANT DAM 4 LEVEL		Analogue PID	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FIC75333	2000-F75-PID-PR-40004	F75PW606/607/608 FLOWRATE		Analogue PID	A_L	20	m³/h	5	F75-PW-606 NOT RUN AND F75-PW-607 NOT RUN AND F75-PW-608 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
LI75364	2000-F75-PID-PR-40004	DD4 AS LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Antiscalant IBC will require replacement shortly. Advise Field Operator to investigate	3
LI75364	2000-F75-PID-PR-40004	DD4 AS LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Immediate (<5 minutes)	High	Antiscalant IBC to be replaced immediately. Advise Field Operator to take replacement IBC to Decant Dam 4.	3
FAL75335	2000-F75-PID-PR-40004	F75PW606 FLOW SWITCH		Digital	Alarm			3	F75-PW-606 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Equipment tripped. Operator to investigate & restart if safe to do so. Alarm enabled delay of 10s for pump start	3
FAL75336	2000-F75-PID-PR-40004	F75PW607 FLOW SWITCH		Digital	Alarm			3	F75-PW-607 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Equipment tripped. Operator to investigate & restart if safe to do so. Alarm enabled delay of 10s for pump start	3
FAL75337	2000-F75-PID-PR-40004	F75PW608 FLOW SWITCH		Digital	Alarm			3	F75-PW-608 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Equipment tripped. Operator to investigate & restart if safe to do so. Alarm enabled delay of 10s for pump start	3
LI75264	2000-F75-PID-PR-40007	F75PS559 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75264	2000-F75-PID-PR-40007	F75PS559 LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75264	2000-F75-PID-PR-40007	F75PS559 LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3

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FI75264	2000-F75-PID-PR-40007	F75PS559 DISCH FLOW		Analogue	A_L	10	m³/h	5	F75-PS-559 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75264	2000-F75-PID-PR-40007	F75PS559 DISCH FLOW		Analogue	A_LL	5	m³/h	5	F75-PS-559 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
LI75265	2000-F75-PID-PR-40007	F75PS593 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75265	2000-F75-PID-PR-40007	F75PS593 LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75265	2000-F75-PID-PR-40007	F75PS593 LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75265	2000-F75-PID-PR-40007	F75PS593 DISCH FLOW		Analogue	A_L	10	m³/h	5	F75-PS-593 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75265	2000-F75-PID-PR-40007	F75PS593 DISCH FLOW		Analogue	A_LL	5	m³/h	5	F75-PS-593 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
LI75365	2000-F75-PID-PR-40007	F75PS560 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75365	2000-F75-PID-PR-40007	F75PS560 LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75365	2000-F75-PID-PR-40007	F75PS560 LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75365	2000-F75-PID-PR-40007	F75PS560 DISCH FLOW		Analogue	A_L	10	m³/h	5	F75-PS-560 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75365	2000-F75-PID-PR-40007	F75PS560 DISCH FLOW		Analogue	A_LL	5	m³/h	5	F75-PS-560 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75730	2000-F75-PID-PR-40008	F75PS801 DISCH PRESSURE		Analogue	A_HH	700	kPa	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75730	2000-F75-PID-PR-40008	F75PS801 DISCH PRESSURE		Analogue	A_H	650	kPa	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
FI75730	2000-F75-PID-PR-40008	F75PS801 DISCH FLOW		Analogue	A_L	5	m³/h	5	F75-PS-801 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75730	2000-F75-PID-PR-40008	F75PS801 DISCH FLOW		Analogue	A_LL	2	m³/h	5	F75-PS-801 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75731	2000-F75-PID-PR-40008	F75PS802 DISCH PRESSURE		Analogue	A_HH	700	kPa	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75731	2000-F75-PID-PR-40008	F75PS802 DISCH PRESSURE		Analogue	A_H	650	kPa	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
FI75731	2000-F75-PID-PR-40008	F75PS802 DISCH FLOW		Analogue	A_L	5	m³/h	5	F75-PS-802 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75731	2000-F75-PID-PR-40008	F75PS802 DISCH FLOW		Analogue	A_LL	2	m³/h	5	F75-PS-802 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75732	2000-F75-PID-PR-40008	F75PS803 DISCH PRESSURE		Analogue	A_HH	700	kPa	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75732	2000-F75-PID-PR-40008	F75PS803 DISCH PRESSURE		Analogue	A_H	650	kPa	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
FI75732	2000-F75-PID-PR-40008	F75PS803 DISCH FLOW		Analogue	A_L	5	m³/h	5	F75-PS-803 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75732	2000-F75-PID-PR-40008	F75PS803 DISCH FLOW		Analogue	A_LL	2	m³/h	5	F75-PS-803 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75733	2000-F75-PID-PR-40008	F75PS804 DISCH PRESSURE		Analogue	A_HH	700	kPa	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PI75733	2000-F75-PID-PR-40008	F75PS804 DISCH PRESSURE		Analogue	A_H	650	kPa	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	High pump discharge pressure. Advise Operator / Maintenance to investigate	3
FI75733	2000-F75-PID-PR-40008	F75PS804 DISCH FLOW		Analogue	A_L	5	m³/h	5	F75-PS-804 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75733	2000-F75-PID-PR-40008	F75PS804 DISCH FLOW		Analogue	A_LL	2	m³/h	5	F75-PS-804 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75793	2000-F75-PID-PR-40009	FIM III SEEPAGE FLOW		Analogue	A_L	20	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FDI75793	2000-F75-PID-PR-40009	FIM III SEEPAGE DIFF FLOW		Analogue	A_H	25	m³/h	**600**	**F75-PS-558 NOT RUN AND F75-PS-801 NOT RUN AND F75-PS-802 NOT RUN AND F75-PS-803 NOT RUN AND F75-PS-804 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential seepage water return line leak. Operator to investigate in field	**5**
FDI75793	2000-F75-PID-PR-40009	FIM III SEEPAGE DIFF FLOW		Analogue	A_L	-25	m³/h	**600**	**F75-PS-558 NOT RUN AND F75-PS-801 NOT RUN AND F75-PS-802 NOT RUN AND F75-PS-803 NOT RUN AND F75-PS-804 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential seepage water return line leak. Operator to investigate in field	**5**
LI75785	2000-F75-PID-PR-40011	F75PS558 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75785	2000-F75-PID-PR-40011	F75PS558 LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75785	2000-F75-PID-PR-40011	F75PS558 LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75785	2000-F75-PID-PR-40011	F75PS558 DISCH FLOW		Analogue	A_L	10	m³/h	5	F75-PS-558 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75785	2000-F75-PID-PR-40011	F75PS558 DISCH FLOW		Analogue	A_LL	5	m³/h	5	F75-PS-558 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
LI75789	2000-F75-PID-PR-40012	F75PS556 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75789	2000-F75-PID-PR-40012	F75PS556 LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75789	2000-F75-PID-PR-40012	F75PS556 LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3

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FI75789	2000-F75-PID-PR-40012	F75PS556 DISCH FLOW		Analogue	A_L	10	m³/h	5	F75-PS-556 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75789	2000-F75-PID-PR-40012	F75PS556 DISCH FLOW		Analogue	A_LL	5	m³/h	5	F75-PS-556 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
LI75790	2000-F75-PID-PR-40012	F75PS557 LEVEL		Analogue	A_H	99	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75790	2000-F75-PID-PR-40012	F75PS557 LEVEL		Analogue	A_L	10	%	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Operator to check pump is in automatic mode, else advise Field Operator / Maintenance to investigate sump area	3
LI75790	2000-F75-PID-PR-40012	F75PS557 LEVEL		Analogue	A_LL	5	%	5	-	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75790	2000-F75-PID-PR-40012	F75PS557 DISCH FLOW		Analogue	A_L	10	m³/h	5	F75-PS-557 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FI75790	2000-F75-PID-PR-40012	F75PS557 DISCH FLOW		Analogue	A_LL	5	m³/h	5	F75-PS-557 NOT RUN	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
FI75792	2000-F75-PID-PR-40012	FIM III U/DRAIN FLOW		Analogue	A_L	20	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FDI75792	2000-F75-PID-PR-40012	FIM III U/DRAIN DIFF FLOW		Analogue	A_H	25	m³/h	**600**	**F75-PS-556 NOT RUN AND F75-PS-557 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential underdrainage return line leak. Operator to investigate in field	**5**
FDI75792	2000-F75-PID-PR-40012	FIM III U/DRAIN DIFF FLOW		Analogue	A_L	-25	m³/h	**600**	**F75-PS-556 NOT RUN AND F75-PS-557 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential underdrainage return line leak. Operator to investigate in field	**5**
FDI75321	**2000-F75-PID-PR-40014**	**F75TK651 INFLOW DIFF**		**Analogue**	**A_H**	**10**	**m³/h**	**600**	**SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN**	**MOSCAD RTU**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential bore water line leak. Operator to investigate in field**	**5**
FDI75321	**2000-F75-PID-PR-40014**	**F75TK651 INFLOW DIFF**		**Analogue**	**A_L**	**-10**	**m³/h**	**600**	**SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN**	**MOSCAD RTU**	**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**	**Rapid (>5 to <15 minutes)**	**High**	**Potential bore water line leak. Operator to investigate in field**	**5**
~~**FDI30115**~~	~~**2000-F75-PR-PID-40025**~~	~~**LINE C TO FIM II F DIFF FLOW**~~		~~**Analogue**~~	~~**A_H**~~	~~**50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30115**~~	~~**2000-F75-PR-PID-40025**~~	~~**LINE C TO FIM II F DIFF FLOW**~~		~~**Analogue**~~	~~**A_L**~~	~~**-50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30116**~~	~~**2000-F75-PR-PID-40025**~~	~~**LINE A TO FIM II F DIFF FLOW**~~		~~**Analogue**~~	~~**A_H**~~	~~**50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30116**~~	~~**2000-F75-PR-PID-40025**~~	~~**LINE A TO FIM II F DIFF FLOW**~~		~~**Analogue**~~	~~**A_L**~~	~~**-50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30117**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE C TO FIM II E DIFF FLOW**~~		~~**Analogue**~~	~~**A_H**~~	~~**50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30117**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE C TO FIM II E DIFF FLOW**~~		~~**Analogue**~~	~~**A_L**~~	~~**-50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30118**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE C TO FIM II G DIFF FLOW**~~		~~**Analogue**~~	~~**A_H**~~	~~**50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30118**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE C TO FIM II G DIFF FLOW**~~		~~**Analogue**~~	~~**A_L**~~	~~**-50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30119**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE A TO FIM II E DIFF FLOW**~~		~~**Analogue**~~	~~**A_H**~~	~~**50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30119**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE A TO FIM II E DIFF FLOW**~~		~~**Analogue**~~	~~**A_L**~~	~~**-50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30120**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE A TO FIM II G DIFF FLOW**~~		~~**Analogue**~~	~~**A_H**~~	~~**50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
~~**FDI30120**~~	~~**2000-F75-PR-PID-40026**~~	~~**LINE A TO FIM II G DIFF FLOW**~~		~~**Analogue**~~	~~**A_L**~~	~~**-50**~~	~~**m³/h**~~	~~**30**~~	~~**-**~~	~~**MOSCAD RTU**~~	~~**Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.**~~	~~**Rapid (>5 to <15 minutes)**~~	~~**High**~~	~~**Potential tails line leak. Operator to investigate in field**~~	**5**
FI75270	2000-F75-PID-PR-40031	FIM II EXT CELL G DECANT FLOW2		Analogue	A_L	20	m³/h	5	F75-PW-664 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FDI75270	2000-F75-PID-PR-40031	FIM II EXT DECANT DIFF FLOW		Analogue	A_H	25	m³/h	**600**	**F75-PS-593 NOT RUN AND F75-PW-664 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential decant water return line leak. Operator to investigate in field	**5**
FDI75270	2000-F75-PID-PR-40031	FIM II EXT DECANT DIFF FLOW		Analogue	A_L	-25	m³/h	**600**	**F75-PS-593 NOT RUN AND F75-PW-664 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential decant water return line leak. Operator to investigate in field	**5**
FI75794	2000-F175-PID-PR-40002	FIM III SEEPAGE FLOW 2		Analogue	A_L	20	m³/h	5	-	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
FDI75794	2000-F175-PID-PR-40002	FIM III SEEPAGE DIFF FLOW 2		Analogue	A_H	25	m³/h	**600**	**F75-PS-558 NOT RUN AND F75-PS-801 NOT RUN AND F75-PS-802 NOT RUN AND F75-PS-803 NOT RUN AND F75-PS-804 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential seepage water return line leak. Operator to investigate in field	**5**
FDI75794	2000-F175-PID-PR-40002	FIM III SEEPAGE DIFF FLOW 2		Analogue	A_L	-25	m³/h	**600**	**F75-PS-558 NOT RUN AND F75-PS-801 NOT RUN AND F75-PS-802 NOT RUN AND F75-PS-803 NOT RUN AND F75-PS-804 NOT RUN**	MOSCAD RTU	Financial 2: Loss equivalent greater than 1 hour and up to 6 hours production. Repairs costing $50k-$500k.	Rapid (>5 to <15 minutes)	High	Potential seepage water return line leak. Operator to investigate in field	**5**
PIC17504	2000-F175-PID-PR-40002	SEGDT01PP1/PP2/PP3 PRESSURE		Analogue PID	A_HH	800	kPa	5	SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Equipment Tripped	Logging	Equipment tripped. Operator to investigate & restart if safe to do so.	3
PIC17504	2000-F175-PID-PR-40002	SEGDT01PP1/PP2/PP3 PRESSURE		Analogue PID	A_H	600	kPa	5	SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3
PIC17504	2000-F175-PID-PR-40002	SEGDT01PP1/PP2/PP3 PRESSURE		Analogue PID	A_L	200	kPa	5	SEGDT01PP1 NOT RUN AND SEGDT01PP2 NOT RUN AND SEGDT01PP3 NOT RUN	MOSCAD RTU	Financial 1: Loss equivalent up to 1 hour production. Repairs costing <$50,000	Prompt (>15 to <30 minutes)	Low	Potential blockage, rupture or pump issues. Advise Field Operator and Maintenance team to investigate immediately	3